Competitive Radicalization among Agents: Evidence from the Holocaust

Luis Bosshart
Matthias Weigand*

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Abstract

We study the dynamics of radicalization under dictatorship. We reconstruct the internal organization of the Nazi regime by linking comprehensive records from the party, state, economy, and security apparatus, to records of victims of persecution and secret governance records. These sources show a system of overlapping jurisdictions in which the ruler rarely intervened and rival organizations acted in his name while competing for his approval. We develop a model in which a ruler trades off control against initiative induced by competition among agents. We test the model using variation in the movements of military and paramilitary troops and the timing and location of five million acts of persecution during the Holocaust. We find that persecution was higher where command authority was less concentrated. Event studies show that killing intensified after units under rival commands arrived, but not after non-rival units arrived. We also examine central directives and high-frequency private information from diaries of central figures in the extermination of European Jews. We find that agents took initiative ahead of documented central instructions, while interpreting cues and competing for approval. These findings suggest that radicalization can arise from organizational structure.

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JEL Classification: P00, P50, D23, D70, D73, D74, H11, H70, N44

*Bosshart: Harvard Academy for International and Area Studies. Email: lbosshart@fas.harvard.edu. Weigand: Harvard University, Department of Economics. Email: mweigand@g.harvard.edu. We thank Melissa Dell, Ed Glaeser, Leander Heldring, Kun Heo, Varun K, Anirudh Sankar, and Raúl Sánchez de la Sierra for helpful comments and encouragement.

1 Introduction

Delegation creates a trade-off between initiative and control. A principal may know the direction in which agents should act without being able to specify the appropriate action in every contingency or fully monitor their choices. Dictatorship provides a consequential setting to study this trade-off. Dictatorship concentrates formal authority in a ruler, but policy depends on agents who interpret and implement the ruler's objectives. A central organizational problem is therefore how to induce initiative while retaining control over the policy that emerges from their interaction.[1]

Many dictatorships combine concentrated formal authority with parallel state structures. Formal and informal state, party, and quasi-state institutions operate in parallel with overlapping responsibilities. Studying these structures and their consequences is difficult because such data are hard to obtain in dictatorships, or may misrepresent how power is exercised in practice.

This project aims to understand the organizational structure and radicalization dynamics in one of the most consequential political regimes in history. The Nazi regime organized violence on a massive scale. Leading accounts attribute the radicalization of policy and scale of persecution to competition among overlapping agencies (Broszat, 1969; Mommsen, 1976; Hüttenberger, 1976).[2]

The main contribution of this paper is to present new data describing the formal and informal organizational structure of the Nazi regime and to provide the first quantitative analysis of its relationship to the Holocaust. Our data cover the principal organizations of state and society: officials, offices, and institutional reorganizations from state directories; a near-universe of party membership registers; all stock corporations and their board members and managers; and military, paramilitary, and police units and their stationing histories. These sources identify potential actors and organizations over time. We complement these sources with secret internal records from the Reich and Party Chancelleries, the central interfaces between state and party organizations and the

[1] A related problem arises when open-ended objectives are delegated to autonomous artificial agents. Recent work identifies selection pressure and destabilizing interaction as risks of multi-agent systems (Hammond et al., 2025).

[2] We examine how this organizational structure contributed to the radicalization of policy that culminated in the Holocaust. Contrary to a view of dictatorship as a system of centralized command, key aspects of Nazi governance relied on decentralized initiatives and overlapping jurisdictions that induced competition among agencies. This institutional design is hypothesized to have incentivized agencies to escalate repressive measures in an effort to outperform rivals and anticipate Hitler's intentions. Some of the earliest measures of persecution, including legal discrimination and the establishment of concentration camps, emerged from decentralized initiatives and were later incorporated into official policy. The extermination of European Jews likewise developed through the central endorsement of mass killings initiated under local competition among agencies. We provide further detail through case studies in Section 3 and examine overlapping authority and its consequences in other contexts in Appendix B, including Stalin's Great Terror, Imperial Japan, and China's Cultural Revolution and contemporary local policy experiments.

leadership. We link these sources to the most comprehensive database of Holocaust victims, which records the dates and places of residence and persecution of 4.8 million people (Yad Vashem, 2026).

We begin by documenting four facts about the organizational structure of the Nazi regime. First, between 1931 and 1936, the number of institutions listed in state directories doubled, while the number of effective institutions identified in internal governance records tripled. Second, many of these institutions were created by extraordinary decree and tied to specific individuals. Third, their jurisdictions overlapped extensively: by 1936, almost no policy area fell under the exclusive authority of a single institution. Fourth, internal records show a reversal: before the dictatorship, the head of government usually attended cabinet-level meetings and was rarely invoked in his absence; under Hitler, almost every decision invoked his authority, yet cabinet meetings with him virtually disappeared. The proliferation of parallel institutions therefore coincided with the dismantling of bodies for collective decision-making, representation, and conflict resolution, including the cabinet and corporatist organizations.

To clarify the organizational incentives of this system, we develop a model of delegation under dictatorship, building on the entrepreneurial administration of Breton and Wintrobe (1986). Formal authority is concentrated in the ruler, but he cannot credibly bind himself to reward initiative. Without a cabinet or other body to settle conflicts, he governs through agents who interpret his objectives and anticipate his later approval. He can assign two agents the same domain and promote the one who acts, or exercise direct control over a single agent. Rivalry elicits action before the ruler learns whether it is appropriate; direct control waits for that information but risks coming too late. Radicalization may therefore emerge endogenously from organizational structure.

We examine the consequences of this organizational structure during the Holocaust. Our baseline analysis uses disaggregated data on the murder of persecution victims across time and space to test whether killing increased after troops under rival commands entered a locality. We first show descriptively that a larger share of residents was murdered where command control was less concentrated. We then estimate a stacked event study of the entry of a second unit in a cell held by one command. We find that cumulative deaths among residents rose steeply after a rival command entered, with no increase before its entry. Importantly, we find that such an increase is absent when a further unit of the same command enters. The rise therefore followed the entry of an independent command rather than additional troop capacity in itself. However, this pattern

may also reflect time-varying differences across locations, such as strategic centrality.

We therefore complement the baseline with two designs that compare locations with similar expected exposure to overlapping commands but different realized troop presence. The first instruments the presence of a second command with distance to the destinations of documented transfers. It compares cells near each destination with cells near places the command could have reached instead. The second compares cells that the invasion plan, drawn up before the campaign, scheduled to fall behind the front in the same month. The identifying assumption is that, conditional on planned exposure, residual variation in realized exposure reflects campaign contingencies rather than local characteristics. The increase after rival entry persists under both designs.

We conclude by examining high-frequency private information from diaries of five senior Nazi officials who played central roles in the extermination of European Jews.[3] These records document intense rivalry within the regime. A senior SS official observes in his diary: "People think that in the 'authoritarian' state everything is ordered from above, and have no idea that up there an unceasing struggle for power prevails." We code 38,331 dated actions from the five diaries. Disputes appear on one in four days with an entry. Among policy-setting actions, 97 percent were agents' own decisions and 3 percent were superiors' orders or prohibitions. Pairwise comparisons within rounds of actions on the same policy support the model's predicted sequence.

Taken together, these results illustrate a general problem of large dictatorships: rulers cannot fully direct the organizations through which they govern. Overlapping authority can elicit initiative, limit dependence on any one agent, and implement policies the center has not specified.

This paper brings an important data point to the theoretical literature on dictatorship and delegation. For example, Wintrobe (1998) and Svolik (2012) study how dictators secure the loyalty of their agents and share power with them; Egorov and Sonin (2011) explore a trade-off between loyalty and competence among subordinates; Aghion and Tirole (1997) distinguish formal from real authority in organizations; and Lazear and Rosen (1981), Green and Stokey (1983), and Tullock (1980) study how rewards based on relative performance elicit effort. Closest to our setting, Dewatripont and Tirole (1999) show how organizations can use competition to elicit noncontractible

[3]They are Hans Frank, governor-general of occupied Poland; Heinrich Himmler, head of the SS and the German police; Alfred Rosenberg, minister for the occupied eastern territories; and Erich von dem Bach-Zelewski, Higher SS and Police Leader in central Russia, Himmler's plenipotentiary for anti-partisan warfare, and Joseph Goebbels, minister of propaganda and Gauleiter of Berlin.

effort.[4] In China, a similar combination of vertical integration and decentralized initiative elicited effort through policy experimentation and produced growth (Xu, 2011; Wang and Yang, 2025). In contrast, we describe the organizational structure and production of violence of a regime comprehensively, across state, party, economy, and armed forces, and trace its consequences for mass killings in Eastern Europe, a subject that has received limited quantitative attention despite its historical importance. Our findings suggest that policy radicalization may arise endogenously from organizational structure.[5] These results qualify accounts of autocracy in which the center issues orders and directs operations, while subordinates obey. Such models therefore implicitly assume that state bureaucracy acts as a monopoly, ruling out competition.[6]

We also contribute to the economics of violence. At the individual level, Ager et al. (2022) find that public recognition of German fighter pilots during World War II increased both victory and death rates among their former peers. At the organizational level, prior work examines how central state institutions shape violence (Heldring, 2021; 2025; Meng et al., 2015; Markevich et al., 2025).[7] We complement this work by showing how organizational structure encourages decentralized initiative in the production of violence.[8] This emphasis on agency also connects our analysis to Dunia et al. (2026) and Marchais et al. (2021), who study the moral foundations of selection into armed organizations. The interaction between organizational incentives and individual discretion raise important questions on responsibility for the exercise of violence. At Nuremberg and elsewhere, perpetrators defend their actions by claiming that they merely followed orders, invoking a view of dictatorship in which control is centralized. Our evidence supports the qualitative argument of Breton and Wintrobe (1986) that officials exercised discretion over the timing and intensity of persecution. Violent radicalization could thus arise from initiative within an entrepreneurial

[4] A closely related literature studies the organization and consequences of direct and indirect rule (Tirole, 1986; Padró i Miquel and Yared, 2012; Banerjee and Iyer, 2005; Dell, 2010; Lowes and Montero, 2021; Henn et al., 2024).

[5] Some theories of dictatorship locate the problem of control in conflicting interests: political actors must be induced to support the ruler, and agents must be induced to pursue his objective. Our mechanism operates even when this alignment problem is moot. A ruler who cannot contract on initiative may use rivalry to elicit it from aligned agents, and rivalry then makes policy an equilibrium among agents rather than a choice of the ruler. A problem is therefore not only how to align agents with the ruler, but how to organize interaction among aligned agents; the absence of instruments to balance competing interests can itself develop into a central problem of dictatorship.

[6] These findings contrast with stylized models of bureaucracy as a system of "fixed and official jurisdictional areas," with authority "distributed in a stable way" and "strictly delimited." (Weber, 1946, p. 196)

[7] In autocracies, coercive institutions fragmented to prevent coups produce more, and less discriminate, violence (Greitens, 2016). In Colombia, rewards for reported kills led army units to kill civilians and report them as combat deaths (Acemoglu et al., 2020).

[8] Ager et al. (2022) and Heldring (2025) focus on the German Reich, home to about 3 percent of Holocaust victims. Our analysis covers millions of victims across Europe.

administration, rather than simply from orders passed down a bureaucratic hierarchy. These incentives help explain officials' initiative without excusing their choices. Responsibility rests both with agents who initiated and escalated persecution and with leaders who set objectives and sustained institutions that rewarded such initiative, even when no explicit orders were specified.

Finally, we contribute new data and a quantitative framework to extensive research on Nazi rule. First, we link data on the regime's actors, institutions, persecution, and internal governance at the level of persons, organizations, places, and time. These data provide a resource for research beyond the questions examined here. Second, we develop a framework that connects two explanations of Nazi rule often treated as alternatives. Intentionalist accounts emphasize Hitler's objectives and decisions, while structuralist accounts emphasize overlapping jurisdictions, conflict among agencies, and radicalization through the policy process (Kershaw, 1997; 1985). Intentionalist accounts are criticized for overstating Hitler's control over policy and understating institutional conflict and initiative from below. Accounts of competitive fragmentation, in turn, must explain how the regime remained effective until its final days despite its divisions. Our framework reconciles these views by combining centralized formal authority with decentralized real initiative.[9]

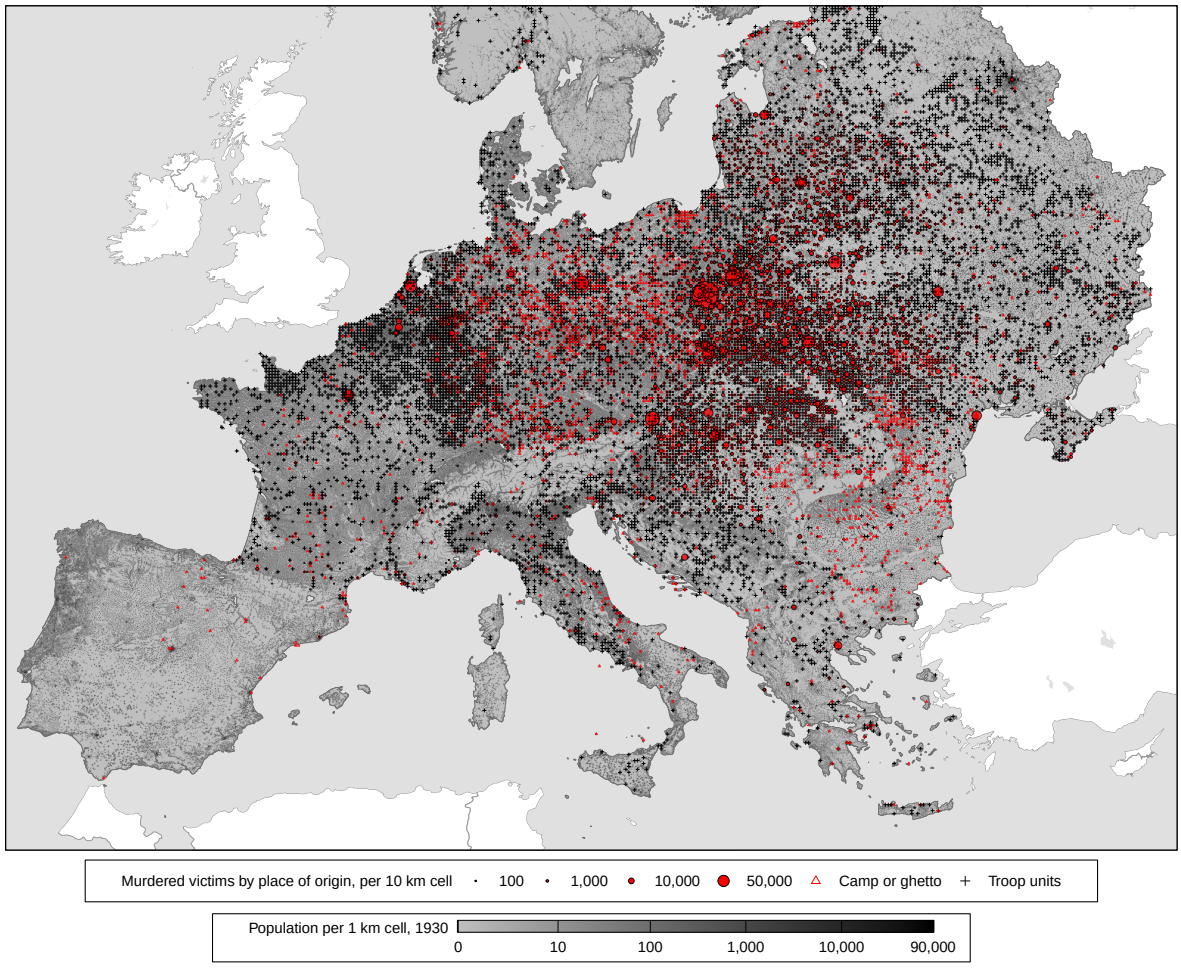
2 Data

Our study concerns the relationship between organizational structure and policy radicalization in Nazi Germany. The Nazi regime was one of the most consequential political regimes in history, transforming institutions and organizing state violence on an unprecedented scale. Approximately 5.7 million Jews were murdered during World War II. The full death toll of Germany's extermination policies remains uncertain. Other victims included approximately 200,000 Sinti and Roma, at least one million non-Jewish Polish civilians, 2.8 million Soviet prisoners of war, three to four million Soviet civilians, and half a million non-Jewish civilians in other German-occupied countries and in Germany itself. In total, an estimated 12–14 million civilians and prisoners of war died outside combat under German rule (Herbert, 2016a, p. 95).

We compile data describing the organizational structure that produced this violence and the victims of its persecution. Our primary data sources are the records of the victims, which we

[9]In our model, the ruler's objectives determine the direction of policy, while organizational structure determines initiative.

Figure 1: The Geography of the Holocaust



This figure overlays murdered victims by place of origin, camps and ghettos, and troop units on a map of the 1930 population. Victims are shown for 10-kilometer cells with at least 10 victims. These cells contain 4.62 million of the 4.81 million murdered victims in the Yad Vashem database (Section 2.1). Camps and ghettos mark the 4,409 cells with a site in the camp query of the European Holocaust Research Infrastructure (EHRI). Troop units mark the 9,399 cells in which the troop-unit panel places a unit at a town or district (Section 2.3). Population data come from the censuses of 34 countries (Appendix A.2). Land outside the grid is white.

describe in Section 2.1, and the directories, internal records, and unit registers of the regime, which we describe in Section 2.3. Section 2.2 describes the population at risk and the population surface.

2.1 Victims

Our main data on the victims are based on the Central Database of Shoah Victims' Names, compiled by Yad Vashem (Yad Vashem, 2026). The database joins 7.96 million person clusters, each built from one or more records of a victim. We collect 7.85 million of these clusters, of which 7.17 million have a located place. For each person, the records state the fate, the place of origin, and the place and month of death where the database records them. The place of origin is the wartime

place, else the prewar residence, else the birthplace. Of the persons with a located place, we identify 4.81 million who were murdered or presumably murdered. Because the database records the victims whose names were reported to it, its coverage differs by source and region. Thus, we compare places with themselves over time wherever we can.

We complement these data with information on the camps and ghettos from the camp query of the European Holocaust Research Infrastructure (EHRI), which merges several registers and lists 8,503 sites (Simons and Purschwitz, 2025).

We measure killing on a grid of 25-by-25-kilometer cells that extends from the Atlantic coast to the Volga bend and covers the places of origin of nearly all murdered victims. We assign each victim to the cell of the prewar residence and call the victims assigned to a cell its residents. We refer to the territories Germany occupied east of the Reich as the East: interwar Poland from September 1939, and the Soviet territory, the Baltic states included, from June 1941. In the East, residence also locates the killing: throughout 1941–1944, most victims from each of the Soviet Union, the Baltic states, and annexed Poland died within 10 kilometers of home.[10] Because coverage differs across regions, we scale deaths by recorded residents: our main outcome is murdered residents per 100 residents of any fate recorded in the database. It measures deaths relative to recorded residents, not to the full prewar population. Many victims have no recorded month of death. Where a record gives no death date, we use the month of deportation or transport. We spread the remaining deaths across months in proportion to dated deaths from the same cell or the smallest larger area with enough observations.[11] To measure the extension of killing beyond adult men, we classify victims as women and children if they were female or under 15 at death.

2.2 Population at Risk

We measure the prewar Jewish population with the JewishGen Communities Database (JewishGen, 2026), which lists 6,359 Jewish communities, their locations, and, where known, their sizes. We collect 6,357 communities, of which 5,703 in Europe report a size, at a median date of 1921. We supplement these data with the Polish census of 1931, by religion, and the Soviet census of 1939, by

[10] Elsewhere, deportation separated the place of death from the place of residence. Across Europe, most victims of 1941 died within 10 kilometers of home, and the great majority of those of 1944 died more than 100 kilometers away (Appendix Figure A15).

[11] Appendix A.1 describes the allocation of deaths recorded only by year or without a date.

nationality. Together, these censuses place 5.8 million Jews in our grid cells. Their counts provide an alternative denominator for the killing of residents.

Our data on the population at large are based on the published censuses of 34 countries, 136,582 municipalities and districts in all. We allocate each unit's population across its square kilometers in proportion to residential built-up volume. This yields a population surface for Europe around 1930 on a grid of one square kilometer, which we release with the paper. Our national totals closely match the published figures for all but two of the 34 countries (Appendix A.2). The surface forms the gray base of Figure 1.

2.3 Organizations

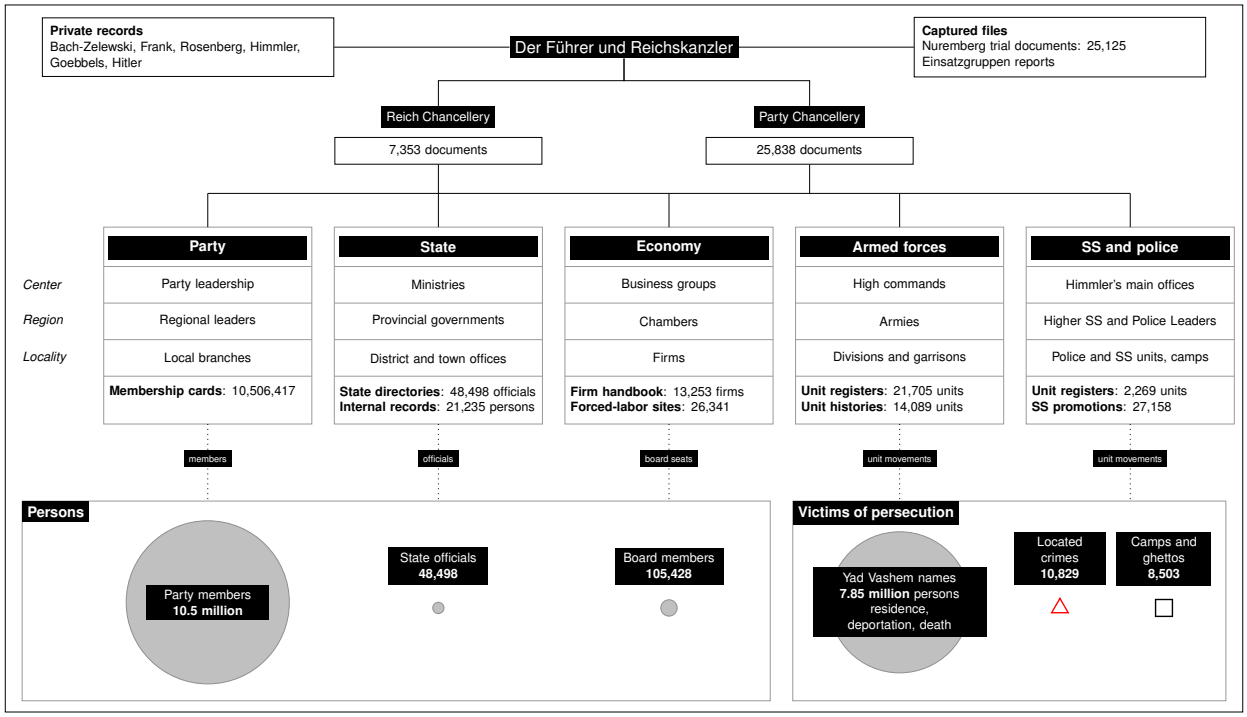
Below the Führer, Adolf Hitler, five hierarchies extended from the center through the regions to the localities: the party, the state, the economy, the armed forces, and the SS (Schutzstaffel) and police. Alongside the victim data and population surface, three datasets document the regime's structure and operations: chancellery records by document, official Reich directories by institution and edition, and armed forces and SS and police units by unit and month.[12] We add registers of party members and company boards. Figure 2 locates these sources within the regime's structure.

Internal records. Our data on the internal records are based on the edited volumes of the two chancelleries that connected Hitler to the state and to the party. We collect the *Akten der Reichskanzlei*, 7,353 documents from 1919 to 1945 (Historische Kommission bei der Bayerischen Akademie der Wissenschaften and Bundesarchiv, 2025).[13] For each document, the edition records its date, the participants and the recorder of a meeting, the persons it names, and an abbreviation register whose expansions name institutions. We use the 5,262 documents from 1925 to 1945 and code five features: whether they record a cabinet-level meeting; whether they list the head of government as present or name him without listing him; whether they invoke a Führer decree, decision, order, or briefing; whether they name an extraordinary mandate—a special commissioner, general

[12] Appendix Table A1 reports summary statistics for each dataset. Appendix A describes the internal records, state directories, policy domains, troop-unit panel, victim data, and population grid.

[13] The documents come from the Weimar edition, the online edition of the Hitler government to 1939, and the five print volumes for 1940 to 1945 (Appendix A.3). The print volumes also hold 990 records of the briefings in which Hans Heinrich Lammers, the head of the Reich Chancellery, reported to Hitler. We leave these records out because the earlier editions have no counterpart to them.

Figure 2: Data Sources



This figure locates the data sources within the regime's organization. Solid lines trace the chain of command from the Führer through the two chancelleries to the five hierarchies. Dotted lines link the hierarchies to records of persons and victims. Counts show records in our data. Circle areas in the lower panel are proportional to these counts.

inspector, or agency named after its head; and which institutions they name.[14] We also collect the *Akten der Partei-Kanzlei der NSDAP* (Heiber, 1983), which reconstructs from other archives the files of the office of Hitler's deputy, renamed the Party Chancellery in 1941. We obtain 25,838 of the edition's 25,842 files, almost all from 1933 to 1945. They record appointments, jurisdictional disputes, proposed rules, and requests for decisions attributed to Hitler.

We add 2,439 police and Security Service situation reports from 1933 to 1945, with an index of 14,126 institutions (Boberach, 1984; 1999); 25,125 of the 30,699 Nuremberg prosecution documents (Institut für Zeitgeschichte, 2006); 6,195 dated entries from Joseph Goebbels's diaries (Fröhlich, 1993); and 1,453 speeches, writings, and orders by Hitler from 1925 to 1933 (Institut für Zeitgeschichte, 1992). These series served different purposes and survived different selection processes. We therefore compare records within sources and treat agreement across sources as corroboration.

[14] We harmonize ministry names across the Weimar and Hitler-era editions, which use different names for institutions. We identify the head of government in the edition's person register and the document text.

State directories. We document the state's formal structure by transcribing the official Reich directories: the *Handbuch für das Deutsche Reich* of 1931 and 1936, the *Behördenführer* of 1939, and the *Taschenbuch für Verwaltungsbeamte* of 1942. They list Reich institutions and their leaders (Reichsministerium des Innern, 1931; Reichs- und Preußisches Ministerium des Innern, 1936; Münz, 1939; Warnack, 1942). We obtain 1,528 institutions with at least one person, counting each institution once per section of an edition. We classify them by hierarchy: state administration; the party, its units, and affiliates; corporatist bodies—chambers, estates, and Reich associations that organized occupations or sectors under public law; and assemblies and churches.[15] Lower offices add 10,807 institutions and 48,498 officials holding 89,704 spells in office. Each edition records assignments at publication; we do not observe changes between editions.

Party and firm registers. We use the NSDAP membership cards held by the Bundesarchiv: 10.5 million entries with name, birth date, entry date, residence, and membership number. Firm data come from the *Handbuch der deutschen Aktiengesellschaften*, which covers 13,253 stock corporations and 105,428 board members and managers from 1930 to 1945.[16] We link both registers to the internal records and directories. Appendix Figure A6 places the persons named in the internal governance records and the commanders of the killing units at their residences on these cards.

Troop units. Our data on armed units cover the military, Waffen-SS, and police. We read the full register of Tessin (1966): 27,702 entries on the founding, subordination, stations, and renamings of army, air force, navy, and Waffen-SS units. We add the Order Police register (Tessin and Kannapin, 2000). We assess coverage using Feldpost numbers, which the field post assigned to every unit it served (Kannapin, 1980). Of the 34,667 numbers naming a battalion, regiment, or division of the army, air force, navy, Waffen-SS, or police, more than nine in ten belong to units in our data.[17]

We then add the sources that record where each unit was active and when. For the military, the *Lexikon der Wehrmacht* gives dated unit stations (Altenburger, 2026), and the Armed Forces High Command war diary locates commands and their sectors by day (Schramm, 1961). For the

[15]The 1942 edition omits central party offices. The 1939 edition lists firms, banks, and sports clubs among corporatist bodies and lists regional branches individually. We exclude firms, banks, and clubs and count each body's regional branches once.

[16]The full panel covers 1871 to 1960, with 27,100 firms and 313,742 persons.

[17]The share is two in three across all 67,454 numbers, including columns, staffs, and construction units that the registers do not list separately.

Waffen-SS, we add National Archives guides to its records (National Archives and Records Service, 1978; 1981). For the police, we add police battalion registers (Arico, 2011), court-based reconstructions of police operations (Curilla, 2006; 2020), and, for the Security Police, Einsatzgruppen reports (Matthäus et al., 2014; Mallmann et al., 2011; Mallmann et al., 2014). When sources locate a unit in the same month, we prioritize the most specific place, then the most precise date, then the source reporting stations directly.[18] The panel contains 24,360 units and 639,767 unit-months from September 1939 to May 1945, recording each unit's location and movements by month.

We measure command structure on the same grid as the victims.[19] Its cells match the precision of the panel, which locates units by town or district (Appendix Figure A3). We assign each unit to the chain of command to which it reported: the army, the Security Police, or the SS and police units of Himmler's regional commanders, the Higher SS and Police Leaders. An Order Police battalion under an army division thus counts with the army.[20] For the descriptive analysis, a Herfindahl–Hirschman index sums the squared shares of the three commands among the units within 100 kilometers of each cell. For the entry analysis, a rival entry is the first month in which a unit of a second command enters a cell held by one command in the preceding month. A non-rival entry is the first month in which another unit of that command enters and no rival command arrives.

Linkage. We link persons using a German National Library authority number, a confirmed historical match, or a party-register match. Otherwise, we use a folded name key only when the surname and all forename tokens identify a unique person in every source joined. We link organizations by name, parent, jurisdiction, and date, retaining a source-specific identifier for every harmonized identifier. Each link retains its matching rule. Each source retains its evidentiary meaning. In Reich Chancellery meeting records, a person is present only if listed as a participant or recorder; a name in the text is a mention. We code policy domains from source headings under

[18] We carry a place forward until a source moves the unit. A unit-month with no place, or only a country or front, takes its parent command's area in the war diary; each reading of that area applies for at most the three following months. Nearly four in five unit-months have a place, and just over half have a town or district. Locations are less precise in the East: more than nine in ten placed unit-months in Germany, Austria, and Czechia identify a town or district, compared with fewer than half in Russia, Ukraine, Belarus, the Baltic states, and Moldova. Appendix A.6 details the sources, reading of scanned volumes, and rules.

[19] Matching deaths to the units present also attributes the killing to potential perpetrators, including units that historians have overlooked. Historians still "know little about the Germans' mass crimes that occurred during their retreat beginning in the summer of 1942" (Herbert, 2016b, p. 44). Appendix Table A7 lists the units with the highest killing rates among those for which no source reports a number of victims, mostly army units. These attributions indicate where further historical investigation could begin; they are associations, not findings of guilt.

[20] Appendix Figure A2 shows every troop structure the panel places in the East and the command each reported to.

pre-specified rules (Appendix A.5).

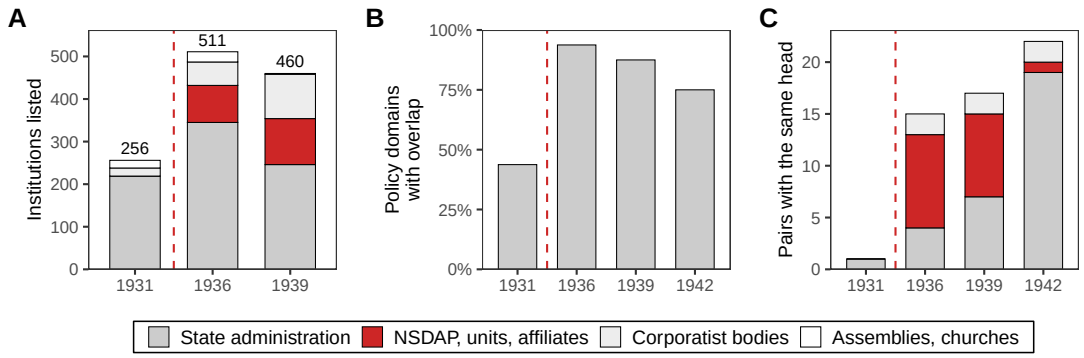
Egodocuments. We assemble high-frequency private records from the diaries of five senior Nazi officials: Joseph Goebbels, Hans Frank, Heinrich Himmler, Alfred Rosenberg, and Erich von dem Bach-Zelewski. We digitize Frank's office diary from U.S. National Archives microfilm (T992, 12 rolls) and Bach-Zelewski's diary from a typed transcript in the Bundesarchiv (R 20/45b). For the other three officials, we use published editions: Goebbels's diaries (Fröhlich, 1993), Rosenberg's diaries (Matthäus and Bajohr, 2015), and Himmler's appointment books (Witte et al., 1999; Uhl et al., 2020). A language model extracts each reported action, recording its date, actor, addressee, type, and policy. This yields 38,331 dated actions (Appendix A.3). These records are subjective and therefore may not be representative, and their authors had incentives to portray themselves favorably. Yet none was written for a superior. We know only partly why they kept these records. Goebbels sold publication rights in 1936 for posthumous release; Frank described a working document that would prove his resolve to future generations; Himmler's staff kept his appointment books; and Rosenberg wrote for himself. Historians use all five records to study the regime's decisions, and four have critical editions. We therefore focus on the more concrete features of reported interactions: who acted toward whom, and in what order.[21]

3 Descriptive Evidence

We first use directories and internal records to describe the regime's organization. We then trace its consequences through three case studies of legal discrimination against Jews, concentration camps, and mass killings in Eastern Europe that motivate the model and establish the historical setting for the quantitative analysis.

[21] The records were kept differently and survived by chance. Goebbels wrote by hand until July 1941, then dictated daily to a stenographer. In 1944, he had his diary copied onto glass plates, which were buried near Potsdam, unearthed by Soviet troops in 1946, and located in a Moscow archive in 1992 (Fröhlich, 1993). Frank's office recorded his meetings daily. He had the 38 volumes with him when arrested in May 1945, and they became evidence at Nuremberg (Geiss and Jacobmeyer, 1980, p. 7). Himmler's adjutants and secretary maintained his appointment books (Witte et al., 1999, p. 103); those for 1943–1945 surfaced in a Russian military archive in 2016 (Uhl et al., 2020). Rosenberg wrote on loose sheets; his diary disappeared after the trials and was recovered in 2013 (Matthäus and Bajohr, 2015, pp. 20, 30–34).

Figure 3: Institutions in State Directories



This figure describes the institutions listed in the state directories (Section 2.3). Panel A counts them by hierarchy in the 1931, 1936, and 1939 editions. Corporatist bodies are the chambers, estates, and Reich associations under public law, with firms, banks, and clubs excluded and regional branches counted once. Panel B plots the share of the sixteen policy domains in Appendix Table A5 with two or more supreme Reich bodies. Panel C counts pairs of bodies with the same printed head by the hierarchy of the body outside the state administration. A pair that includes a party office counts as a party pair. The dashed line marks 30 January 1933. The 1942 edition omits the central party offices.

3.1 Organization of the Regime

First, the number of institutions rose after 1933. Figure 3, Panel A counts the institutions listed in the official directories of the Reich in the editions of 1931, 1936, and 1939, by hierarchy. The directories of 1936 list twice as many institutions as those of 1931. The state administration grew by more than half, and the party, with its units and affiliates, which the edition of 1931 does not list, accounts for a third of the increase. We also measure the institutions that entered the record of government with the Reich Chancellery documents. Figure 4, Panel A plots the number of distinct institutions named per 100 documents, and Figure 4, Panel B the share of mentions that go to institutions absent from the record of 1925. Both series step up after 1933. By 1944, the documents name almost three times as many distinct institutions per 100 documents as in any Weimar year. Institutions absent from the record of 1925 receive fewer than one in ten mentions in 1932 and more than two in three in 1944. Party offices are among them. The office of the deputy of Hitler appears in no document of 1933 and in more than two of five documents of 1937, and it is the third most central institution in the network of institutions that the Hitler-era documents name together (Appendix Figure A7). The new agencies can be traced back to the first years of the regime: in a crisis, Hitler set up new agencies with special powers that bypassed the existing ministries (Kershaw, 1985, p. 98).

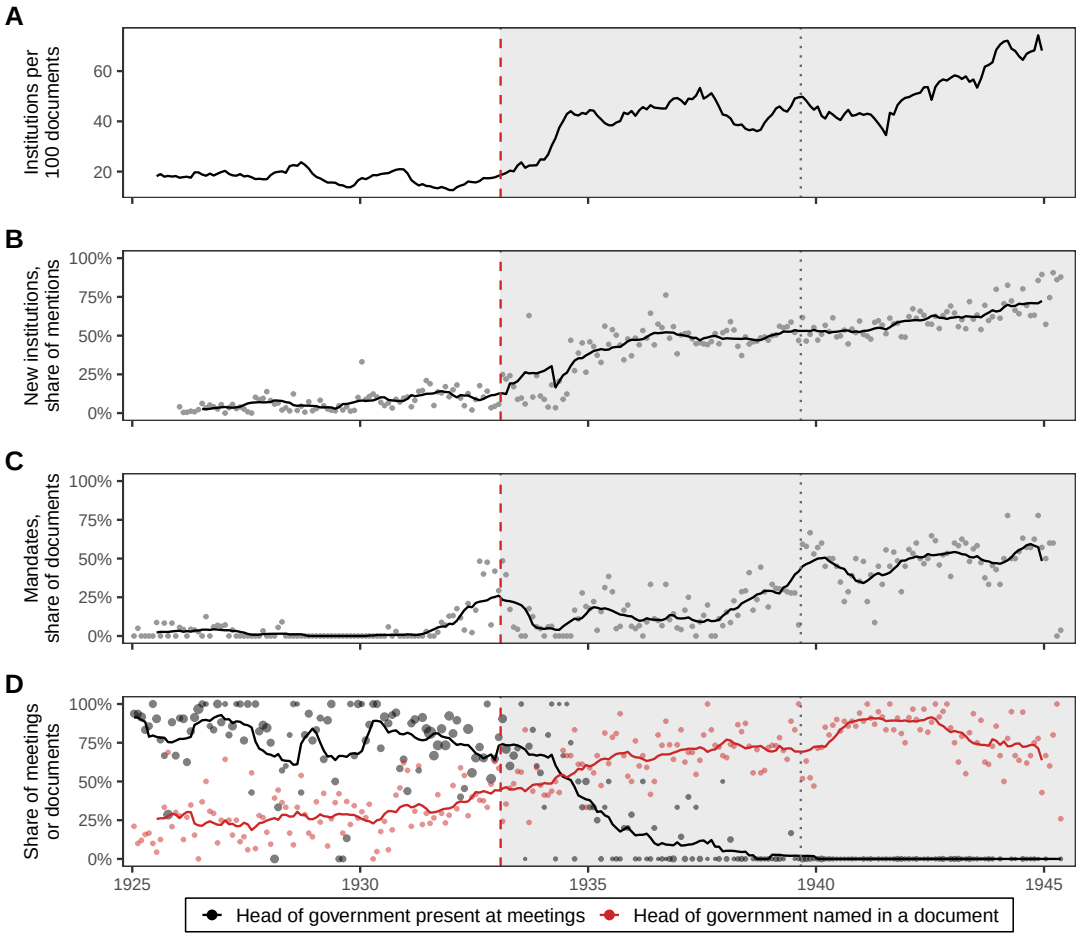
Second, extraordinary mandates appear in a growing share of the documents. We identify

documents naming a special commissioner, general inspector, or agency named after its head. Figure 4, Panel C plots their share, which rose in two steps. Such documents were rare in each year from 1925 to 1931, accounted for one in five in 1933, two in five in 1939, and over half in 1944. Most in 1933 name Reich commissioners for the Prussian ministries. We also identify persons heading two or more supreme Reich bodies. Figure 3, Panel C plots pairs of bodies with the same listed head. One person headed two or more bodies in 1931, compared with eleven in 1936; most of the fifteen pairs in 1936 included a party office. Hitler appointed special commissioners for specific tasks, responsible to him alone, producing "a jungle of competing and overlapping agencies" (Kershaw, 1997, p. 96).

Third, by 1936, almost every policy domain had two or more supreme Reich bodies. We assign these bodies to the sixteen policy domains in Appendix Table A5; Figure 3, Panel B plots the share of domains with two or more bodies. Fewer than half had multiple supreme bodies in 1931. At the series' peak in 1936, all but one did. By 1942, the share had declined to three in four domains. In the model of Section 4, a domain with two or more supreme bodies is under overlap, so each edition records a cross-section of arrangements; the decline to 1942 partly reflects that the 1942 edition omits the central offices of the party. Appendix Figure A8 lists the bodies in each domain. A classic account describes Nazi rule as government by "a multitude of largely autonomous rulers competing with one another" (Hüttenberger, 1976, p. 422, our translation). Sovereign authority rested with Hitler, but ministries, party offices, special commissioners, and the SS and police divided the administration below him (Broszat, 1969). The regime was thus not a unified Führer state. During the war, at least 23 state, party, and military institutions had a say in employing foreign workers in the Reich (Herbert, 2022, p. 4).

Fourth, under Hitler, the head of government stopped attending cabinet-level meetings, while the documents increasingly named him without listing him as present. Figure 4, Panel D plots the share of cabinet-level meetings that list the head of government in office among the participants, and the share of documents that name him without listing him as present. The two series cross. The Weimar chancellors attended most cabinet-level meetings in each year. The head of government in office attended most of them in 1933 and none from 1940. Documents that name the head of government without listing him as present made up less than a third of the documents in each year from 1925 to 1931 and more than four in five from 1940 to 1942. We also identify the documents that

Figure 4: Institutions in Internal Chancellery Records



This figure plots monthly series from the 5,262 documents of the *Akten der Reichskanzlei*, 1925 to 1945. Lines show centered twelve-month windows that do not cross 30 January 1933. Dots show single months. Panel A counts distinct institutions per 100 documents. Panel B plots the share of mentions of institutions absent from the 1925 record. Panel C plots the share of documents naming a special commissioner, general inspector, or agency named after its head. Panel D plots the share of cabinet-level meetings listing the serving chancellor as a participant (black, with dot area proportional to the number of meetings) and the share of documents naming him without listing him as present (red, Section 2.3). Shading begins on 30 January 1933. The dotted line marks 1 September 1939.

invoke a Führer decree, decision, order, or briefing, and Appendix Figure A9 plots their share by month. The share rose from almost none in 1934 to two in five documents in 1936 and stayed close to half from 1937 to 1944. These measures record attendance and attribution; they do not show that Hitler ceased to influence decisions through other channels. The cabinet stopped meeting in early 1938 (Kershaw, 1985, p. 98). With the parties and parliament gone, a dispute was no longer decided by the merits of the arguments of a ministry but by proximity to Hitler, whom ministers, party officials, and civil servants now sought above all to win over (Herbert, 2016a, pp. 35–36). Hitler arbitrated among them: he let rivals fight and then sided with the winner (Kershaw, 1985,

p. 98). His agents described this arbitration in their diaries.[22]

Agencies competed for authority over the same domain, while parallel units competed for a superior's approval. Officials acted toward Hitler's goals without waiting for orders, anticipating future approval.[23] Historians describe this process as cumulative radicalization (Mommsen, 1997, p. 82).[24]

3.2 Competitive Radicalization and the Holocaust

We examine how competition across and within jurisdictions elicited decentralized initiative and radicalized policy through three case studies (Appendix B) that inform the model in Section 4, with the third, the Holocaust in the East, providing the setting for the quantitative analysis in Sections 5 and 6.

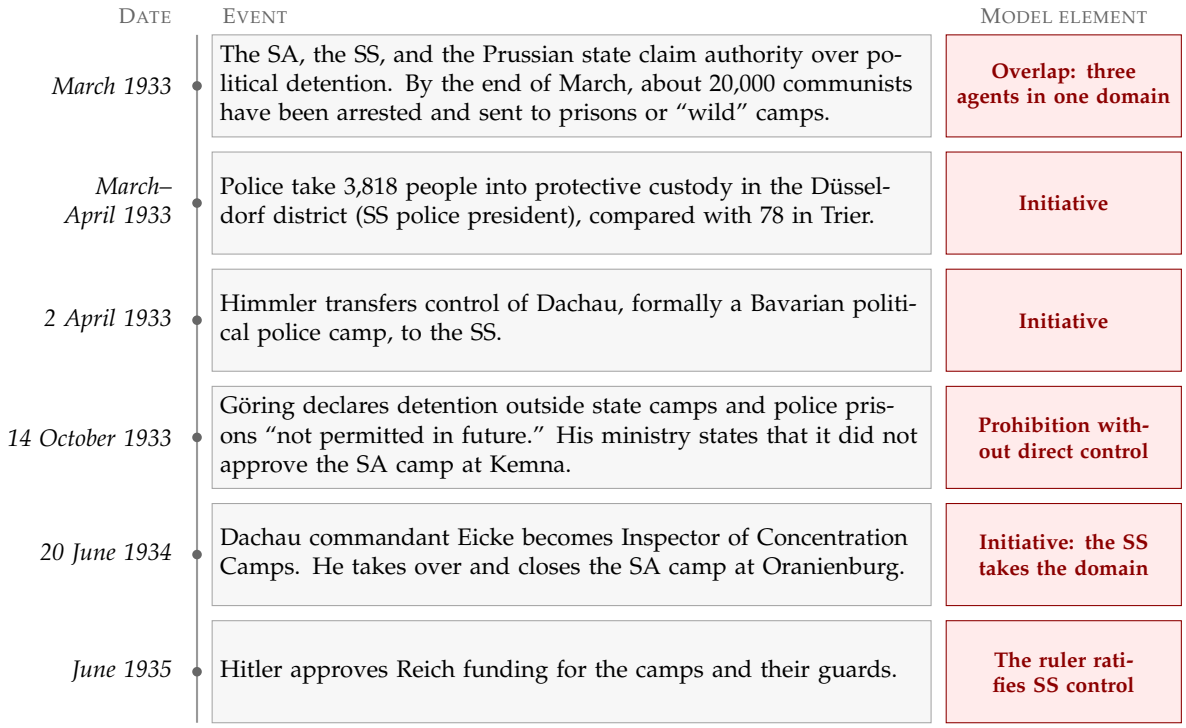
First, competition among towns drove early legal discrimination against Jews. Local officials adopted discriminatory measures before central rules required them, outdoing one another in a "dynamic of outbidding" (Herbert, 2016a, p. 38, our translation). Many towns were "a step ahead not only of the Reich authorities but even of the party leadership" (Gruner, 2000, p. 94, our translation). A September 1933 party decree ordering the repeal of local measures not sanctioned by law had "very little effect" (Gruner, 2005, p. 273). In summer 1938, the central government began preparing central legislation that towns had long demanded (Gruner, 2000, p. 121).

[22] In May 1940, Hans Frank, governor-general of occupied Poland, told police leaders that Hitler "lets the healthy competition of the various endeavors and institutions persist" and that "every authority corresponds to a counter-authority" (Appendix Table A9). Frank added that he would be "a bad chief" if he failed to exploit their "healthy ambition." In January 1943, after Himmler's police chief in the General Government conducted a mass police action on Himmler's orders, Frank protested: "I am supposed to bear the political responsibility in this area, and someone else gives the orders: that is impossible. ... If I do not agree, then, according to the wording of the decree, the Führer himself decides." In August 1943, Alfred Rosenberg, minister for the occupied eastern territories, explained the influence of Martin Bormann, head of the Party Chancellery: "no one who received letters from him could tell whether this was a personal order of the Führer or B.'s view." He called Bormann's methods "the most wretched antechamber politics." In January 1944, the economics minister, Walther Funk, complained that other agencies meddled in his domain and extracted from Hitler "a decision of the Führer" that was "in no way founded" (Herbert, 2022, p. 1, our translation).

[23] In February 1934, a state secretary in the Ministry of Food told officials: "Everyone who has the opportunity to observe it knows that the Führer can hardly dictate from above everything which he intends to realise sooner or later. On the contrary, up till now everyone with a post in the new Germany has worked best when he has, so to speak, worked towards the Führer. Very often and in many spheres it has been the case—in previous years as well—that individuals have simply waited for orders and instructions. Unfortunately, the same will be true in the future; but in fact it is the duty of everybody to try to work towards the Führer along the lines he would wish. Anyone who makes mistakes will notice it soon enough. But anyone who really works towards the Führer along his lines and towards his goal will certainly both now and in the future one day have the finest reward in the form of the sudden legal confirmation of his work" (Kershaw, 1997, p. 104).

[24] Each office-holder "tried to gain the special sympathies of the Führer" by appearing as a fanatical fighter for the goals of the movement (Mommsen, 1997, p. 82).

Figure 5: Control over Political Detention, 1933–1935



This figure traces the transfer of control over political detention from the SA and the Prussian state to the SS between 1933 and 1935. The right column identifies the corresponding model elements in Section 4. The prize is future control over the domain, here political detention. Sources: Herbert (2016a, p. 33), Broszat (1968, pp. 407, 410–412), and Longerich (2012, pp. 151, 183).

Second, concentration camps arose from competition among party, state, and paramilitary institutions. The SA (the party's storm troopers), the SS (the party's paramilitary formation), and the Prussian state each claimed authority over political detention. The Reichstag Fire Decree of 28 February 1933 provided the pretext for the suspension of basic rights and suppression of political opposition. By the end of March, about 20,000 communists had been arrested and sent to prisons or the SA's "wild" camps (Herbert, 2016a, p. 33). The Prussian Interior Ministry sought to reassert state control by banning detention outside state camps and police prisons, but the SS ultimately gained control of the camps (Broszat, 1968, pp. 410–411).[25] In June 1935, Hitler approved Reich funding for the camps and their guards (Longerich, 2012, p. 183). Established to suppress political opposition, the camps later became central institutions of the Holocaust. Figure 5 traces this sequence and links it to the model in Section 4.

Third, mass killing in Eastern Europe combined competition across and within jurisdictions.

[25]Prussian Interior Minister Göring wrote in his circular of 14 October 1933 declared detention outside state camps and police prisons "not permitted in future" (Broszat, 1968, pp. 410–411). Referring to the SA camp at Kemna, in Wuppertal, his ministry wrote: "I have not approved the establishment of this camp" (Broszat, 1968, p. 412).

Historical accounts suggest that these killings "cannot be understood as the implementation of a single order" (Longerich, 2010, p. 235) and that "Hitler himself may never have signed an order to kill the Jews" (Hilberg, 2003, p. 1061). Before the invasion of the Soviet Union, the center assigned overlapping authority behind the front to three major commands: the army, the Security Police, and Himmler's Higher SS and Police Leaders.[26] Within his command, Himmler compared units by their reported killings.[27] The "decisive step on the way towards a policy of racial annihilation" (Longerich, 2010, p. 252) came when competing SS and police units expanded their killings from mainly Jewish men of military age to women, children, at different times and initially without a clear order.[28] The four commanders under Himmler whose units first escalated the killing, in late July and early August, received promotions or decorations within a year. Officers who held back lost their commands.[29] Appendix B provides a fuller account.

[26]Numerous military, paramilitary, and police units operated in the occupied East. Appendix Figure A2 maps these units and their command structures. Here, we summarize the main actors. The army's rear-area commands held security divisions, field and local commandants' offices, the Secret Field Police, and attached police battalions, and raised militias and eastern volunteer units. Himmler's Higher SS and Police Leaders in the army group rear areas commanded SS brigades, Order Police regiments and battalions, and the locally recruited Schutzmannschaften, while Heydrich's Einsatzgruppen (mobile Security Police units) and their Kommandos followed the armies and, from late 1941, became stationary Security Police offices. The Armed Forces High Command gave Himmler special responsibilities in army territory in March 1941, and the army authorized independent Security Police operations there in April (Longerich, 2010, p. 182; Longerich, 2012, p. 517). As the front moved east, civil administrations under Rosenberg's new ministry, with their general and district commissars, took over army territory (Longerich, 2012, pp. 527–531; Pohl, 2008; Berkhoff, 2004) and contested the security commands' jurisdiction. In October 1941, Rosenberg complained to Reich Chancellery head Lammers that police leaders ignored the Führer decree of 17 July on jurisdiction (Dean, 2005, p. 91). Berlin overruled Hinrich Lohse, head of Ostland's civil administration, after he banned the shooting of Jews at Liepāja (Longerich, 2010, p. 298; Hilberg, 2003, p. 392). On 31 July 1942, Wilhelm Kube, commissioner general (*Generalkommissar*) for White Ruthenia, reported: "in the last 10 weeks we have liquidated about 55,000 Jews" (Longerich, 2010, p. 348).

[27]Within his command, Himmler acted as the ruler and his commanders as agents competing for rewards. In August 1941, he rebuked the 1st SS Brigade for failing to report, citing the SS Cavalry Brigade's daily reports (Cüppers, 2005, pp. 174–175, 182). The next day, 12 August, he met Friedrich Jeckeln, the Higher SS and Police Leader for southern Russia, and appeared "indignant" about his activities. Jeckeln then intensified the killing (Longerich, 2012, p. 533). He announced that he would complete the "liquidation" of Jews expelled from Hungary to Kamenets-Podolsk before the 1 September transfer to civil administration (Pohl, 2009, p. 163, our translation). His men shot 23,600 people, mostly Jews, in three days (Longerich, 2012, p. 533), and his units reported more than 40,000 killings in August (Mallmann et al., 2011). British codebreakers concluded from police radio traffic that the three Higher SS and Police Leaders competed for the best results (Cüppers, 2005, p. 186).

[28]Heydrich's Security Police Kommandos followed Himmler's units in the same sectors, extending killing to women and children between early August and October. Several escalated after Himmler broadened the order in mid-August (Cüppers, 2005, pp. 182–184; Longerich, 2012, p. 533). At Bila Tserkva in August 1941, a divisional staff officer alerted by army chaplains delayed the shooting of 90 Jewish children by a day. The Sixth Army commander then ordered its completion (Friedländer, 2007, pp. 216–217).

[29]For example, Friedrich Jeckeln, the Higher SS and Police Leader responsible for killings in southern Russia, subsequently also gained command in the Baltic states. Appendix Figure A13 relates the timing of escalation under 26 commanders to their careers. It dates escalation by the first killing of Jewish women and children alongside men and records commanders who left before this shift. Appendix Table A8 lists the commanders, their reported victim counts, and their careers. Two officers of the 2nd SS Cavalry Regiment who held back lost their commands (Cüppers, 2005, pp. 164, 179). In February 1943, Bach-Zelewski wrote in his diary that "an ambitious commander is tempted to fake successes not won in combat by shooting suspected partisans." Three months earlier, he wrote of a commander Himmler had appointed to lead an operation: "Unfortunately, he has soiled the honor of arms with wild persecutions of Jews."

These cases suggest that competition encouraged decentralized initiative that intensified persecution. They do not cover all forms of Nazi violence or all victim groups. Section 2 lists other victim groups, including the Sinti and Roma, non-Jewish Polish civilians, Soviet prisoners of war, and Soviet civilians. Overlapping authority was not unique to Nazi Germany. Appendix B describes similar arrangements elsewhere. We model this mechanism in Section 4 and test its implications for the Holocaust in the East in Sections 5 and 6.

4 Model

Formal authority in a dictatorship is concentrated in the ruler, who cannot credibly bind himself.[30] He can state an objective but cannot specify the appropriate action in advance, and he cannot commit to reward the agents who take it. He can offer only the prospect of a future appointment. Under rivalry, he gives two agents overlapping jurisdiction and appoints the one who shows more initiative. Under direct control, an apparatus independent of the agents certifies compliance, so he can prohibit initiative and enforce his orders. Rivalry produces false positives, implementation he would reject; direct control produces false negatives, appropriate policies left unimplemented.

4.1 Setup

A ruler sets an objective in a policy domain and authorizes one or both of two agents, $i \in \{1, 2\}$, to act on his behalf. He retains formal authority but cannot specify the appropriate action in advance. Agents may interpret his objective and act before receiving an order; we call this *initiative*.

All players are risk neutral. The agents are identical and derive no utility from the policy. The ruler cannot commit to contractual rewards for initiative. The only reward he can offer is a future *prize*: appointment to a position carrying rent $R > 0$, which he awards or withholds after learning whether implementation is appropriate. Appointment expresses his subsequent approval and may confer exclusive jurisdiction over the domain.[31]

Appendix Table A9 gives the German originals and sources of these diary passages. Allied and West German investigations found no case in which refusing to shoot endangered an individual subordinate. Members of the 1st SS Brigade who refused received other duties (Cüppers, 2005, pp. 109–110, 169).

[30] In Nazi Germany, a prerogative state that acted without legal constraint coexisted with a normative state of statutes and courts (Fraenkel, 1941).

[31] Figure 5 records such a transfer: the Prussian state and the SA lost jurisdiction over political detention to the SS.

Whether implementation is appropriate for the ruler depends on a state $\theta \in \{0,1\}$: it is appropriate at $\theta = 1$ and inappropriate at $\theta = 0$, with $\Pr(\theta = 1) = \pi \in (0,1)$. Neither the ruler nor the agents know θ, or have a signal of it, until it becomes common knowledge after the initial period. Implementation is irreversible, so subsequent disapproval cannot undo an agent's initiative.

Implementation in the initial period is *early*, and early implementation is always timely. Waiting allows the ruler to learn θ and specify an act in an *order*, but, if executed, the order is timely only with probability $\delta \in (0,1)$. We call $1-\delta$ *time pressure*: the share of the policy's value lost by waiting for information. This loss reflects the time available to act, so a second agent does not improve δ. A *false positive* is implementation at $\theta = 0$; a *false negative* is failure to implement in time at $\theta = 1$.

Implementation can therefore come early and uninformed, from an agent's initiative, or late and informed, from the ruler's order. Direct control, defined below, makes the late route available: it lets the ruler forbid early action and enforce his order once he learns θ.

Before the agents act, the ruler publicly chooses their jurisdiction and whether to exercise direct control over their conduct. Authorizing both agents creates *overlap*, allowing him to allocate the prize by their relative initiative. Two agents suffice for this comparison, and each further rival would dilute the expected prize that rewards initiative.[32] Overlap costs $K > 0$: the value of the task the second agent would otherwise perform for the ruler, together with duplicated effort, conflict between the rivals, and lost coordination. The ruler can also establish *direct control*, an apparatus independent of the agents, at cost $m > 0$. Direct control certifies compliance with, or defiance of, instructions, allowing him to reward obedience and punish defiance by $P \geq 0$. These two choices produce four arrangements:

	Direct control	
Agents	No	Yes (cost m)
One	Routine administration	Control
Two (cost K)	Rivalry	Both instruments

These instruments make different aspects of conduct verifiable. Overlap produces a *ranking* of initiative. The two agents operate in the same domain under the same conditions, report their conduct, and can contest each other's reports. An agent who takes the initiative ranks above

[32] The exchange of initiative for posts and authority formalizes the account of Breton and Wintrobe (1986); the comparison of agents' initiative builds on Maskin et al. (2000).

one who does not; identical choices produce a tie. Direct control certifies compliance or defiance and thus makes instructions enforceable. The ranking provides information from below at no additional cost to the ruler; direct control provides information from above at cost m. Without direct control, the ruler is *absent* from the domain: agents act in his name, but he observes only their ranking, when one exists.[33] Prohibitions are then unenforceable, whether issued by the ruler or by a rival.[34] An additional unit under an existing agent shares that agent's claim to the prize and position in the ranking; it therefore creates no separate ranking.

A *jurisdiction grant* chooses the arrangement: it authorizes agents in a domain without specifying an action. *Direction* states the policy's goal and announces the appointment rule. An *instruction* specifies, restricts, or approves an act. An *order* specifies an act; a *prohibition* restricts action by forbidding initiative. Awarding the prize approves, or *ratifies*, the agent's conduct: initiative under rivalry and compliance under control. *Initiative* is action that advances the stated goal beyond the latest instruction.

The timing is as follows.

1. The ruler chooses the arrangement and announces the appointment rule and any prohibition. The agents observe these choices. Before acting, an agent may report his plan, and the ruler may veto it.

2. The authorized agents simultaneously choose initiative $a_i \in \{0, 1\}$ in the *initial-period game*. No one observes θ or a signal of it. The policy is implemented early if at least one agent takes the initiative.

3. Everyone learns θ. Under direct control, the ruler may issue an order. If executed, it is timely with probability δ.

4. The ruler assigns the prize and punishes certified defiance. Payoffs are realized.

Agents. Each authorized agent chooses whether to take the initiative and whether to execute an order. Each action costs $c > 0$; the execution cost is incurred even if implementation is late. The

[33] Figure 4, Panel D, documents this absence: Hitler stopped attending cabinet-level meetings, while documents increasingly named him without recording him as present.

[34] Section 3 records two examples. Göring's circular against detention outside state camps did not preserve the Prussian state's jurisdiction (Figure 5), and Bormann's decree requiring the repeal of unsanctioned local measures had little effect.

agent observes the arrangement, the appointment rule, and any instruction, and learns θ after the initial period. His payoff is

$$U_i \;=\; R\,\mathbb{1}\{i \text{ appointed}\} \;-\; c\,a_i \;-\; c\,\mathbb{1}\{i \text{ executes}\} \;-\; P\,\mathbb{1}\{i \text{ punished}\}. \tag{1}$$

The prospect of the prize is his only motive for initiative.

Ruler. The ruler chooses the arrangement, the appointment rule, any instructions, and the recipient of the prize. He observes the ranking under overlap, certified compliance or defiance under direct control, and θ after the initial period. He observes no other aspect of conduct. His payoff is

$$\begin{aligned} U_0 \;=\; & b\,\mathbb{1}\{\theta = 1,\ \text{timely implementation}\} \;-\; h\,\mathbb{1}\{\theta = 0,\ \text{implementation}\} \\ & -\; K\,\mathbb{1}\{\text{overlap}\} \;-\; m\,\mathbb{1}\{\text{direct control}\}, \end{aligned} \tag{2}$$

where $b > 0$ is the value of timely implementation when appropriate, and $h > 0$ is the *false-positive cost*. The prize is an existing position; its rent does not enter (2).

Assumption 1 (No Commitment)**.** *The ruler cannot commit to transfers or outcome-contingent promises. He announces an appointment rule before the initial period. The rule is followed whenever its condition is verifiable under Assumption 2. Ties in the ranking are broken by a fair lottery. The agents cannot contract with one another.*

Following the announced rule is an equilibrium-selection convention. Because the rent does not enter (2), every assignment of the prize is sequentially rational. The model therefore assumes, rather than derives, adherence to the rule. This convention is costless to the ruler, who is indifferent among the agents ex post. When the prize confers jurisdiction, appointment can also be interpreted as ratifying the leading agent's existing control of the domain. The restriction on contracts between agents excludes collective bodies, such as a cabinet or chamber, through which they could allocate jurisdiction among themselves.[35]

Assumption 2 (Verifiable Events)**.** *Appointment and punishment may depend only on two verifiable events: the ranking of initiative, available only under overlap, and certified compliance with, or defiance of,*

[35]These bodies ceased to perform this role under the Nazi regime: the cabinet stopped meeting in early 1938, and parliament and the corporatist bodies lost their functions (Section 3).

an instruction, available only under direct control. Neither defiance nor a lower rank is rewarded. Appointment is monotone in rank: a higher rank never reduces an agent's probability of appointment. Nothing else about conduct is verifiable.

The level of initiative is unverifiable because the policy's content cannot be specified in advance. Outcomes are also unverifiable: they combine initiative with local conditions and do not identify who acted. Monotonicity excludes rewarding sabotage of the ruler's policy or penalizing an agent for outranking his rival.

Assumption 3 (Late Orders)**.** *An order must specify an act using information available only after θ is learned. A prohibition requires no such information. It can be issued from the outset, or as a veto of a plan that an agent reports before he acts; it is enforceable under direct control.*

The ruler can order an act only once he can describe it. Early implementation therefore requires initiative. A veto is a prohibition aimed at a reported plan: once an agent reports what he intends to do, the ruler can forbid it before it is carried out.

Assumption 4 (Valuable Prize)**.** $c < R/2$.

The expected rent after a tied ranking, $R/2$, exceeds the cost of initiative. The need for overlap or direct control rests on the commitment and verifiability restrictions. Without Assumptions 1 and 2, the ruler could pay a single agent for timely initiative.

The solution concept is subgame-perfect equilibrium, subject to the convention in Assumption 1. The ruler chooses the arrangement and the *protocol*—the appointment rule and instructions—to maximize expected payoff in (2). There is one domain and no discounting across stages. Proofs appear in Appendix E.

4.2 Arrangements

Lemma 1 (Arrangement Outcomes)**.** *Under Assumptions 1–4:*

(i) Under routine administration, the agent takes no initiative and the policy is never implemented.

(ii) Under rivalry, the tournament rule "appoint the higher-ranked agent, with ties broken by a fair lottery" (Lazear and Rosen, 1981) induces both agents to take the initiative. Implementation is early in both states.

Table 1: Arrangements, Payoffs, and Errors

Arrangement	Ruler's payoff	Probability of False positive	Probability of False negative
Routine administration	0	0	π
Rivalry	$\pi b - (1-\pi)h - K$	$1-\pi$	0
Control	$\pi\delta b - m$	0	$\pi(1-\delta)$
Both instruments			
Rivalry protocol	$\pi b - (1-\pi)h - K - m$	$1-\pi$	0
Control protocol	$\pi\delta b - K - m$	0	$\pi(1-\delta)$

The table reports expected payoffs and unconditional error probabilities under the protocols in Lemma 1. With both instruments, the ruler pays both costs and uses the protocol with the higher payoff.

(iii) Under control, a prohibition and the rule "appoint unless defiance is certified" induce the agent to refrain from initiative and execute any order. The ruler orders implementation only when $\theta = 1$.

(iv) With both instruments, the optimal protocol yields $\max\{\pi b - (1-\pi)h,\ \pi\delta b\} - K - m$: the higher payoff from the rivalry and control protocols, net of both costs.

Table 1 reports the corresponding payoffs and error probabilities.

Under rivalry, restraint forfeits an agent's expected share of the prize. There is no punishment without direct control. Rivalry induces implementation before any order; control induces implementation in response to an order. Routine administration leaves the domain to a single agent who has no incentive to implement the policy.

Proposition 2 (Two-Way Choice)**.** *Using both instruments is never optimal. Routine administration is strictly suboptimal if either rivalry or control yields a positive payoff, and is weakly optimal otherwise. The ruler therefore chooses between rivalry and control, with routine administration as the outside option.*

The instruments are substitutes. With both in place, the ruler uses one protocol and gains nothing from the other instrument. Routine administration amounts to forgoing the policy when neither instrument yields a positive net payoff. In an optimal arrangement, overlap therefore entails no direct control, and direct control entails a single agent.

4.3 Rivalry versus Control

Proposition 3 (Error Tilt)**.** *Under rivalry, implementation occurs in both states. The false-positive probability is $1-\pi$, and the false-negative probability is zero. Under control, timely implementation occurs only*

when $\theta = 1$ and an order is executed in time. The false-positive probability is zero, and the false-negative probability is $\pi(1-\delta)$. When the optimal arrangement switches from control to rivalry, the false-positive probability rises from 0 to $1-\pi$, and the false-negative probability falls from $\pi(1-\delta)$ to 0.

Choosing between rivalry and control thus means choosing between two types of error. We call the early, uninformed implementation induced by overlap *competitive radicalization*. The part that the informed ruler would reject is the false positive. The entry prediction in Remark 2 below concerns this implementation and holds in both states. Binary initiative makes this error pattern stark. With continuous initiative, inappropriate implementation could also vary in magnitude. Whether error probabilities vary smoothly would depend on how that extension is specified.

Define the payoff difference between rivalry and control as

$$D \;\equiv\; \pi(1-\delta)\,b \;+\; m \;-\; K \;-\; (1-\pi)\,h. \tag{3}$$

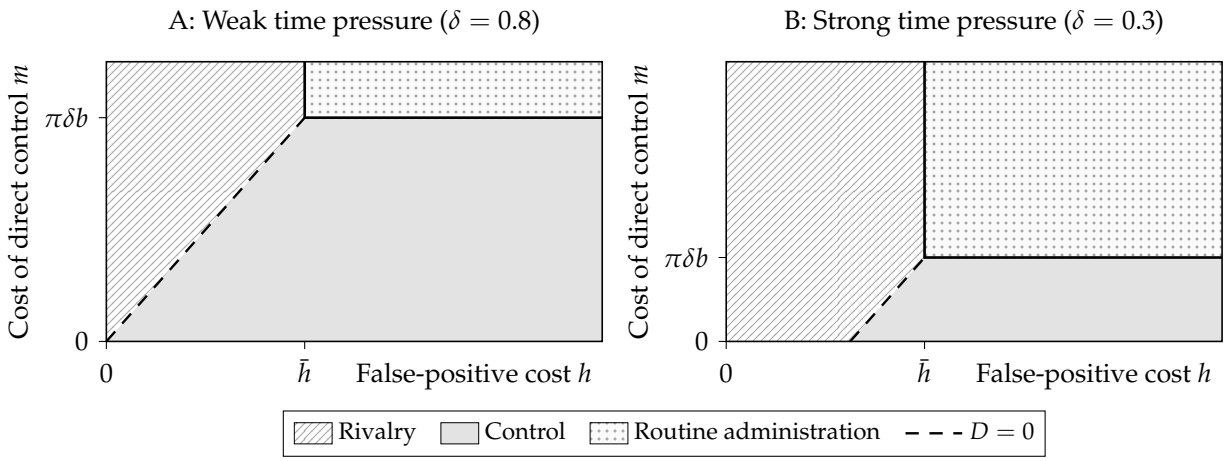
Proposition 4 (Rivalry Boundary)**.** *Between rivalry and control, rivalry is optimal if and only if $D \geq 0$, and uniquely optimal if $D > 0$. The payoff difference D is strictly increasing in time pressure $1-\delta$, the cost of direct control m, the value of timely implementation b, and the prior π. It is strictly decreasing in the false-positive cost h and the overlap cost K.*

These comparative statics follow directly from (3). Time pressure reduces the payoff from control, which delays implementation until the state is known. The cost of direct control favors rivalry, whereas the cost of overlap favors control. A lower false-positive cost favors rivalry because only rivalry produces false positives. A higher prior increases the value of early implementation and reduces the probability that it is inappropriate. Figure 6 shows the optimal arrangement as a function of the false-positive cost and the cost of direct control.

Remark 1 (Polarization under Time Pressure). As time pressure rises, the region in which rivalry is optimal weakly expands, and the region in which control is optimal weakly contracts. Where rivalry yields a negative payoff, a domain that leaves control moves to routine administration.

Time pressure reduces the payoff from control, $\pi\delta b - m$, without changing the payoff from rivalry. The same shock can therefore move some domains into rivalry and others into routine administration, shrinking the set of domains under direct control. Figure 6 illustrates this con-

Figure 6: Optimal Arrangement



This figure shows the optimal arrangement as a function of the false-positive cost h (horizontal axis) and the cost of direct control m (vertical axis), with $\pi = 0.5$, $b = 1$, and $K = 0.1$. Panel A sets $\delta = 0.8$ and Panel B sets $\delta = 0.3$. Rivalry is optimal when $D \geq 0$ and $h \leq \bar{h} = (\pi b - K)/(1 - \pi)$. Control is optimal when $D \leq 0$ and $m \leq \pi\delta b$. Routine administration is optimal when neither instrument yields a positive payoff. Boundary equalities allow ties.

traction from Panel A to Panel B. Appendix Figure A14 traces the error probabilities across the rivalry–control boundary.

Remark 2 (Entry of a Rival). In a domain held by one agent without direct control, the entry of a second agent under the tournament rule of Lemma 1(ii) changes routine administration into rivalry. Implementation changes from never to early in both states. The false-negative probability falls from π to 0, and the false-positive probability rises from 0 to $1 - \pi$. An additional unit under the incumbent agent leaves the arrangement unchanged.

Both remarks follow from Lemma 1 and Proposition 4. Section 5 uses the troop-unit panel to test the prediction in Remark 2: rival entry increases implementation, but an additional unit under the incumbent agent does not.[36]

4.4 Discussion

Delegation trades initiative against control. The ruler cannot contract on initiative: he lacks commitment, and initiative is unverifiable. Nor can he order early implementation, because he can specify an act only after the initial period (Assumption 3). Without these restrictions, he could induce timely implementation directly. Under the restrictions, each instrument makes a different

[36]The model's prediction of no implementation under routine administration is a normalization. Killing also occurred in cells held by one command. The test concerns the change at entry relative to that level.

event verifiable. Rivalry uses relative initiative to induce implementation before θ is known, producing false positives when implementation is inappropriate. Control uses certified compliance or defiance to delay implementation until θ is known, producing false negatives when execution is late. The boundary in (3) balances these errors and the costs of the two instruments.

Radicalization in the model requires competition: without a rival, no agent takes the initiative (Lemma 1(i) and (iii)). The absence of an early order makes competition attractive. Because the ruler can specify an act only after agents act (Assumption 3), direct control delays implementation and forgoes $\pi(1-\delta)b$, the time-pressure term in (3). Under rivalry, agents act before anyone knows whether implementation is appropriate, and some of what they implement the ruler would reject. Remark 2 holds the domain fixed and varies the arrangement.

Remark 3 (Relative Fear). Relax Assumption 2 and the restriction of punishment to direct control as follows, without imposing Assumption 4. Under rivalry with the tournament rule of Lemma 1(ii), the agent ranked lower also loses $L \geq 0$, with ties broken by the same lottery, and an agent who takes the initiative at $\theta = 0$ is sanctioned by $X \geq 0$ with probability $q \in [0,1]$. Both enter the agent's payoff (1) when they apply, and the ruler imposes them under the convention of Assumption 1. Initiative is strictly dominant in the initial-period game if and only if

$$\frac{R+L}{2} \;>\; c + (1-\pi)\, q\, X. \tag{4}$$

If the reverse strict inequality holds, no agent takes the initiative in either state.

The left side of (4) is the stake of falling behind a rival; the right side adds to the cost of initiative the expected sanction for implementation the ruler would reject. Agents do not observe θ, so a sanction cannot make initiative depend on the state: under either strict inequality, it leaves initiative unchanged or deters it in both states. A ruler who chooses rivalry therefore gains nothing from sanctioning implementation he would reject, and his agents fear falling behind a rival more than going too far. Discretion over rewards and punishments widens this gap: removal from office raises L, and the absence of a rule that defines excess lowers q.

Under rivalry, the ruler retains formal authority but does not choose what agents implement before he becomes informed (Aghion and Tirole, 1997). The stated goal directs initiative, while π, b, and h determine his evaluation of implementation. The arrangement determines whether

implementation precedes information and, consequently, which error arises.

The ranking in the model arises whenever a superior compares agents on a common scale. It can emerge from overlapping assignments in one domain or from parallel branches reporting to one superior, as when Himmler compared killing reports in the occupied East (Section 3). Parallel branches can thus generate a ranking without the overlap cost K of placing two agents in one domain. A coarse goal encourages initiative while leaving the ruler discretion over what to reward and which actions to disown. This discretion complements Assumption 1: a ruler who cannot commit benefits from leaving the terms of reward unspecified.

The overlap of party and state offices admits two readings. In the account of the dual state, a prerogative state of party and police wrested power from the statutes, courts, and civil administration of the normative state (Fraenkel, 1941). In the model, the same overlap places parallel agents in one domain, where they compete for the prize.

The parameters capture differences across domains and regimes. Direct control is less costly where courts, the press, or an established bureaucracy already certify conduct, and more costly in new territory without such an apparatus. Overlap is more costly when tasks require coordination, so K rises with task interdependence. The false-positive cost measures the loss the ruler assigns to inappropriate implementation. It can vary across domains and places even for the same ruler.

A positive false-positive cost does not imply opposition to the policy's stated goal. The state $\theta = 0$ can describe cases in which implementation harms the ruler's other objectives more than it advances the policy. This interpretation leaves the payoff in (2) unchanged, and h measures the net loss. Agents ignore these costs because their appointment depends on initiative, not on diffuse and delayed harm that cannot be attributed to either agent. Rewards for the measured task can divert effort from unmeasured tasks (Holmström and Milgrom, 1991). For a given arrangement, implementation does not depend on h: rivalry induces early implementation in both states, control induces implementation by order only when $\theta = 1$, and routine administration induces none. These conditional predictions do not identify the ruler's preferences.[37]

[37] Civil administrators in occupied Soviet territory sought to preserve labor for the war economy. In the Ostland, they halted the shooting of Jewish workers in autumn 1941 until Berlin ordered in December that economic concerns be disregarded (Longerich, 2010, pp. 297–298; Pohl, 2009, pp. 182–184). In the model, this response placed the false-positive cost near zero in that domain. Frank later acknowledged that the killings had caused "the most enormous difficulties" for labor (Geiss and Jacobmeyer, 1980, p. 137). Appendix Table A9 gives the passage. The killing continued. We do not claim that Hitler counted these deaths as false positives. The model requires only a positive h.

The baseline assumes identical agents and contains no ability or learning process. However, rivalry may also serve selection. In an extension with heterogeneous agents, initiative could reveal an ability to act without instructions, while ordered execution might reveal less about independent judgment. If a higher rank predicts better performance in future tasks, selection could support the same appointment rule as incentives. Appointing the leading agent might then be optimal even after inappropriate implementation, because the appointment reflects expected future performance. With noisy local outcomes, rival initiatives could also provide information for selecting among experiments (Wang and Yang, 2025).

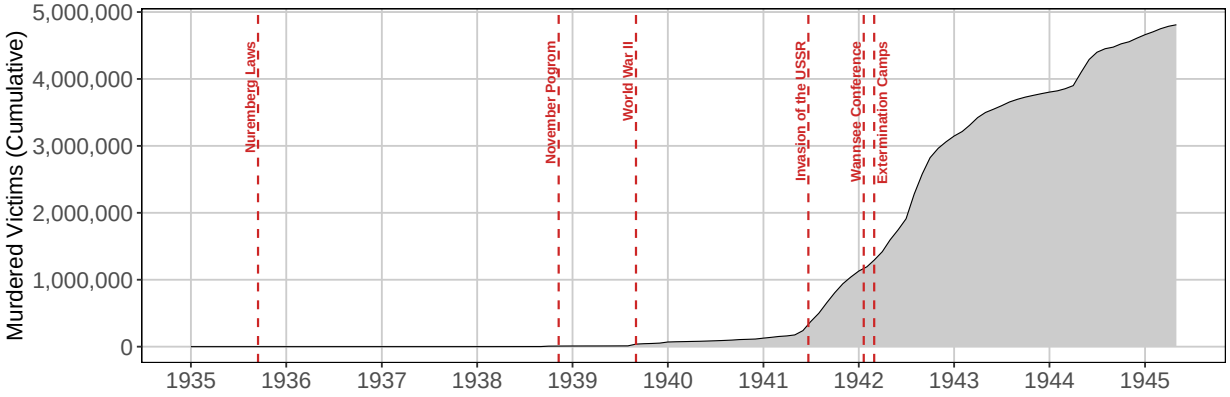
The analysis treats the ruler as choosing the arrangement. The boundary in (3) also suggests a selection interpretation: arrangements may emerge incrementally and persist where they yield higher payoffs. Such a process could favor rivalry under time pressure, and direct control where it is inexpensive. To the extent that selection follows these payoffs, it generates the same sorting across domains. Comparisons of domains with and without overlap must therefore account for differences in priors, policy values, time pressure, false-positive costs, and instrument costs. Neither interpretation associates overlap with a weak ruler. In the model, overlap is an organizational response to the domain's environment.

5 Empirics: The Holocaust in the East

We study the Holocaust in the East, where overlapping commands operated alongside one another and where the records of the victims locate the killing. We begin with descriptive comparisons across cells, then examine changes around rival entry, and finally use documented transfers and the invasion plan to assess selection into entry. We show that killing was generally higher where command structure was less concentrated and that killing rose after a rival entered, with no comparable increase after another unit of the existing command arrived. The pattern persists in comparisons based on documented transfers and the invasion plan. Section 6 then uses central directives and private diaries to trace the sequence of initiative and instruction.

Germany occupied the East in two steps: interwar Poland in September 1939 and the Soviet territory in June 1941. Figure 7 plots the cumulative murders that the Yad Vashem database records by month from 1935 to the end of the war. The database records few murders before the war.

Figure 7: Murdered Victims, 1935 to 1945



This figure plots cumulative murders recorded in the Yad Vashem database by month from January 1935 to May 1945. Deaths recorded only by year or without a date are spread over months as in Section 2.1. Dashed lines mark the Nuremberg Laws (September 1935), the November pogrom (1938), the invasions of Poland (1 September 1939) and the Soviet Union (22 June 1941), the Wannsee Conference (20 January 1942), and the start of killing in the extermination camps of occupied Poland (March 1942) (Herbert, 2016a, pp. 39, 62, 93–94; Longerich, 2010, pp. 29, 305; Friedländer, 2007, p. 204).

Killing rose after the invasion of the Soviet Union on 22 June 1941, peaked in summer 1942, slowed in 1943, and rose again in 1944.

5.1 Concentration of Commands

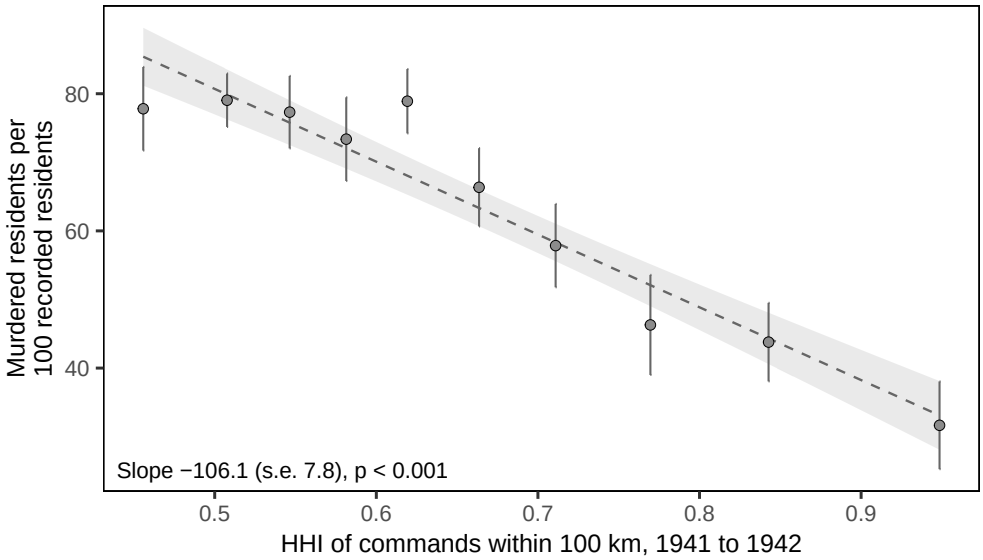
We first establish the association between shared authority and killing. We regress the share of recorded residents murdered by May 1945 on the concentration index,

$$s_c = \beta \overline{H}_c + X_c'\gamma + \mu_{z(c)o(c)} + \varepsilon_c, \tag{5}$$

where s_c is murdered residents per 100 recorded residents in cell c and $\overline{H}_c$ is the mean Herfindahl–Hirschman index from June 1941 to December 1942. The controls X_c include the logs of recorded residents and nearby units and, in the full specification, the log of the prewar Jewish population. The fixed effects $\mu_{z(c)o(c)}$ group cells by occupation regime and occupation month.

By the end of the war, about three quarters of recorded residents had been murdered in the least concentrated tenth of cells, compared with about one third in the most concentrated tenth (Figure 8). The association persists after accounting for occupation timing, recorded residents, nearby units, and the prewar Jewish population (Appendix Table A10). Killing also rises as concentration falls within cells, including cells held by one command (Appendix Table A11). These associations hold when the index distinguishes more fine-grained command structures within the

Figure 8: Concentration of Commands and the Killing of Residents



This figure plots residents murdered by May 1945 per 100 recorded residents by decile of the mean Herfindahl–Hirschman index from June 1941 to December 1942. The sample comprises the 1,661 cells of Table 2, Panel A, with at least 20 recorded residents. The index measures the concentration of the units within 100 kilometers of a cell across the three commands (Section 2.3). Bars show 95 percent confidence intervals. The dashed line shows the fit of equation (5) without controls, with its 95 percent band (Appendix Table A10, column 1).

three main commands. These patterns could reflect deployments toward anticipated killing. We therefore turn to changes around the arrival of a second command.

5.2 Entry of a Rival Command

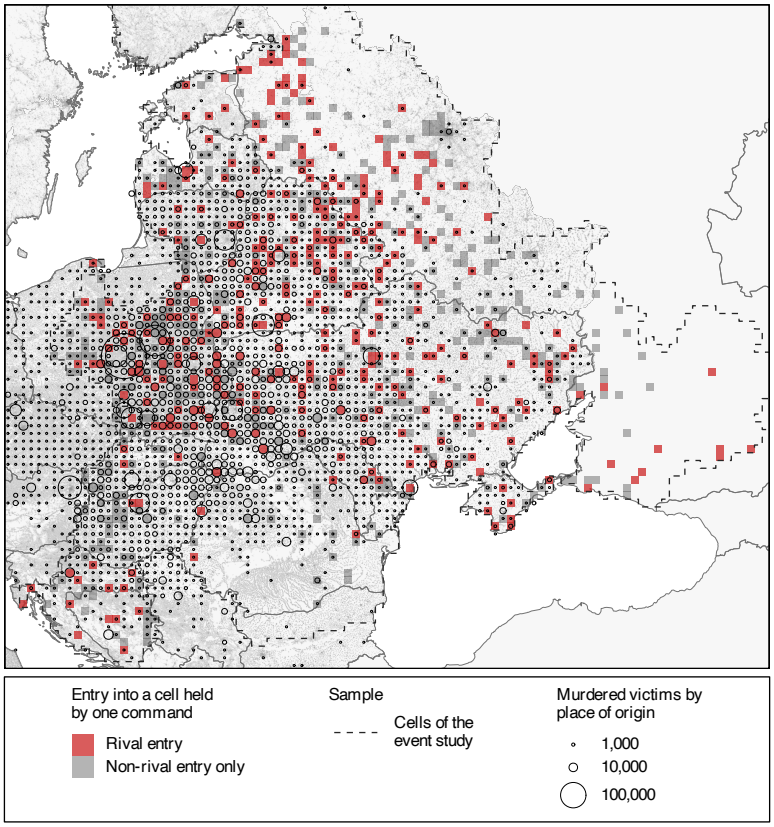
We next use the timing of troop movements to examine how killing responded to the arrival of a unit with rival and non-rival command structures. Figure 9 maps the cells of the East with each kind of entry. We compare changes in cells receiving a second unit with changes in cells that do not receive the same kind of entry. We stack cell-month observations by entry month and estimate

$$y_{cst} = \sum_{k \neq -1} \beta_k \, \mathbb{1}[t - g_s = k] \times T_{cs} + \alpha_{cs} + \lambda_{zst} + \varepsilon_{cst}, \tag{6}$$

where y_{cst} is cumulative deaths per 100 recorded residents in cell c, month t, and stack s; g_s is the stack's entry month; and T_{cs} identifies treated cells. Comparison cells had one command in the preceding month and no entry of the same kind from the entry month through the next nine months. Within each stack, they receive the same total weight as treated cells.[38] Stack-by-cell

[38] Appendix Figure A17 illustrates the design for a Security Police unit entering beside an army unit.

Figure 9: Rival and Non-Rival Entries



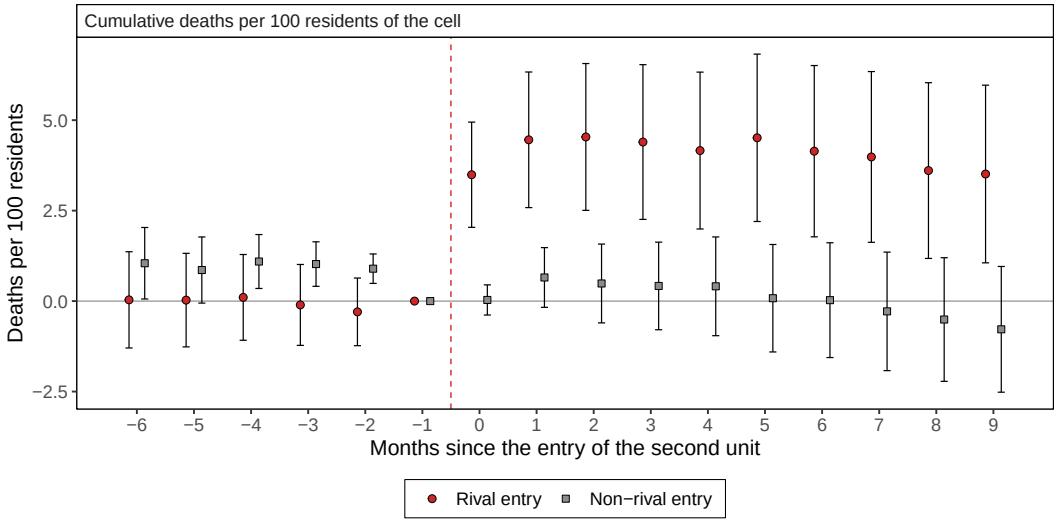
This figure maps eastern grid cells that received a second unit between September 1939 and May 1945 against the 1930 population (Appendix A.2). Cells measure 25 by 25 kilometers and had one command in the preceding month (Section 2.3). Circles show murdered victims by prewar residence, aggregated over cells with at least 10 victims, as in Figure 1. Borders are modern.

fixed effects (α_{cs}) absorb fixed differences across cells, while stack-by-zone-by-month fixed effects (λ_{zst}) absorb common changes within occupation regimes. Standard errors are clustered in 100-kilometer blocks.

The design assumes that, without entry, deaths in treated and comparison cells of the same stack and zone would have followed parallel trends. Comparing rival and non-rival entries asks whether the increase accompanies a change in command structure or also follows the arrival of another unit within the same command. Units within each chain reported through a common superior who compared their performance, so each command corresponds to an agent in the model (Section 3). The model predicts earlier implementation after a rival enters a domain held by one agent, but not after another unit of the incumbent agent arrives (Remark 2).

Killing rose after a rival command entered. Figure 10 plots the event-study estimates for rival and non-rival entries. Relative to comparison cells, cumulative deaths after rival entry rose by an

Figure 10: Entry of a Rival Command



This figure plots estimates of equation (6) using the 4,384 eastern grid cells from September 1939 to May 1945 for rival entry (350 cells) and non-rival entry (695 cells), as defined in Section 2.3. The outcome is cumulative resident deaths per 100 recorded residents (Section 2.1). The month before entry is the reference period. Bars show 95 percent confidence intervals based on standard errors clustered in 100-kilometer blocks.

average of about four per 100 recorded residents over the entry month and the next nine months. No comparable increase followed non-rival entry. Before rival entry, the coefficients are jointly indistinguishable from zero. For non-rival entry, they differ jointly from zero, indicating that treated and comparison cells were already on somewhat different paths.

We also estimate the association between command presence and subsequent killing,

$$y_{ct} = \beta D_{ct} + \alpha_c + \lambda_{zt} + \varepsilon_{ct}, \tag{7}$$

where y_{ct} is deaths per 100 recorded residents over the six months beginning in month t. For rival presence, D_{ct} indicates at least two commands. In a separate regression, it indicates multiple units of one command with no other command present. The regressions include cell and zone-by-month fixed effects.

Table 2 reports the presence and entry estimates. Rival presence is associated with about two additional deaths per 100 recorded residents over six months. The corresponding association for non-rival presence is small and statistically indistinguishable from zero. The entry estimates show the same contrast.

To test whether the increase included the extension of killing beyond men, columns (2) and (3)

Table 2: Rival Presence and the Killing of Residents

	Rival entry			Non-rival entry		
	All residents, deaths per 100	Women and children, deaths per 100	Women and children, in excess of men	All residents, deaths per 100	Women and children, deaths per 100	Women and children, in excess of men
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Presence, OLS						
Presence within 25 km	1.99***	1.85***	1.65***	0.29	0.32	0.40*
	(0.50)	(0.51)	(0.42)	(0.25)	(0.27)	(0.24)
Outcome mean	5.66	5.57	8.10	5.66	5.57	8.10
Observations	92,822	92,822	92,822	92,822	92,822	92,822
Panel B: Entry, event study						
Treated × post-entry	4.12***	3.97***	4.85***	−0.74	−0.56	0.86
	(1.21)	(1.21)	(1.26)	(0.72)	(0.75)	(0.85)
Leads, joint p	0.616	0.221	0.563	0.001	0.003	0.055
Outcome mean	34.17	31.92	53.35	37.56	35.17	53.13
Treated cells	350	350	350	695	695	695
Observations	160,700	160,700	160,700	117,007	117,007	117,007
Cell FEs	✓	✓	✓	✓	✓	✓
Zone-by-month FEs	✓	✓	✓	✓	✓	✓
Control for men's deaths			✓			✓

This table compares rival presence or entry (columns 1 to 3) with non-rival presence or entry (columns 4 to 6), as defined in Sections 2.3 and 5.2. The outcomes are murdered residents per 100 recorded residents, murdered women and children (female or under 15 at death) per 100 recorded women and children, and the percentage of murdered women and children already dead. Estimates for the last outcome control for the same percentage among murdered men aged 15 to 64. Panel A estimates equation (7) for deaths over the six months beginning in the current month, using 2,322 cells of occupied Poland and the occupied Soviet territory, June 1941 to May 1945. Panel B estimates equation (6) with one post-entry coefficient for cumulative outcomes in the 4,384 cells of the eastern grid and reports the joint test of the leads. Standard errors in parentheses are clustered in 100-kilometer blocks. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

examine women and children. The increase extended to women and children. Their deaths per 100 recorded women and children rose by about as much as deaths among recorded residents overall. Among women and children who were eventually murdered, the share already dead also rose after rival entry, controlling for the corresponding share among men (columns 2 and 3). We detect no comparable changes after non-rival entry (columns 5 and 6). These patterns are consistent with rival entry accelerating killing and its extension beyond adult men—the form of initiative described in Section 4.

These estimates persist with prewar Jewish population or total population as the denominator (Appendix Table A12) and in annual estimates that do not allocate deaths recorded only by year across months.[39] The entry estimates persist under alternative report dates, assumptions about how long units remained in a location, occupation-cohort controls, and distance thresholds (Appendix Figures A18 and A19).

[39]The annual estimates use deaths recorded to a month or a year, cumulated to the end of each year. Relative to the year before entry, deaths in the entry year rise by 2.7 per 100 recorded residents after rival entry (standard error 1.0) and change by −0.1 after non-rival entry (0.7).

5.3 Documented Transfers and the Invasion Plan

We next use two designs to assess whether rival commands entered places where killing would have risen anyway.

First, we instrument the presence of a second command with distance to the destinations of documented transfers. The transfer design compares actual destinations with places the commands could plausibly have reached. The Armed Forces High Command war diary records the moves of army formations, and the Einsatzgruppen documents and reports and Himmler's appointment books those of the SS and police: 892 executed moves to a town or district of the East between June 1941 and May 1945 (Appendix A.6). A rival transfer is a move into a place that one other command held in the preceding month. For each of the 107 rival transfers, we identify five nearby counterfactual destinations held by one other command in the preceding month, using army group areas to restrict the choice.

The instrument Z_{ct} is the inverse hyperbolic sine of the distance from cell c to the nearest destination of a documented move in month t, capped at 300 kilometers. Cells along the commands' routes may differ systematically from other cells. We therefore recenter the instrument (Borusyak and Hull, 2023), subtracting its expected value $\bar{Z}_{ct}$ when each rival transfer is assigned with equal probability to its actual destination or one of its five counterfactual destinations, holding other moves fixed. The recentered instrument $\tilde{Z}_{ct} = Z_{ct} - \bar{Z}_{ct}$ compares cells near actual destinations with cells near feasible alternatives. The first stage is

$$D_{ct} = \psi \tilde{Z}_{ct} + \alpha_c + \lambda_{zt} + u_{ct}, \tag{8}$$

and the second stage is equation (7) with D_{ct} instrumented.[40] The IV interpretation requires that, conditional on the fixed effects, each transfer's destination be as good as randomly drawn from its actual destination and five alternatives; the recentered instrument is then unrelated to otherwise expected changes in killing. It also requires that destinations affect killing only through rival presence. The recentered instrument is balanced on prior killing, with standardized differences below 0.01, and predicts rival presence with a first-stage F-statistic of 44 (Appendix Tables A14 and A15).

The IV estimate is 16.4 additional deaths per 100 recorded residents over six months (standard

[40] Appendix D.1 details the classification of transfers, the counterfactual destinations, and balance.

error 7.7), compared with 2.0 in OLS (Appendix Table A15).[41] The effects measured per entry are closer: a rival transfer 25 rather than 100 kilometers from a cell adds 2.8 deaths per 100 recorded residents over six months, compared with 4.1 in the main event study. The IV coefficient divides this death effect by the corresponding increase in recorded rival presence, only 0.17 (Appendix Table A15). Appendix D.2 discusses how measurement error in recorded presence and the concentration of transfers in 1941, when entry effects were largest, contribute to the gap between IV and OLS.

Second, we compare cells that the invasion plan scheduled to fall behind the front in the same month. The planned-cohort design addresses differences in the anticipated military advance. Before the campaign, the invasion plan specified when each cell was to fall behind the front (Appendix Figure A22). Cells scheduled for the same month form a planned cohort. We estimate equation (6) using only comparison cells in the treated cell's stack and planned cohort. This holds planned occupation timing fixed. Among these cells, rival entry is followed by about two additional deaths per 100 recorded residents. The estimate is similar after controlling for destinations recorded in plans and orders (Appendix Table A17, Panel B).[42] Fewer than one in twenty rival entries match moves documented as planned or ordered beforehand.

The increase is not confined to the entry of Security Police units beside the army: the other rival entries, among them army units entering beside the SS and police, show a similar rise (Appendix Figure A20). We observe unit counts but not troop strength, however, and cannot rule out that rival commands brought greater armed force.

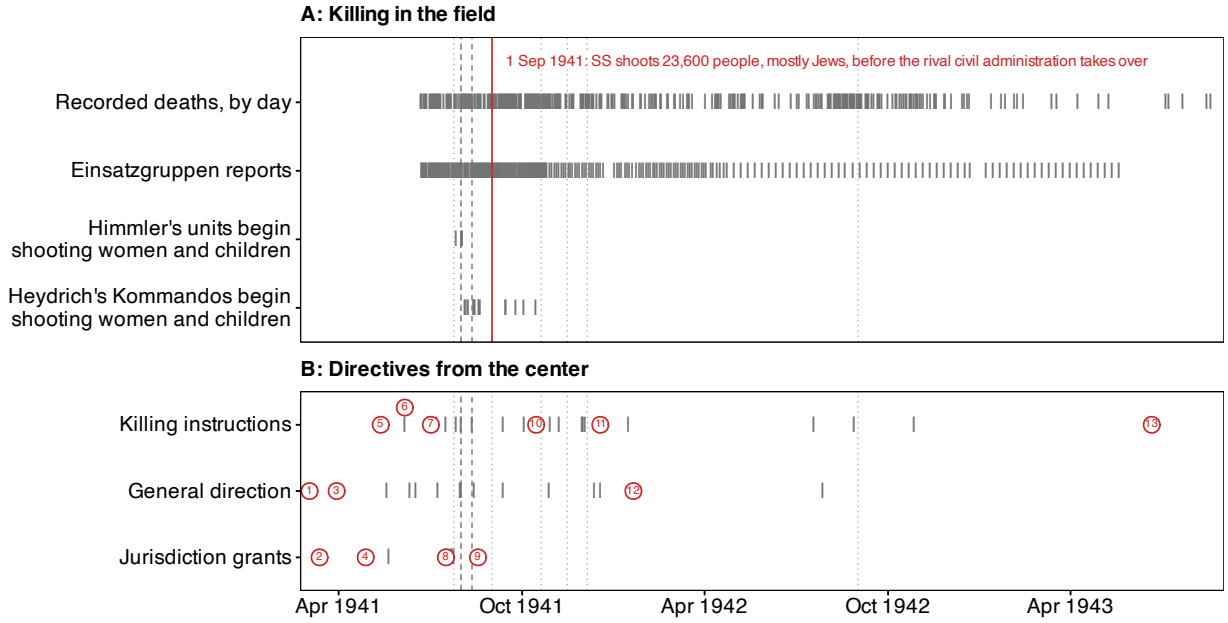
These comparisons place the rise in killing after rival entry but do not establish how it was set in motion. We therefore turn to central directives and senior officials' private records.

6 Mechanism: Initiative

We use central directives and private records to trace the sequence of direction, initiative, and approval. After the war, SS leaders on trial invoked a Führer order, but "no direct witness of such an order survived the war" (Kershaw, 1985, p. 135). We first compare documented directives with the start and extension of killing in occupied Soviet territory. We then use the diaries of five senior

[41] For women and children, the estimates are 16.3 and 1.9 deaths per 100 recorded women and children, respectively.
[42] The estimates are 1.9 without destination controls and 2.3 with them, both with a standard error of 1.1.

Figure 11: Killing in the Field and Directives at the Center, 1941 to 1943



This figure compares killing in the field (Panel A) with central directives (Panel B) from March 1941 to December 1943. Panel A marks days with at least 100 recorded deaths among residents of the occupied Soviet territory (Appendix A.1), the issue dates of 249 Einsatzgruppen reports, and the dates when units under 21 of the 26 commanders in Appendix Figure A13 began killing women and children, separately for Himmler's units and Heydrich's Kommandos. Panel B classifies 50 directives as jurisdiction grants, direction, or instructions (Section 6 and Appendix Table A13). Numbers identify directives discussed in the text. The red line marks the transfer of Kamenets-Podolsk to civil administration on 1 September 1941, Jeckeln's deadline for the shootings. Dashed lines mark Himmler's order of 1 August and his meeting with Jeckeln on 12 August. Dotted lines mark decreed transfers to civil administration.

officials to study the sequence of action and approval across other areas of policy.

We classify records using the terms of Section 4.1. A jurisdiction grant authorizes an agency to act in a domain or divides authority among agencies. Direction states a goal or specifies a procedure. An instruction orders, prohibits, or approves an act. Initiative advances a stated goal beyond the latest instruction.

6.1 Killing before Orders

In the records we examine, the killing of Jews in occupied Soviet territory began before the central instructions explicitly ordering it. Some units also began killing women and children before orders extended to them. Figure 11 compares killing in the field with central directives from 1941 to 1943. None of the directives is a written order by Hitler to murder all Jews.

Before the invasion, central directives granted jurisdiction and stated goals. The first recorded

instructions to kill targeted civilian suspects and political commissars.[43] The victim series begins on 22 June, and the first Einsatzgruppen report dates to the following day. Explicit instructions concerning Jews followed: Heydrich's telex on inciting pogroms on 29 June, Himmler and Heydrich's approval of border-zone massacres on 30 June, and Heydrich's letter of 2 July, which informed the Higher SS and Police Leaders of the instructions to the Einsatzgruppen and named Jews in party and state posts among those to be executed (item 7). Hitler divided civil and police authority between Rosenberg and Himmler later that month (item 8).

The extension of killing followed a similar sequence. Jeckeln's units began murdering women and children on 27 July, before Himmler's order of 1 August instructing the SS cavalry to kill all Jews. The cavalry followed on 2 August. Several more commanders' units escalated in mid- to late August, during and after Himmler's extended order of 12–15 August; the last did so in mid-October. The initial escalation before the documented order is consistent with the model's prediction that rivalry can induce implementation before instructions arrive (Lemma 1(ii)). Transfers of jurisdiction could also create deadlines, as at Kamenets-Podolsk, where Jeckeln set the transfer to civil administration as his deadline for killing Jews expelled from Hungary (Section 3; Figure 11, Panel A).

Central directives sometimes restricted killing. Lohse's provisional guidelines in August and his October ban on shootings at Liepāja set limits for the civil administration (items 9 and 10). On 30 November, Himmler's telephone log recorded that a transport of Berlin Jews to Riga was not to be liquidated; they had already been shot that morning. The next day, he warned Jeckeln that unauthorized acts would be punished (Longerich, 2010, p. 299). On 18 December, the ministry for the occupied East told Lohse to disregard economic interests (item 11), and Hitler told Himmler to extirpate Jews as partisans. The records provide no evidence of a system to verify compliance with these limits, and killing elsewhere continued while they stood.

Most directives in our records date from the first months of the killing. More than four fifths precede the Wannsee Conference of January 1942, where Heydrich outlined the mass murders

[43] Hitler called for eliminating the "Jewish-Bolshevist intelligentsia" and described the coming war as one of annihilation (items 1 and 3). The Armed Forces High Command granted Himmler special responsibilities in March, and the army authorized Security Police units to act on their own responsibility in April (items 2 and 4). In May, the armed forces issued guidelines for troops, and Himmler placed SS and police units under the Higher SS and Police Leaders. Instructions to kill targeted civilian suspects on 13 May and political commissars on 6 June (items 5 and 6). Himmler and Heydrich briefed their field commanders in mid-June.

to senior officials (item 12). The order to dissolve the Ostland ghettos in June 1943 followed the large-scale killing of the previous year (item 13).[44]

6.2 Approval across the Regime

The diaries of Goebbels, Frank, Himmler, Rosenberg, and Bach-Zelewski extend the evidence beyond the Holocaust in the East. We use them to examine how officials understood competition and whether approval generally followed action. Jurisdictional disputes appear on about one quarter of days with an entry (Figure 12, Panel A).[45]

Nearly all of the more than 13,000 recorded policy-setting actions were the writers' own decisions. Orders and prohibitions received from superiors accounted for only a small fraction (Figure 12, Panel B). This describes the decisions preserved in the diaries; it does not establish how all policy was made.

Approval usually followed action. We group each diary's actions on the same policy within 30 days into policy rounds, separately for writers under their superiors and subordinates under the writers. In more than 500 rounds, the diary records both an agent's action and the superior's approval. The agent acts first in about four fifths. Because rounds often contain several actions and one approval, this ordering could arise by chance: with three actions and one approval, a random ordering puts an action first three times in four.

We therefore randomly reassign actions to the recorded dates within each round. Agents act first in just over half of the reshuffled rounds, on average. None of the reshufflings reaches the observed share (Figure 12, Panel C). This sequence is consistent with the model's prediction that superiors approve after agents act (Lemma 1).[46] Because actions include proposals and requests, this ordering alone does not establish that implementation preceded authorization.

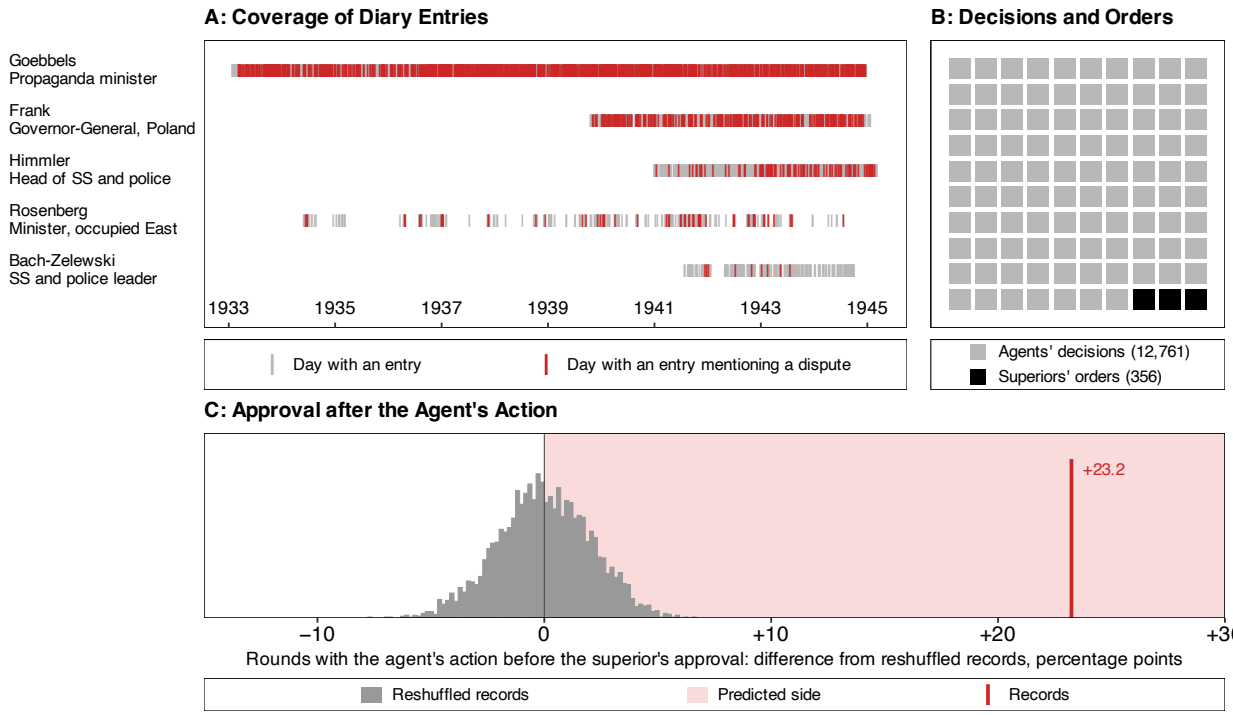
The diaries are selective records. Writers may describe their own actions more fully and fa-

[44] Month-dated deaths among residents of occupied Soviet territory fell by more than four fifths between 1942 and 1943.

[45] Goebbels wrote of "Endless disputes over money and jurisdiction" in June 1933, adding, "But Hitler is on my side." In 1937, he described ministries that "fight like warring powers. And nothing happens." Rosenberg wrote in July 1943 "that we have no government," only "groups like the Diadochi, but no war cabinet." Bach-Zelewski described having to "fight hard for every bit of power to be able to act."

[46] Appendix Figure A21 tests the full sequence in Lemma 1 and Assumption 3: superiors prohibit and order after agents report but before they decide, and approve after agents decide and report. All six predicted orderings occur more often than under reshuffling, both for writers under their superiors and for subordinates under the writers. No reshuffling reaches the observed average excess.

Figure 12: Diary Entries, Decisions, and Approvals



This figure summarizes actions coded from the diaries of Goebbels, Frank, Himmler, Rosenberg, and Bach-Zelewski (Appendix A.3). Panel A marks days with entries recording actions (gray) and disputes (red). Panel B reports the shares of policy-setting actions that are the writers' own decisions (gray) and their superiors' orders or prohibitions (black). Superiors are Hitler and, for Bach-Zelewski, Himmler. Panel C groups actions on one policy within 30 days into rounds for writers under their superiors and for subordinates under the writers. It shows the share of rounds in which the agent's first action preceded the superior's first approval, minus the share under 4,000 reshufflings of actions across the round's dates (Appendix equation (10)). The red line marks the records, gray bars the reshufflings, and the pink area the side the model predicts.

vorably than orders received, inflating the share of decisions attributed to them.[47] They may also emphasize personal quarrels and thus overstate conflict. Historians nevertheless document jurisdictional disputes through the end of the war. Such disputes also occur in parliamentary democracies. What distinguished the Nazi regime was the absence of institutional and legal channels for communication, negotiation, and compromise that could contain them (Herbert, 2019, pp. 253–254). In the model, this absence is captured by the assumption that agents cannot contract with one another (Assumption 1). Competition for the prize resolves contested domains and induces action before the ruler can assess its appropriateness.

[47] Rosenberg's diary barely mentions the extermination despite his knowledge of it (Matthäus and Bajohr, 2015, p. 26). While his forces suppressed the Warsaw Uprising in August 1944, Bach-Zelewski claimed that Himmler had given him full authority and that he would "save women and children" (transcript, p. 108).

7 Conclusion

We examine the organization of the Nazi regime and document four facts about it. First, the institutions listed in the directories of the Reich doubled between 1931 and 1936. Second, many of these institutions were extraordinary mandates tied to persons. Third, by 1936 almost every policy domain had two or more supreme bodies. Fourth, the head of government stopped attending cabinet-level meetings while decisions were increasingly attributed to him in his absence. We then find that the entry of a rival command in a locality in the East was followed by more killing, while the entry of a further unit of the same command was not. An instrument built from the documented transfers of the commands also gives a rise in killing. The killing began before the central instructions that named its victims, the commanders whose units escalated first were rewarded, and, across the regime, superiors approved after their agents acted. Together, these results suggest that the organization of the regime, and not only the aims of its ruler, could shape the pace of its violence: where rival agents acted for a future prize, the killing came before the orders.

Dictatorship operates through agents the ruler cannot fully direct. The arrangement of a domain decides which error he bears: rivalry elicits initiative before information arrives and can produce implementation he would reject, while direct control waits for information and risks implementing too late. The findings do not fit the image of one ruler above a single chain of command that executes instructions. Overlapping authority could produce action beyond what the ruler would choose where, as in Nazi Germany, agents acted for a future prize.[48]

Our findings point toward a trade-off in the division of authority: rivalry elicits initiative at the cost of control over what is implemented. A ruler who chooses rivalry accepts acts he would reject in exchange for acts that come in time. The ruler remains responsible for the direction he chooses and the rivalry he creates, even when he does not issue every instruction the system produces.

[48]The same organization appears beyond Nazi Germany (Appendix B). Under Stalin, regional party secretaries and NKVD chiefs jointly administered the mass operations of 1937–38, and regions bid against one another for higher execution limits. In China, provincial leaders compete for promotion, and the center has recentralized control over local policy experiments.

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Supplementary Appendix: For Online Publication

A Data

Table A1 reports summary statistics for each dataset. The following sections describe the victims, the population grid, the internal records, the state directories, the policy domains, and the troop units.

Table A1: Summary Statistics

Variable	Mean	SD	Min	Max	N	Enters as
Panel A: Reich Chancellery documents, 1925 to 1945 (document)						
Cabinet-level meeting	0.42	0.49	0.00	1.00	5,262	(0/1)
Hitler listed as present	0.03	0.17	0.00	1.00	5,262	(0/1)
Hitler named, not present	0.39	0.49	0.00	1.00	5,262	(0/1)
Führer decision invoked	0.20	0.40	0.00	1.00	5,262	(0/1)
Extraordinary mandate named	0.19	0.40	0.00	1.00	5,262	(0/1)
Institutions named	5.13	4.00	0.00	38.00	5,262	(count)
Persons named	13.42	10.97	0.00	81.00	5,262	(count)
Panel B: Party Chancellery files, 1933 to 1945 (file)						
Persons indexed	2.87	8.65	0.00	700.00	25,656	(count)
Summary names Hitler	0.18	0.39	0.00	1.00	25,656	(0/1)
Panel C: State directories, 1931 to 1942 (institution in an edition)						
State administration	0.70	0.46	0.00	1.00	1,528	(0/1)
NSDAP, units, affiliates	0.14	0.35	0.00	1.00	1,528	(0/1)
Corporatist body	0.13	0.34	0.00	1.00	1,528	(0/1)
Persons listed	17.68	60.11	1.00	760.00	1,528	(count)
Panel D: Troop units, September 1939 to May 1945 (dated place statement)						
Army	0.82	0.38	0.00	1.00	176,727	(0/1)
Air force	0.04	0.19	0.00	1.00	176,727	(0/1)
Navy	0.01	0.09	0.00	1.00	176,727	(0/1)
Waffen-SS	0.05	0.22	0.00	1.00	176,727	(0/1)
Order Police	0.06	0.23	0.00	1.00	176,727	(0/1)
Security Police and SD	0.02	0.15	0.00	1.00	176,727	(0/1)
Placed	0.97	0.17	0.00	1.00	176,727	(0/1)
Placed at a town or district	0.66	0.48	0.00	1.00	176,727	(0/1)
Dated to the month	0.71	0.46	0.00	1.00	176,727	(0/1)
Panel E: Yad Vashem persons with a located place (person)						
Murdered or presumably murdered	0.67	0.47	0.00	1.00	7,170,091	(0/1)
Place of origin located	0.98	0.13	0.00	1.00	7,170,091	(0/1)

This table summarizes each dataset. Panel titles give the unit of observation, and indicator means are shares. State directories cover institutions listed with at least one person in the 1931, 1936, 1939, and 1942 editions (Appendix A.4). Troop units cover unit-months with a reported location (Appendix A.6). Yad Vashem persons are database clusters with a located place. The occupied Soviet territory panel covers cells with a recorded victim or unit. The share of victims dead is calculated for cells with at least 20 dated murdered victims by origin.

A.1 Victims

Procedure. We locate each place string of the Yad Vashem database (Yad Vashem, 2026) at the coordinate that most of its records give, because the records of one person can give the same place different coordinates. We exclude country names from this exercise. We use two places for each person: the place of origin, which assigns the victim to a cell, and the place and month of death, which locate and date the recorded deaths. The cell-month panel of the occupied Soviet territory counts the persons whose recorded death falls in a cell and month, from June 1941 to July 1944.

Dating every victim. For the event study around the entry of a second unit, we date every murdered victim by the cell of prewar residence. A victim whose death is recorded to the day or the month keeps that month; where no death date is recorded, the month of deportation or transport serves instead. A victim recorded to the year only is spread over the months of that year in proportion to the month-dated deaths of the smallest reference group with at least 20 month-dated deaths in that year: the victim's own cell, else its present-day province, else its wartime territory, else the whole frame. A victim with no date is spread over all months of the war in the same way. Most murdered residents of the frame carry a year or no date. The victim's own cell serves as the reference group for most of them, and the province for nearly all others. The procedure assumes that the undated victims of a reference group died in the months in which its dated victims died.

Coverage. Recorded deaths with a place and a month are a minority of the murdered victims placed by origin in the grid, and their share differs by region. Because deportations moved victims away from home, the place of death and the place of origin measure different events. Appendix Figure A15 plots the distance from home at death by territory and half-year.

Dating to the day. Figure 11 dates the murdered residents of the occupied Soviet territory of June 1941, eastern Galicia included, by the day of death where a record gives one, wherever they died. The day survives only in the raw records; a death dated 1 January stands for an unknown day and is left out. From May 1941 to December 1943, 151,114 residents carry a day of death, between half and three quarters of those dated to the month. The month-dated deaths of residents fall from 109,961 in 1942 to 16,339 in 1943, and the day-dated from 81,557 to 9,923. The figure marks the days with 100 or more.

A.2 Population

We build a population surface for Europe around 1930 on a grid of one square kilometer in the European equal-area projection. The surface holds 391.8 million people in 6.22 million cells across 34 countries. A cell inside the covered area with no population is recorded as zero, so a cell absent from the file lies outside the covered area. We release the surface with the code that builds it from the published sources and with the validation statistics below.

Sources. Table A2 lists the sources of the counts and of the boundaries for each country. For 22 countries, the counts come from the census closest to 1930 at the level of the municipality, the commune, the parish, or the settlement; for Finland, they are the parish registers of 31 December 1930. The other 12 report counts for districts or counties, among them the Soviet censuses of 1926 and, for western Russia and Crimea, of 1939. The boundaries are those of the census year where a map of that year exists. Elsewhere, we match the census units to later boundaries, and Table A2 gives the year of each.

Procedure. We build the surface in four steps. First, we keep only the units observed at the census nearest 1930. The units of different censuses cover overlapping ground, and mixing them would count the same people twice; in Poland, it would add five million people.

Second, we move each unit to 1930. We interpolate a unit observed at several censuses log-linearly and move a unit observed once by the growth of its country between that census and 1930. Where 1930 itself is observed or lies between two observed censuses, the level is the published count, which we do not rescale. Where neither holds, as for the Soviet census of 1926 and the

Lithuanian census of 1923, the published national figure for 1930 sets the level, and the released file flags these countries. Western Russia is the one region built from a census after 1930. We give each district of the census of 1939 a value for 1926 by scaling its towns and its villages by the ratio of the urban and of the rural population of 1926 to that of 1939 in its group of gubernias; twelve cities, Moscow and Leningrad among them, carry their own counts of 1926 (TsSU SSSR, 1928). We then interpolate the two values as for any unit observed twice.

Third, we give each unit a territory. For Germany, we assign the present-day cadastral districts (Vermessungsverwaltungen der Länder, 2026; GUGiK, 2026; Rosreestr, 2022) to the municipalities of 1933, within the county boundaries of 1935 from the Mosaic collection of historical German boundaries (MPIDR and CGG, 2011; Hubatsch and Klein, 1975). Where a country reports units that were never mapped, we give the seat of each unit a territory by tessellation: the communes of Poland within the county boundaries of 1932, the districts of western Russia within the present-day region of their seat, and the units of Greece, Albania, and Subcarpathian Ruthenia within present-day boundaries. The seats come from the GeoNames gazetteer and, for Poland, also from the Geschichtliches Ortsverzeichnis (GeoNames, 2026; Verein für Computergenealogie, 2026); the Greek seats come from the lists of the eparchies. The tessellated territories of Poland, western Russia, and Subcarpathian Ruthenia yield to the mapped boundaries of any other country. The districts of Soviet Belarus of 1926 were never mapped either; through their seats, they join the district polygons of 1940–41 of the pre-1939 republic. A town that a source records as a point receives a disc that holds its population at 5,000 people per square kilometer.

Fourth, we spread the population of each unit over the cells it covers in proportion to a weight, and we scale the cells to sum to the unit total. The weight of a cell is its residential built-up volume in 1975, the earliest epoch of the Global Human Settlement Layer (Pesaresi and Politis, 2023; Pesaresi et al., 2024), multiplied by $\exp(-d)$, where d is the distance in kilometers to the nearest populated place in GeoNames. Where the 1975 layer shows no building in a unit, as across much of northern Norway, the distance term alone sets the weight. We rasterize the units at 250 meters before aggregating to one kilometer, so that a unit smaller than a square kilometer keeps its population and a unit nested in another, such as a Soviet town in its district, keeps its own territory. The allocation assumes that within each unit the settlement pattern of 1975 reflects that of 1930.

Validation. We validate the surface in three ways, and the released file reports each. First, we compare the total of each country with its published national figure. Thirty-two of the 34 totals agree within 1.5 percent. For eight countries without a census in 1930, the published figure for 1930 sets the level. For six of them, Lithuania, the Memel Territory, Yugoslavia, Bulgaria, Soviet Ukraine, and Soviet Belarus, which hold 15 percent of the population, the units sum exactly to the census total, so the agreement holds by construction; the released file flags them. The same holds for Venezia Giulia and the Free City of Danzig, whose units partition the published total and which hold 0.26 percent of the population. For the other 26 countries, the comparison is an independent check. For western Russia, the benchmark is the census of 1926 for the same gubernias, moved to 1930 by the growth of the Russian Soviet republic (Andreev et al., 1998); the surface is 1.25 percent below it. The cells of Spain hold 572,273 fewer people than its units, the population of the Canary Islands and the African enclaves, which lie outside the grid.

Second, we measure the error of the allocation. We aggregate the 50,183 German municipalities of 1933 to coarser administrative levels, reallocate the aggregates, and compare the result with the surface built from the municipalities themselves (Table A3). With units the size of a Kreis, comparable to a Soviet district, 24.9 percent of the population falls in a different square kilometer, and the correlation with the municipal surface is 0.93 in logs; with units the size of a German

state, the two figures are 35.1 percent and 0.91. Spreading the population evenly over each unit instead misplaces 60.0 and 69.7 percent. Countries with municipal counts therefore carry little allocation error, and countries with district counts carry the error of the Kreis row. In the German test, tessellating the seats of the municipalities, the method used for Poland and western Russia, misplaces 16.9 percent; the median tessellated unit covers 84 square kilometers in Poland and 1,011 in western Russia, against 8 in the test.

Third, we count the people placed in a cell that another country also claims: 499,580 people, or 0.13 percent of the total, almost all where boundaries of different years meet, with the largest concentration in Upper Silesia.

Coverage and special cases. The surface does not cover Great Britain, Ireland, or European Russia east of Moscow and Voronezh. Romania enters at its present-day extent. The lands it held in 1930 and ceded in 1940, Bessarabia, northern Bukovina, and southern Dobruja, with 3.7 million people, enter separately on the interwar map of the plasa. Five territories whose borders differ from those of today enter as units of their own: these lands, Subcarpathian Ruthenia, the Free City of Danzig, the Memel Territory, and the territory Italy held east of its present border until 1947, from Istria and the Kvarner to the Slovene Littoral, Fiume, and Zara.

Two national totals depart from the published figure by more than 1.5 percent. The total of Denmark is 5.4 percent below it. The Danish census reports the country both as parishes and as municipalities, and only the municipal table includes the market towns; the shortfall is the 114 of its 1,394 units that match no present-day parish. The total of Finland is 3.5 percent above the published figure. Its counts are the parish registers of 31 December 1930, because the census of 1930 covered towns only. The registers count 3,667,067 people, 286,319 of them absent from their parish, and they still carry the emigrants that the official total of 3,462,700 leaves out.

Eleven German cities whose municipal polygon is too small for their population, such as Hamburg on 8 square kilometers and Leipzig on 9, take their county polygon of 1933 from the Mosaic collection (MPIDR and CGG, 2011) and spread their population by the weight. Any other polygon that implies a density no European city reached becomes a disc at 5,000 people per square kilometer, filled evenly, and the released file lists these units. The densest cell holds 90,092 people. The released file also marks the cells of Poland, western Russia, and Subcarpathian Ruthenia, whose territories are tessellated; they hold 61.8 million people, 15.8 percent of the total.

Table A2: Sources of the 1930 Population Grid

			Counts		Boundaries			
Country	Unit	Units	Census	Source	Year	Source	Population	Diff. (%)
Germany	Gemeinde	50,183	1933, 1935	Statistisches Reichsamt (1936)	1933	Vermessungs-verwaltungen der Länder (2026), GUGiK (2026), Rosreestr (2022), and MPIDR and CGG (2011)	64,494,192	−0.63
France	commune	37,987	1921–1936	Litvine et al. (2026)	1921–1936	Litvine et al. (2026)	41,575,548	−0.08
Italy	comune	8,076	1921–1936	ISTAT (1994)	2001	Eurostat (2026a)	40,775,899	−0.03
Poland	gmina	3,671	1921, 1931	GUS (1933)	1932[a]	IH PAN and WGiK PW (2022), GeoNames (2026), and Verein für Computergenealogie (2026)	31,457,137	0.18
Soviet Ukraine	raion, town	868	1926	TsSU SSSR (1928)	1927	HURI (2026)	31,200,000	0.00
Western Russia and Crimea	raion, town	912	1926, 1939	TsSU SSSR (1928) and RGAE (1939)	2022[a]	GeoNames (2026) and Natural Earth (2022)	29,569,970	−1.25
Spain	municipio	8,115	1920–1940	Goerlich Gisbert et al. (2015)	2011	Goerlich Gisbert et al. (2015)	24,026,571	0.00
Romania	commune	2,902	1930	Institutul Central de Statistică (1938)	2021	Eurostat (2026b, 2026c)	14,220,741	−0.42
Yugoslavia	srez	378	1931	Opšta državna statistika (1937)	1931	Popović and Popović (2020)	13,700,000	0.00
Czechia	obec	6,250	1921, 1930	ČSÚ (2017)	2021	Eurostat (2026b)	10,672,792	−0.01
Hungary	settlement	3,150	1920–1941	KSH (1923, 1932, 1947)	2021	Eurostat (2026b)	8,685,109	−0.04
Belgium	municipality	581	1920, 1930	Administration de la Statistique Générale (1922) and Office Central de Statistique (1934)	2021	Eurostat (2026b)	8,091,997	0.00
Netherlands	gemeente	1,078	1930	CBS (2006)	1930	Boonstra (2007)	7,935,567	0.00
Austria	Gemeinde	2,122	1923, 1934	Statistik Austria (2018)	2017	Statistik Austria (2017)	6,668,445	−0.17
Greece	eparchy	142	1928	Genikē Statistikē Hypēresia tēs Hellados (1935)	2022[a]	GADM (2022)	6,390,000	0.00
Portugal	concelho	273	1930	Direcção Geral de Estatística (1933)	2022	GADM (2022)	6,335,041	−0.40
Sweden	county	25	1920–1940	SCB (1969)	1930	Junkka (2022)	6,142,191	0.00
Bulgaria	okoliya	84	1934	Glavna direktsiia na statistikata (1938)	2021	Eurostat (2026b, 2026c)	5,900,000	0.00
Soviet Belarus	raion, town	205	1926	TsSU SSSR (1928)	1940	Zhukov (2017) and GeoNames (2026)	5,300,000	0.00
Switzerland	Gemeinde	2,889	1920–1941	BFS (2020)	2000	BFS (2025)	4,066,400	0.00
Romania, ceded lands	plasa	81	1930	Institutul Central de Statistică (1938)	1930	Buterez (2016)	3,721,834	0.08
Finland	kunta	569	1930	Tilastollinen päätoimisto (1933)	1930	Hyvönen et al. (2011) and Lahti et al. (2023)	3,585,015	3.53
Denmark	sogn	1,280	1930	Det Statistiske Departement (1931)	2014	Geodatastyrelsen (2014)	3,357,406	−5.44
Slovakia	obec	2,847	1921, 1930	Majo et al. (2017)	2021	Eurostat (2026b)	3,320,728	−0.27
Norway	municipality	229	1920, 1930	Sikt (2026)	2024	Kartverket (2024) and SSB (2026)	2,814,194	0.00
Lithuania	valscius	282	1923	Centralinis statistikos biuras (1926)	1929	SSVA (2018)	2,208,000	0.00
Latvia	pagasts, town	529	1925–1935	Valsts statistiskā pārvalde (1925, 1930, 1936)	1926	Kaftan et al. (2025)	1,900,045	0.00

continued on next page

Table A2 continued

			Counts		Boundaries			
Country	Unit	Units	Census	Source	Year	Source	Population	Diff. (%)
Estonia	county, town	46	1922, 1934	Riigi Statistika Keskbüroo (1924, 1935)	1938	Rahvusarhiiv (2026)	1,107,226	−1.14
Albania	district	40	1930	Ministria e Punëve të Brendshme (1930)	2022[a]	GeoNames (2026) and GADM (2022)	1,003,097	0.00
Subcarpathian Ruthenia	obec	400	1930	Státní úřad statistický (1937)	2022[a]	GeoNames (2026) and GADM (2022)	725,357	0.00
Venezia Giulia and Zara	naselje, comune	279	1931	DZS (2026) and ISTAT (1932)	2026	DGU (2026), GURS (2026), GADM (2022), and GeoNames (2026)	599,448	0.00
Free City of Danzig	city, Kreis	5	1929	Statistisches Landesamt der Freien Stadt Danzig (1930)	1929	OpenHistoricalMap (2026) and Natural Earth (2022)	407,517	0.00
Luxembourg	commune	100	1922–1935	STATEC (2026)	2023	Administration du cadastre et de la topographie (2023)	299,872	0.03
Memel Territory	Kreis	4	1925	Statistisches Reichsamt (1939)	1929	SSVA (2018)	145,900	0.00

This table lists the sources of the 1930 population grid, one row per country or territory, ordered by population. Unit is the census unit, and Units counts the units observed at the census nearest 1930. Census gives the years of the counts; Germany includes the Saar at its census of 25 June 1935, and the counts of Finland are the parish registers of 31 December 1930. Year gives the date of the boundaries, and [a] marks territories built by tessellation around the seats of the units within them. Population is the total for 1930 after interpolation, and Diff. is its deviation from the published national figure for the same territory, in percent.

Table A3: Accuracy of the Population Grid

			Population misplaced (%)		
Source units	Units	Median area (km^2)	By built-up	By area	Correlation (logs)
Gemeinde (the source data)	50,183	7	0.0	44.0	1.000
Kreis	806	519	24.9	60.0	0.931
Regierungsbezirk	65	5,768	31.7	66.4	0.923
Land	17	5,976	35.1	69.7	0.911
Gemeinde seats, tessellated	50,183	8	16.9	47.4	0.960

This table compares population surfaces from coarser units with the benchmark, which allocates the 50,183 German municipalities of 1933 to the one-kilometer grid by built-up volume. Municipal populations are aggregated to each coarser level and reallocated by built-up volume (By built-up) or evenly over area (By area). The last row tessellates the municipal seats. Population misplaced is the share of people placed in the wrong cell. Correlation is between the logs of the built-up allocation and the benchmark over all cells.

A.3 Internal Records

Sources. We use two editions of the *Akten der Reichskanzlei* (Historische Kommission bei der Bayerischen Akademie der Wissenschaften and Bundesarchiv, 2025): the online edition, with 4,519 Weimar cabinet documents and 1,601 Hitler government documents through 1939, and the five print volumes for 1940 to 1945, whose 1,233 documents we photograph and read page by page. We use the Institut für Zeitgeschichte's online edition of the *Akten der Partei-Kanzlei* (Heiber, 1983) and online editions of the situation reports (Boberach, 1984; 1999), Nuremberg catalog (Institut für Zeitgeschichte, 2006), Goebbels diaries (Fröhlich, 1993), and Hitler's texts (Institut für Zeitgeschichte, 1992).

Procedure. We identify institutions from abbreviation-register entries that expand to organization names. We exclude gazettes, currencies, ranks, and editorial terms and harmonize ministry

names across the Weimar and Hitler-era registers. We count a person as present if the edition lists him as a participant or recorder; a name in the text counts as a mention. We code policy domains from each source's headings under rules fixed before analysis.

Selection. Editorial selection differs between the Weimar and Hitler-era editions. The reconstructed Party Chancellery files draw on recipient offices' records, so domain coverage reflects those offices. We therefore compare records within each source.

Diaries. A language model (Gemini 3.7 Flash) reads each page or entry of the five diaries and records every action it reports, with its date, actor, counterpart, type, and policy, a one-sentence summary close to the text, and a quotation of at most 40 words. It also lists the persons each entry names, with the writer's relation to each. Table A4 gives the definitions of the thirteen types of action and of the relations, with examples. Listing A1 gives the prompt.

Listing A1: Prompt Submitted with Each Page or Entry of the Diaries

```
### ROLE
You read one page or entry of a German diary, appointment book or office journal written between 1925 and 1945, opened by
     a marker [volume, page or entry, date].

### SCHEMA
- **acts** (array of objects):
  - **date** (string): date of the act, ISO 8601 at the text's precision (1941-08-12, 1941-08, 1941)
  - **kind** (string, one of):
    the act
    * direction: sets a goal or a priority, also in a speech, assigns jurisdiction or fixes a procedure, without naming an
      act
    * order: specifies an act for a subordinate
    * prohibition: forbids or restricts an act
    * approval: approves, praises, ratifies or rewards conduct: a post, a promotion, a decoration, a jurisdiction,
     resources, personnel
    * decision: decides or acts on its own authority
    * proposal: puts a scheme, plan or policy forward for a superior to adopt, unasked
    * request: asks a superior for a decision, permission or ruling
    * report: reports conduct or results to a superior
    * dispute: contests, obstructs or negotiates another person's or office's action, claim or jurisdiction
    * defiance: ignores, delays, evades or sabotages an order received
    * trust: expresses trust, loyalty, confidence or distrust toward a superior, a subordinate or another office
    * information: learns of another person's or office's action or plan
    * assessment: judges an act, a person or an office
  - **actor** (string): who acts, with the office, as written
  - **counterpart** (string): the other party: the superior addressed or answering, the subordinate ordered, the person or
      office contested, trusted, learned of or judged; as written
  - **policy** (string): the policy and measure concerned, as the text renders it
  - **content** (string): what is set, ordered, decided, proposed, asked, reported, approved, contested, learned or judged
     , in one sentence close to the text
  - **reason** (string): the reason the text gives, in its words; empty if none
  - **alternative** (string): another course the text names: proposed, weighed, rejected, postponed, cancelled, or what
     could or should have been done instead; empty if none
  - **stake** (string): what the actor stands to gain or lose, as the text names it: a post, a jurisdiction, resources,
     standing, a sanction; empty if none
  - **quote** (string): the passage's words, verbatim, at most 40 words
  - **inferred** (string): fields filled by inference rather than from the words of the text, separated by semicolons;
     empty if none
- **persons** (array of objects):
  - **name** (string): a person named in the entry, as written
  - **office** (string): the office or role the entry gives or implies; empty if none
  - **relation** (string, one of):
    the writer's relation to the person
    * superior: above the writer
    * subordinate: below the writer
    * equal: an equal in the regime
    * rival: a competitor for a task, a jurisdiction or favor
    * ally: an ally or protege
    * family: family or private acquaintance
    * foreign: a foreign statesman or official
    * other: another relation or unclear
  - **attitude** (string): the writer's judgment of the person in this entry, in his words; empty if none
- **themes** (list of strings): the subjects of the entry as an edition's index would list them, two to six words each
- **places** (list of strings): the places named in the entry, as written

### NOTES
- Dates in ISO 8601 at the precision of the text; never pad a day or month.
- Names and places as the text spells them; read through OCR errors.
- Return an empty array when the entry holds no such act.
- Every field is a string.

Output: a single JSON object {"rows": [ ... ]}, where each element of 'rows' is one record with the fields above; use an
     empty 'rows' array if none. Return ONLY valid JSON.

Input text:
SOURCE: <edition>, <volume>, <page or entry>, <date>

TEXT:
[<volume>, <page or entry>, <date>]
<text of the page or entry>
```

Table A4: Actions and Peers: Examples from the Diaries

	Definition or rule	Example
Panel A: Types of action		
Direction	sets a goal or a priority, also in a speech, assigns jurisdiction or fixes a procedure, without naming an act	Himmler, 4 Sept. 1941: Himmler discusses with Koppe the planned deportation of the Jews from the Reich and, as a fallback, their reception in the Litzmannstadt ghetto.
Order	specifies an act for a subordinate	Goebbels, 19 Aug. 1941: Hitler instructs Goebbels to prepare a mass meeting in the Sportpalast to reopen the Winter Relief.
Prohibition	forbids or restricts an act	Frank, 20 Oct. 1939: Hitler forbids the army, the interior minister, and all central offices in Berlin to interfere in the affairs of the General Government.
Approval	approves, praises, ratifies or rewards conduct: a post, a promotion, a decoration, a jurisdiction, resources, personnel	Himmler, 24 Sept. 1943: Himmler approves Rauter's proposal to use police summary courts instead of ordinary courts against opponents.
Decision	decides or acts on its own authority	Himmler, 16 Feb. 1943: Himmler rejects Kaltenbrunner's proposal to deport 5,000 Jews over 60 who are unable to work from Theresienstadt to Auschwitz.
Proposal	puts a scheme, plan or policy forward for a superior to adopt, unasked	Frank, 27 June 1940: Daluege proposes a better-paid Polish special police and a works police at large industrial plants.
Request	asks a superior for a decision, permission or ruling	Goebbels, 5 Mar. 1937: Ribbentrop asks to be allowed to speak on the radio more often.
Report	reports conduct or results to a superior	Himmler, 23 Dec. 1942: Berger discusses newly raised units with Himmler by telephone.
Dispute	contests, obstructs or negotiates another person's or office's action, claim or jurisdiction	Frank, 21 Nov. 1942: Krüger objects that Bühler reported the incidents to the governor-general instead of turning to him directly.
Defiance	ignores, delays, evades or sabotages an order received	Himmler, 12 Aug. 1943: Steiner refuses the transfer of an officer cadet to Himmler's command staff that Himmler had ordered.
Trust	expresses trust, loyalty, confidence or distrust toward a superior, a subordinate or another office	Frank, 25 Mar. 1944: Frank states that Struve still has his full confidence.
Information	learns of another person's or office's action or plan	Himmler, 13 Jan. 1943: Wolff discusses with Himmler the raising of two new SS divisions and passes on Hitler's instruction to proceed with the utmost haste.
Assessment	judges an act, a person or an office	Frank, 20 Oct. 1941: Frank welcomes more police patrols, since German police on the streets give cities such as Kraków a German character.
Panel B: The writer's relation to a person named		
Equal	an equal in the regime	Göring, Heß (Goebbels); Bormann, Speer (Himmler); Himmler, Lammers (Frank); Göring, Lammers (Rosenberg); Prützmann, Jeckeln (Bach-Zelewski)
Rival	a competitor for a task, a jurisdiction or favor	Rosenberg, Ribbentrop (Goebbels); Rosenberg, Frank (Himmler); Himmler (Frank); Himmler, Ribbentrop (Rosenberg); Prützmann, Globocnik (Bach-Zelewski)
Ally	an ally or protege	Kaufmann (Goebbels); Greiser (Himmler); Siebert (Frank); Quisling, Darré (Rosenberg); Fegelein (Bach-Zelewski)
Subordinate	below the writer	Hanke, Funk (Goebbels); Wolff (Himmler); Bühler (Frank); Lohse, Koch (Rosenberg); Dirlewanger (Bach-Zelewski)
Superior	above the writer	Hitler (Goebbels, Himmler, Frank, Rosenberg); Himmler (Bach-Zelewski)
Other	a foreign statesman or official, family, or another relation	Churchill, Roosevelt (Goebbels); Mussolini (Rosenberg); Stalin (Bach-Zelewski)
Panel C: Entries mentioning a dispute		
A dispute with a peer	counts	Bach-Zelewski, 15 July 1942: Bach-Zelewski rejects guard duties for the police and demands combat tasks in a stormy meeting with General von Schenckendorff.

	Definition or rule	Example
A dispute with an office outside the writer's line	counts	Goebbels, 10 Sept. 1941: Goebbels opposes the Luftwaffe's plan to shut down radio transmitters in the occupied territories earlier.
A dispute among the writer's subordinates	counts	Frank, 12 May 1944: Bühler objects to deploying the Special Service only in the Biłgoraj forest or only against partisans.
A dispute brought to the superior	counts	Goebbels, 22 Aug. 1942: Goebbels writes to Hitler to protest against the personnel policy of Dietrich, the Reich press chief, in the press departments.
A rival's dispute with the writer	counts	Goebbels, 6 Mar. 1935: Rosenberg again stirs up a quarrel over the Furtwängler case, and Goebbels resists him.
An action without a dispute	does not count	Himmler, 22 Aug. 1943: Lammers telephones Himmler about the takeover of the Reich Ministry of the Interior.

This table illustrates the diary coding. Panel A lists the thirteen action types with the definitions given to the language model and one coded example in our translation. Panel B lists the writer's relations to persons named in entries and the persons named most often. Panel C shows which actions make a day count as having an entry mentioning a dispute in Figure 12, Panel A.

A.4 State Directories

We use the 1931 and 1936 *Handbuch für das Deutsche Reich*, the 1939 *Behördenführer*, and the 1942 *Taschenbuch für Verwaltungsbeamte* (Reichsministerium des Innern, 1931; Reichs- und Preußisches Ministerium des Innern, 1936; Münz, 1939; Warnack, 1942). Figure A1 shows every institution listed with at least one person, by directory section and hierarchy. Figure A8 lists supreme Reich bodies by policy domain.

A.5 Policy Domains

We use two classifications of policy domains. The first assigns subject headings in the situation reports, Party Chancellery files, and Goebbels diaries to sixteen domains. A heading enters a domain if it contains any defining term and may enter several domains. We exclude person headings and classify country headings as foreign policy before occupation and occupation thereafter. Panel A of Table A5 lists the domains, their first terms, and record counts by source. The second classification manually assigns each supreme Reich body in the state directories to one of sixteen domains. Panel B lists these domains, examples, and body counts by edition.

A.6 Troop Units

Sources. Table A6 lists the panel's sources, their contents, and the unit-months each locates. The panel builds on Tessin (1966), whose register covers the army, air force, navy, and Waffen-SS. We add the Order Police register (Tessin and Kannapin, 2000), the police battalion encyclopedia (Arico, 2011), court-based reconstructions of police operations (Curilla, 2006; 2020), the Einsatzgruppen reports (Matthäus et al., 2014; Mallmann et al., 2011; Mallmann et al., 2014), the Armed Forces High Command war diary (Schramm, 1961), the *Lexikon der Wehrmacht* (Altenburger, 2026), the National Archives guides to Waffen-SS records (National Archives and Records Service, 1978; 1981), and Feldpost numbers (Kannapin, 1980).

Procedure. We read each scanned page or text with a multimodal language model (Gemini 3.7 Flash). The model returns each unit's dated stations, subordinations, and commanders; each place's level (town, district, region, country, or front) and coordinates; and each date's precision

Figure A1: Institutions by Section of the State Directories



This figure shows one mark per institution listed with at least one person in the four editions of the state directories, by directory section. Rows follow the parts of the 1936 edition. Mark area is proportional to the institution's share of persons listed in the edition. A branch network is one mark with its number of offices. Bars below the headers count bodies in each edition by hierarchy, and gray numbers count bodies in each row. Corporatist bodies are counted as in Figure 3. The 1939 edition lists heads only, and the 1942 edition omits the central party offices.

Table A5: Policy Domains

A: Domains of the subject headings

Domain	Heading terms (first six)	Reports	Files	Diary
Jews and race	*Juden, Judentum, jüdisch, Antisemit, Arisierung, Mischling*	258	1,211	975
Churches	*Kirche, kirchlich, Kirchenpolitik, Katholi, Evangel, Protestant*	798	1,525	859
Repression and political policing	*Gestapo, Geheime Staatspolizei, Schutzhaft, Konzentrationslager, KZ, Festnahme*	1,215	1,609	2,123
Party and party-state relations	*Verhältnis zu, Zuständigkeit, Kompetenz, Partei, Staat, Parteikanzlei*	559	10,044	5,136
State administration and personnel	*Beamte, Beamten, Öffentlicher Dienst, Ernennung, Personal, Besoldung*	279	4,493	2,893
Military and war	*Wehrmacht, Heer, Luftwaffe, Marine, OKW, Oberkommando*	532	3,744	2,611
Foreign policy	*Außenpolit, Auswärtiges Amt, Auslands, Ausland, Diplomat, Botschaft*	1,389	4,115	3,714
Occupation and annexation	*Besetzt, Eingegliedert, Generalgouvernement, Protektorat, Ostgebiete, Ostmark*	1,684	3,299	1,826
Economy and finance	*Wirtschaft, Firmen, Industrie, Handel, Preis, Devisen*	770	4,394	2,693
Supply, mood, and home front	*Stimmung, Öffentliche Meinung, Gerüchte, Versorgung, Lebensmittel, Rationierung*	573	2,223	2,498
Local matters and petitions	*Eingaben, Gesuche, Sammlungen, Spenden, Stiftung, Veranstaltung*	174	2,131	715
Labor	*Arbeit, Arbeits, Fremdarbeiter, Zwangsarbeit, DAF, Arbeitsfront*	318	2,175	1,116
Propaganda, press, and culture	*Propaganda, Presse, Rundfunk, Film, Kultur, Theater*	1,369	3,812	5,705
Education and youth	*Schule, Erziehung, Hochschule, Universität, Wissenschaft, Student*	268	2,608	877
Welfare and health	*Fürsorge, NSV, Wohlfahrt, Gesundheit, Erbgesundheit, Euthanasie*	201	1,105	2,320
Law, courts, and constitution	*Justiz, Recht, Rechts, Gesetz, Verordnung, Gericht*	345	3,183	2,146
Records with a heading		2,270	25,385	6,183
Records in a domain		2,054	22,354	6,108

B: Domains of the supreme Reich bodies

Domain	Examples of bodies	1931	1936	1939	1942
Chancelleries, cabinet, party leadership	Reich President; Reich Chancellery	2	7	6	5
Foreign affairs	Foreign Office; Foreign Policy Office	1	4	3	1
Interior, administration	Interior Ministry; Municipal Office	1	3	1	2
Police, security	RFSS and Chief of Police; SS	0	2	2	1
Justice	Justice Ministry; Legal Office	1	3	3	1
Finance	Finance Ministry; Audit Office	5	4	4	3
Economy, prices, armaments	Economics Ministry; Chambers of Commerce (DIHT)	3	4	5	6
Labor, welfare	Labor Ministry; Labor Exchange	3	4	4	4
Food, agriculture	Food Ministry; Agricultural Council	3	4	4	4
Transport, post	Post Ministry; Transport Ministry	3	4	4	4
Building, planning, energy	Spatial Planning Office; Technology Office	0	2	3	4
Propaganda, press, culture	Government Press Office; Civic Education Office	2	6	5	3
Science, education, youth, church	Education Ministry; Ideology Commissioner	0	5	5	3
Armed forces	Defense Ministry; War Ministry	1	3	4	3
Occupied territories	Ministry for the East; RK Ostland	0	0	0	10
Länder, Reich governors	Reich Governors; RK Saarland	0	2	4	1

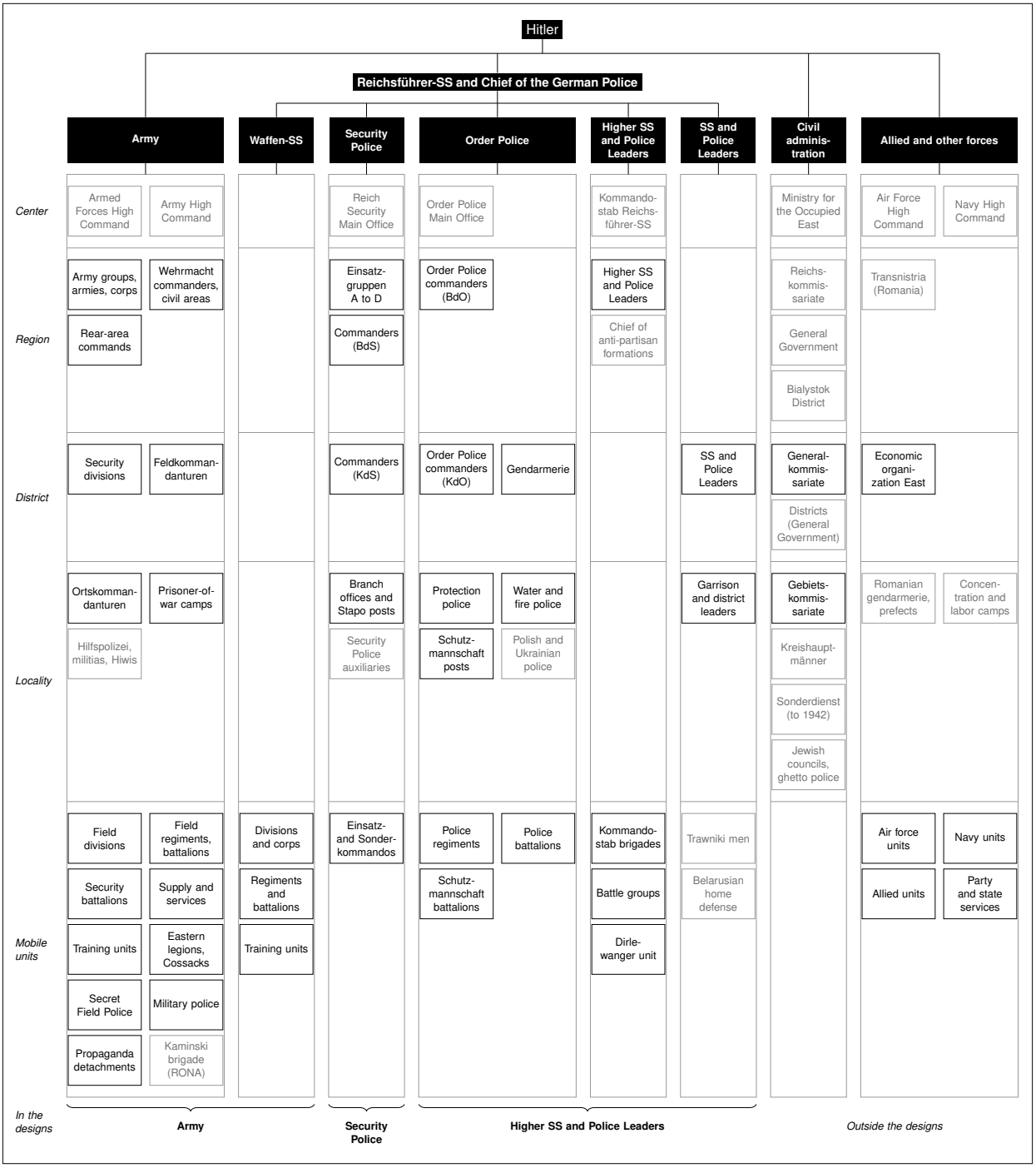
This table defines the policy domains. Panel A lists the domains of subject headings in the situation reports, Party Chancellery files, and Goebbels diaries, with defining terms and the number of records each source assigns to each domain over all years. A record can fall in several domains. Panel B lists the domains of the supreme Reich bodies in the state directories, with examples and the number of bodies in each edition.

Table A6: Troop-Unit Sources

Source	Records	Entries	Unit-months placed
Tessin, *Verbände und Truppen*, volumes 2 to 14	Founding, subordination, stations, and renamings of army, air force, navy, and Waffen-SS units	27,702	206,233
Tessin, *Verbände und Truppen*, volume 16	Stations of army, air force, navy, Waffen-SS, police, and Labor Service units by place, with dates of founding and relocation	36,986	93,172
Lexikon der Wehrmacht	Unit histories that follow Tessin, with dated stations	14,103	170,822
War diary of the Armed Forces High Command	Command posts and sectors of commands by day, 1940 to 1943	6,217	11,685
Tessin and Kannapin, Order Police	Stations, home stations, and Feldpost numbers of police units	1,464	6,224
Neufeldt, Huck, and Tessin, Order Police	Seats of police staffs and dated stations of police regiments and battalions	2,079	6,750
Arico, police battalions	Battalion histories with stations to the company, to July 1942	1,914	2,712
Curilla, Order Police	Police, gendarmerie, and Security Police operations in the Baltic states and Belarus, 1941 to 1944, and in the occupied west and north, 1940 to 1945	2,525	3,992
Einsatzgruppen reports	Stations of the Einsatzgruppen, Kommandos, and Higher SS and Police Leaders by report day	2,637	969
Wikipedia and the literature, SS and police offices	Seats of the regional Security Police and SS and police offices in the occupied East, with the tenures of their holders	126	1,832
Braham, Hungary	German commands in the towns of Hungary, 1944	65	22
National Archives, Waffen-SS guides	Histories of Waffen-SS units with dated locations	713	448
Feldpost register	Field-post numbers of every formation with a military postal address, a benchmark for the panel's coverage; places no unit	237,386	–
All sources			504,861

This table lists the sources for the troop-unit panel. Entries is the number of records from each source: register entries, unit pages, located command posts, dated statements or events, station records, tenures, or Feldpost designations. Unit-months placed counts unit-months from September 1939 to May 1945 whose retained place comes from the source, including places carried forward. The panel contains 639,767 unit-months.

Figure A2: Troop Structures and Commands in the East

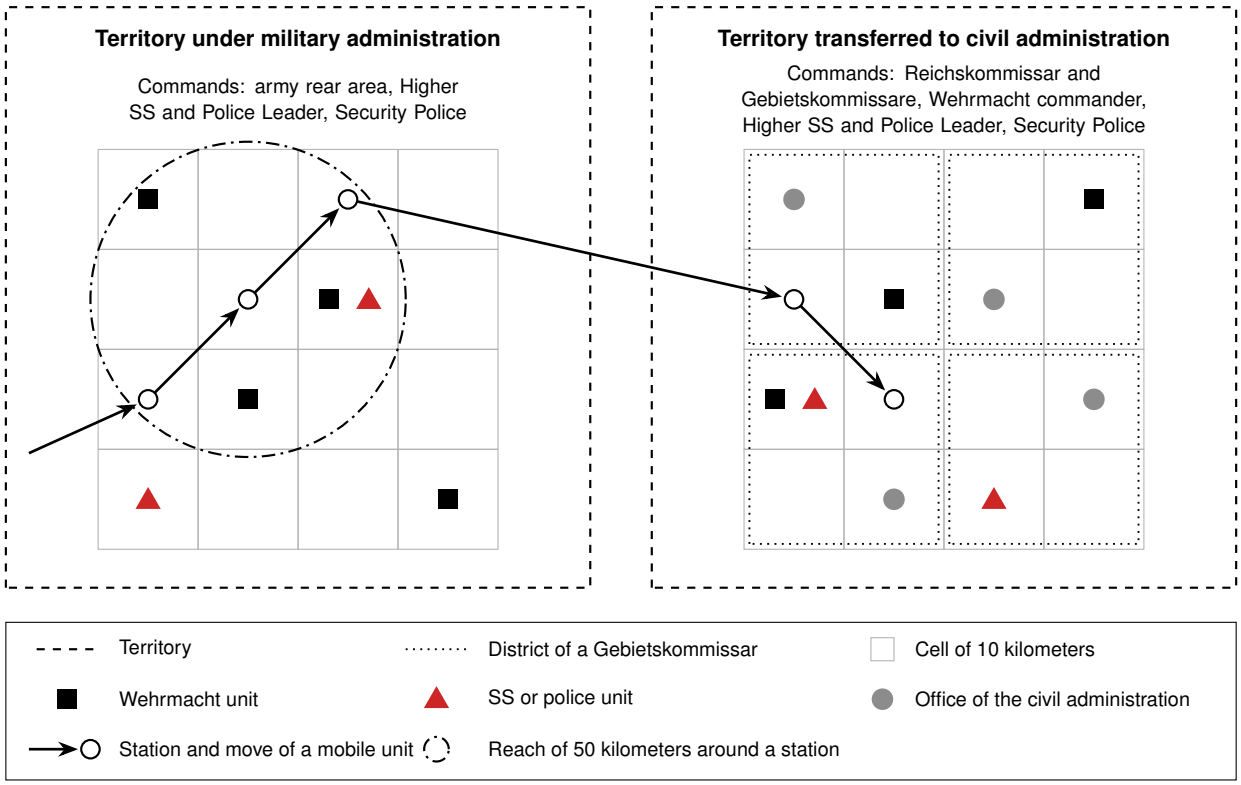


This figure shows the troop structures active from June 1941 to June 1944 in the occupied Soviet territory and the General Government, by command and level. Columns represent the six commands in the data (Appendix A.6), the civil administration, and the allied and other forces. Rows represent levels from the center to the locality, with mobile units below. Each box is one structure; grey boxes mark forces the sources document but the unit registers do not list, so the troop-unit panel holds none of their units. Braces mark the three commands of Section 2.3: the army with the Waffen-SS, the Security Police, and the Higher SS and Police Leaders with the Order Police and the SS and Police Leaders.

in the source. We link units in supplementary sources to Tessin by name, number, and type. The Einsatzgruppen reports appear a median of 18 days after the action they report, so we move a statement dated only by the day of its report back by 18 days. For each unit-month, we retain one place, prioritizing geographic specificity, then date precision, then the source that reports stations most directly. We carry a place forward until a source moves the unit. Unit-months without a location more precise than a country or front take their parent command's area from the war diary; each reading applies for at most the three following months.

Validation. Across 100 manually checked pages, links from *Lexikon* pages to register entries have precision and recall of 0.97. Feldpost numbers agree with the page's unit in 88 percent of 5,883 checks. Of the 34,667 Feldpost numbers naming an army, air force, navy, Waffen-SS, or police battalion, regiment, or division, 92 percent belong to a panel unit; the respective branch shares are 91, 95, 97, 67, and 99 percent. Across all 67,454 numbers, the share is 67 percent. Locations cover 79 percent of unit-months. Maps and cell-level measures exclude region, country, and front locations. The panel records realized presence, not planned movements. Figure A3 shows the structure of the panel in stylized form, and Table A7 lists the units with the highest killing rates among those without a killing report.

Figure A3: Moves of a Mobile Unit



This figure shows the data structure for occupied Soviet territory in stylized form. Ten-kilometer cells belong to territories. A territory remains under military administration or passes on a decreed date to civil administration, which divides it into districts of Gebietskommissare. The troop-unit panel records the units present in each cell and month. The figure also shows a mobile unit's stations, moves, and 50-kilometer reach.

Table A7: Units without a Killing Report

Unit	Command	Commander	Period	Rate	Deaths
Gebietskommissariat Lida	Civil administration	Hermann Hanweg	1942-03–1942-05	1.96	11,000
2nd Reserve Police Company Munich	Order Police	*Herbert Böttcher*	1942-08–1942-12	0.89	16,000
1st Panzer Division	Army	Rudolf Schmidt	1939-09–1944-06	0.82	3,100
170th Infantry Division	Army	Walter Wittke; Erwin Sander	1941-08–1944-06	0.78	1,200
Infantry Regiment 481	Army	Friedrich Weber	1941-06–1944-06	0.77	1,800
BdO Ukraine	Order Police	Otto von Oelhafen	1941-09–1944-02	0.64	25,000
132nd Infantry Division	Army	Rudolf Sintzenich	1941-07–1944-06	0.59	36,000
SS Flak Battalion 1	Waffen-SS	Bernhard Krause	1941-07–1943-12	0.50	4,600
XXX Army Corps	Army	Hans von Salmuth	1941-04–1944-03	0.48	6,400
13th Panzer Division	Army	Walther Düvert	1941-06–1944-03	0.46	7,400
LVI Panzer Corps	Army	Ferdinand Schaal	1942-01–1944-06	0.46	710
XXXIX Panzer Corps	Army	Rudolf Schmidt	1941-07–1944-06	0.44	1,900
14th Infantry Division	Army	Friedrich Fürst	1939-09–1944-06	0.44	3,100
Infantry Regiment 513	Army	Görner	1941-07–1941-10	0.42	37,000
83rd Infantry Division units	Army	Alexander von Zülow	1942-01–1944-06	0.40	710
198th Infantry Division	Army	Albert Buck; Otto Röttig	1941-09–1944-02	0.38	5,900
60th Infantry Division	Army	Friedrich-Georg Eberhardt	1941-07–1942-06	0.38	10,000
19th Infantry Division	Army	Günther Schwantes	1939-09–1939-09	0.38	4,900
XXXXI Army Corps	Army	Georg-Hans Reinhardt; Josef Harpe	1941-06–1942-02	0.37	1,200
SS Police Regiment 24	Order Police	Gassler	1942-07–1944-06	0.37	8,600

This table lists the 20 units with the highest killing rates among the units to which no source attributes a victim count, out of the 942 units present in at least three of the 843 cells of the German-administered East from 1939 to 1944. Rate is the increase in monthly deaths per 100 cell residents associated with the unit's presence. Rates are estimated jointly for all units with cell, zone-by-month, and cohort-by-month fixed effects and shrunk toward the average unit. Period gives the unit's first and last months of presence. Deaths counts the murdered residents of its cells in those months. Commander names the commander who held the post in the months with most of the unit's deaths. An italic name identifies the commander under whom a unit served. KdS and BdO are the regional commanders of the Security Police and the Order Police.

Documented transfers. The war diary of the Armed Forces High Command records the transfers of army formations; the Einsatzgruppen documents and reports and Himmler's appointment books record those of the Security Police and of the units of the Higher SS and Police Leaders. Between June 1941 and May 1945, these sources record 892 executed moves to a town or district in the East, 203 of army formations and 689 of SS and police units; 851 end within 25 kilometers of a cell of the eastern grid. A transfer's counterfactual destinations are drawn from the 289 places within the eastern grid that the sources name at least twice, one per cell.

Chains of command. Figure A2 shows each troop structure the panel places in the East and its command. Reporting lines follow the historical literature and the Einsatzgruppen documents (Pohl, 2008; Hilberg, 2003; Curilla, 2006; Cüppers, 2005; Longerich, 2010; 2012; Angrick et al., 2013). The figure distinguishes six commands: the army, the Waffen-SS, the Security Police, the Order Police, the Higher SS and Police Leaders with their own troops, and the SS and Police Leaders. The designs group these into three commands: Waffen-SS field formations fought under army command, and the Order Police and the SS and Police Leaders reported to the Higher SS and Police Leaders. A structure with one sourced reporting line retains the same command every month: the Kommandanturen report to the army, the Einsatzkommandos to the Security Police, and the Order Police commands and offices to the Higher SS and Police Leaders. A structure with a sourced second line follows the superior named in the registers for that month: the Order Police battalions under the army's security divisions, the Schutzmannschaft battalions attached to army formations, and the SS brigades of Himmler's command staff at the front. The air force, the navy, the allied armies, the civil administration, and the economic organization are outside the three commands.

B History

Many dictatorships combine vertical integration with decentralized initiative: the ruler holds formal authority at the top, but officials act independently in his name. Party, state, and political police organizations often share jurisdiction (Arendt, 1951, ch. 3). Overlapping authority is also common in other political systems (Herbert, 2022, p. 12). In the model of Section 4, a ruler who cannot specify a policy in advance chooses between rivalry, which implements early and risks implementation he would reject, and control, which waits for information and risks implementing too late. This section examines these conditions in Nazi Germany and other historical and contemporary settings.

B.1 Nazi Germany

Hitler held sovereign authority, while ministries, party offices, special commissioners, and the SS and police shared administrative power (Broszat, 1969). Section 3 documents the proliferation of institutions and personal mandates after 1933. By 1936, almost every policy domain had multiple supreme Reich bodies. Hitler withdrew from cabinet-level meetings as official documents increasingly invoked his authority. Central agencies drew on local initiatives and selected among them (Allen, 2005). A 1934 speech promised subsequent legal confirmation for officials who "worked towards the Führer" (Kershaw, 1997, p. 104).

Municipalities often introduced discriminatory measures against Jews before central rules required them. Local officials appeared eager to outdo one another in a "dynamic of outbidding" (Herbert, 2016, p. 38, our translation). On 18 March 1933, Cologne's mayor excluded Jewish firms from municipal contracts (Gruner, 2005, p. 271). Martin Bormann's decree of 12 September 1933 ordered the repeal of local measures not sanctioned by law but had "very little effect" (Gruner, 2005, p. 273). On 19 July 1935, the *Völkischer Beobachter* reported that Breslau's mayor had barred Jews from several public baths. Such reports "soon had a massive copycat effect" (Gruner, 2005, p. 273); Stettin followed the next day (Gruner, 2000, p. 94). The Deutscher Gemeindetag, the municipal association, coordinated these policies, in part "systematically" (Gruner, 2005, p. 271; Gruner, 2000, p. 77, our translation). Its survey found that, by early 1938, 20 of the 59 large cities with municipal pawnshops had excluded Jews from pawnbroking (Gruner, 2005, p. 275). Many towns were "a step ahead not only of the Reich authorities but even of the party leadership" (Gruner, 2000, p. 94, our translation). Until 1938, towns "very seldom" clashed with other agencies over jurisdiction (Gruner, 2005, p. 270). In summer 1938, the central government began preparing legislation that towns had long demanded (Gruner, 2000, p. 121).

Concentration camps emerged amid competition among party, state, and paramilitary institutions. The SA (the party's storm troopers), the SS (the party's paramilitary formation), and the Prussian state each claimed authority over political detention in 1933. The Reichstag Fire Decree of 28 February 1933 provided the pretext for the suspension of basic rights and suppression of political opposition. By the end of March, about 20,000 communists had been arrested and sent to prisons or the SA's "wild" camps (Herbert, 2016, p. 33). In March and April 1933, police took 3,818 people into protective custody in the Düsseldorf district, which had an SS police president, compared with only 78 in the neighboring district of Trier (Broszat, 1968, p. 407). Dachau was formally a Bavarian political police camp. On 2 April 1933, Himmler, who headed both the Bavarian political police and the SS, transferred control to the SS (Longerich, 2012, p. 151). Prussian Interior Minister Göring sought to reassert state control. His circular of 14 October 1933 declared detention outside state camps and police prisons "not permitted in future" (Broszat, 1968, pp. 410–411). Referring to the SA camp at Kemna, in Wuppertal, his ministry wrote: "I have not approved the

establishment of this camp" (Broszat, 1968, p. 412). Control nevertheless passed to the SS. On 20 June 1934, Dachau commandant Theodor Eicke became Inspector of Concentration Camps. He took over and closed the SA camp at Oranienburg. In June 1935, Hitler approved Reich funding for the camps and their guards (Longerich, 2012, p. 183). The concentration camps, initially established during the regime's consolidation to suppress political opposition, later became central institutions of the Holocaust. Figure 5 summarizes this sequence and identifies the corresponding elements of the model in Section 4.

In annexed Poland, the Gauleiters combined party leadership with civil authority. Hitler instructed them to make their territories German and promised not to ask about their methods (Kershaw, 1985, p. 136). Forster and Greiser competed to report the first fully German province. Their rivalry became a "competition in brutality" that Hitler reportedly had not intended (Kershaw, 1997, p. 103). When Himmler challenged Forster's authority over Germanization, Hitler avoided resolving the dispute and reassured both sides (Hüttenberger, 1969, p. 181). Greiser later invoked Hitler's permission to exercise his own judgment concerning the Jews to justify proceeding without fresh approval for a plan to kill Poles with incurable tuberculosis (Kershaw, 1997, p. 103). In occupied Poland, Himmler and Frank were "enemies and rivals" who "competed only in ruthlessness" (Hilberg, 2003, p. 205).[49]

In the occupied Soviet territory, security authority overlapped among Heydrich's Security Police and Einsatzgruppen, Himmler's Higher SS and Police Leaders and their units, and army rear-area commands. The center set this overlap before the invasion: in March 1941, the Armed Forces High Command gave Himmler special responsibilities in the army's area, to be carried out independently of the army (Longerich, 2010, p. 182; Longerich, 2012, p. 517). In July 1941, Hitler assigned civil administration to Rosenberg and police security to Himmler (Longerich, 2012, pp. 527–531). Subsequent territorial transfers placed large areas under civil administration while leaving others under military rule (Pohl, 2008; Berkhoff, 2004). Civil commissioners shared authority with the Security Police, the Higher SS and Police Leaders, and the army (Hilberg, 2003, p. 357; Longerich, 2010, p. 215).

The killing expanded in stages; no single order fixed its scope (Longerich, 2012, p. 532). In postwar trials, SS leaders referred to a Führer order, but "no direct witness of such an order survived the war" (Kershaw, 1985, p. 135). On 1 August 1941, Himmler ordered the SS cavalry to shoot all Jews (Longerich, 2012, p. 531). From mid-August, Einsatzgruppen and Waffen-SS units systematically murdered Jewish women and children (Herbert, 2000, p. 31). Berlin responded to regional reports by ordering or retrospectively approving increasingly radical measures (Herbert, 2000, p. 38). Longerich (2010, p. 304) places these regional initiatives within a centrally directed policy.

Contemporary observers identified competition among Himmler's commanders. In August 1941, British analysts of decrypted police radio traffic concluded that the three Higher SS and Police Leaders were competing for the best results (Cüppers, 2005, p. 186). Jeckeln increased the killings after a meeting at which Himmler appeared dissatisfied with his activity (Longerich, 2012, p. 533). Before the transfer of the Kamenets-Podolsk area to civil administration, Jeckeln promised to complete the "liquidation" of Jews expelled there from Hungary (Pohl, 2009, p. 163, our translation). His men killed 23,600 people, mostly Jews, in three days—the largest massacre to that date (Longerich, 2012, p. 533). Himmler later assigned him to Riga, where his men killed more than 26,000 Jews at Rumbula (Curilla, 2006, pp. 251, 872).

Civil authorities also claimed control over the killing. Lohse banned shootings at Liepāja until

[49] Frank complained in his diaries that the Party Chancellery, police, army, and other "power blocs" imposed conflicting demands on his administration "by the famous elbow system," an "anarchy of authorizations."

Berlin replied that economic concerns should not constrain them (Hilberg, 2003, p. 392; Longerich, 2010, p. 298). In Minsk, Kube, the civil commissioner for White Ruthenia, disputed the Security Police's methods while accepting the killings themselves (Friedländer, 2007, p. 362). In July 1942, he reported the killing of about 55,000 Jews in ten weeks (Longerich, 2010, p. 348).

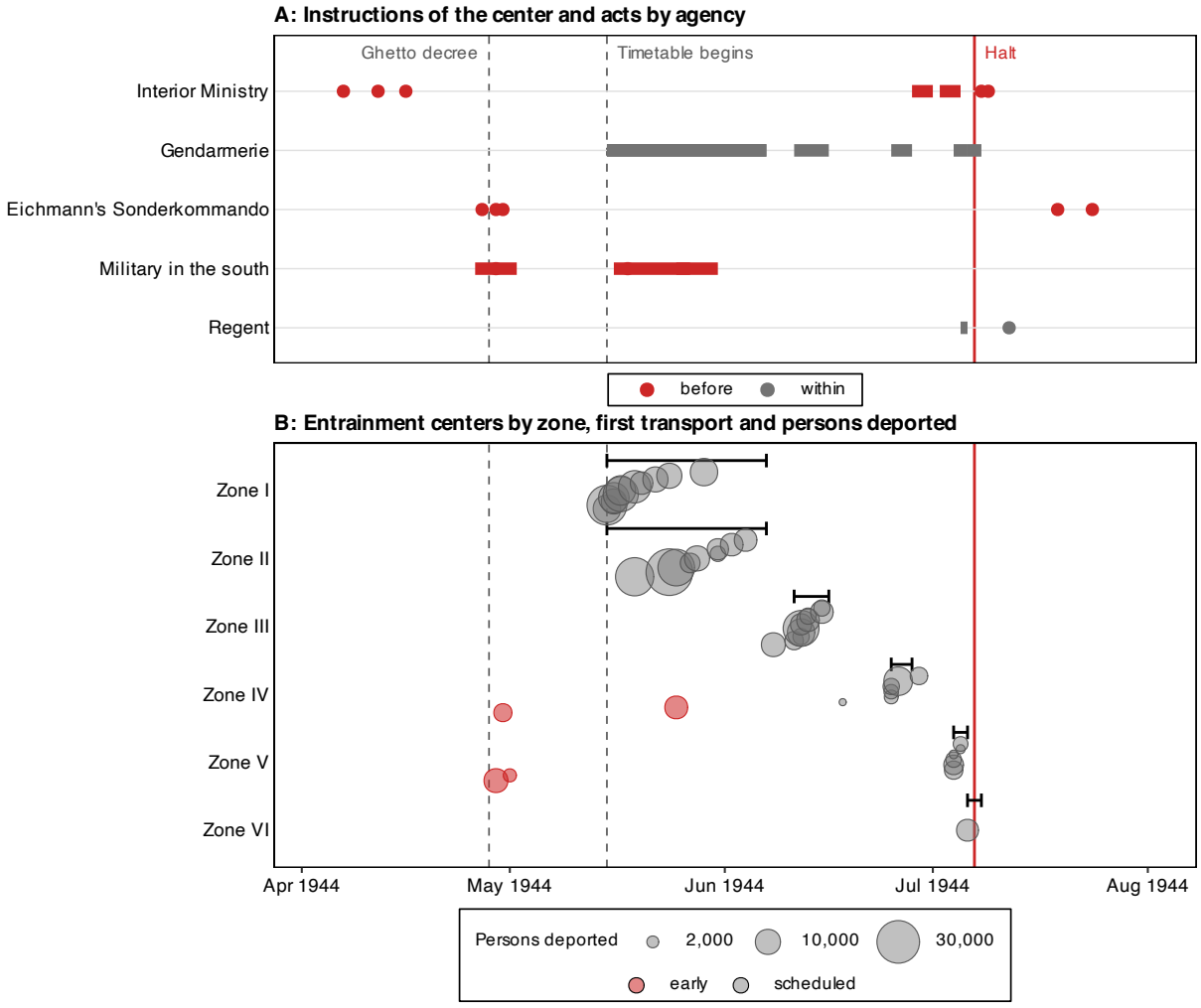
Competition among agencies also shaped the Holocaust in Hungary, an ally of Germany whose own police and military forces implemented the deportations (Braham, 1981, vol. 1, p. 605; vol. 2, pp. 608, 612, 923). Eichmann, who arrived with the occupation of 19 March 1944, competed with the Higher SS and Police Leaders, the Foreign Office, the army commanders in the East, and the Gauleiters, "each of whom wanted to be the first to declare his territory *judenrein*" (Arendt, 1963, pp. 151–152). In Hungary, three SS officers held orders "to help in 'the solution of the Jewish problem,' " and each "stood in a different chain of command": Veesenmayer for the Foreign Office, Winkelmann as Higher SS and Police Leader, and Eichmann for the RSHA (Arendt, 1963, pp. 139–141). Winkelmann "soon emerged as a bitter rival of Veesenmayer," and Eichmann "refused to take orders from either of the others" (Braham, 1981, vol. 1, p. 412; Arendt, 1963, p. 141). The Hungarian right-radical leaders "had outdone even the Germans in the eagerness with which they pushed the roundup, ghettoization, and deportation of Jews," and Eichmann's Hungarian partners "wanted their capital to take the lead in becoming *judenrein*" (Braham, 1981, vol. 1, p. 489; Arendt, 1963, p. 140). Several agencies acted before the center issued its instructions, and some acted after the Regent's halt.[50] The roundups in Subcarpathian Ruthenia and the south began "several days before the official ghettoization decree had been issued," the first trains left for Auschwitz from Kistarcsa and from the SS camp at Topolya on 29 and 30 April, two weeks before the timetable began, and the Hungarian and German military authorities deported about 21,700 Jews from the districts along the southern border "in an independent action" (Braham, 1981, vol. 1, pp. 604, 664–665; vol. 2, pp. 608, 641). After Horthy halted the deportations on 7 July, Eichmann deported the 1,220 internees of Kistarcsa on 19 July "in order to uphold his authority and reputation, and to put something over on the Regent" (Braham, 1981, vol. 2, pp. 762, 773). Appendix Figure A4 places the instructions of the center and the acts of each agency on one date axis. Of the residents of the military's zone whose deaths the records date to April through August 1944, 71 percent died in April or May, against 8 percent in the other cells of the same counties (Appendix Figure A5).

B.2 Other Settings

The Great Terror of 1937–38 combined centrally ordered violence with competition among regional officials. Repeated purges made administrative failure politically costly (Montagnes and Wolton, 2019, p. 1047). Under NKVD Order No. 00447, regional party secretaries proposed execution and imprisonment limits and requested increases. NKVD officers made arrests and dominated the extrajudicial troikas, which also included the regional party secretary and procurator (Gregory, 2009, pp. 184, 186, 219). Stalin placed security officials in competing positions, and regional leaders faced pressure to match others' requests for higher limits. Failure to seek increases could lead to removal or arrest (Gregory, 2009, pp. 49, 220, 234). Combined execution and imprisonment limits nearly doubled in four fifths of regions (Gregory, 2009, pp. 184, 233). The NKVD also authorized large increases without Politburo approval (Gregory, 2009, p. 189). Seeking increases offered no guarantee of protection: Ukraine's NKVD chief made repeated requests but was removed for ex-

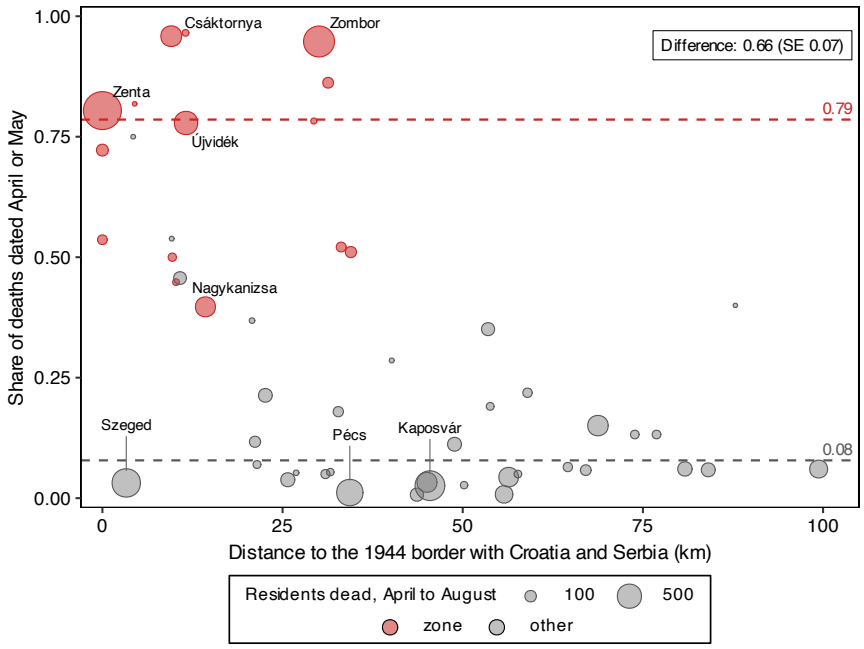
[50] In July 1941, two officials of the alien control office, critics of their minister's "benevolent attitude toward the Jews," devised the expulsion of "alien" Jews, and the deputy prefect of Máramaros County issued his own order four days before the government's decree, an order that "served as model for officials of other counties" (Braham, 1981, vol. 1, pp. 207–210). They were the Jews whose "liquidation" Jeckeln promised at Kamenets-Podolsk; about 2,000 of the 16,000 to 18,000 deportees survived (Braham, 1981, vol. 1, pp. 212–213).

Figure A4: Instructions and Acts in the Deportations from Hungary, 1944



This figure shows the 1944 deportations from Hungary recorded by Braham (1981). Panel A plots three central instructions in red on a common date axis: the ghettoization decree of 28 April, the start of the deportation timetable on 15 May, and the Regent's halt of 7 July. Agency acts appear below, with points marking acts on one day and bars marking acts over a period. Red indicates that the agency acted before the center's instruction or after the halt. Gray indicates that it acted under an instruction or enforced one. Panel B plots each entrainment center in its operational zone on the date of its first transport. Centers are red if that transport left more than a week before the zone's scheduled window, shown by the bracket above each row. Point area is proportional to the number of persons deported from the center.

Figure A5: Deaths Dated April or May by Cell of Residence, Southern Hungary, 1944



Each point represents a 25-kilometer cell of residence in the counties along Hungary's southern border of 1944. The figure plots the share of murdered residents whose deaths the Yad Vashem record dates to April or May 1944 among those dated April through August against distance to the border with Croatia and occupied Serbia. Point area is proportional to the number of residents counted. Cells with at least ten residents are shown, and the largest are labeled with the name of their ghetto or camp site. Red cells lie in the military operational zone described by Braham (1981, vol. 1, pp. 604–605; vol. 2, pp. 608, 617, 666). Dashed lines show the two shares pooled over cells. The box reports the difference between the zone and other cells, estimated from a person-level linear probability model with county fixed effects and standard errors clustered by cell. The sample includes 8,003 residents in 57 cells, including 2,862 in the 21 zone cells.

posing too few enemies and later executed (Gregory, 2009, pp. 233–234). By November 1938, the regime had convicted about 1.4 million people and executed nearly half (Gregory, 2009, p. 187). Montagnes and Wolton (2019, pp. 1045, 1057) model purges as a form of accountability: violence raises effort, but punishing every failure makes success less informative about loyalty. Stalin ended the mass operations that autumn. The Central Committee gained authority over NKVD appointments, and the Politburo prohibited further mass operations and abolished the troikas (Gorlizki and Mommsen, 2009, p. 70).

China has combined broad local economic authority with hierarchical control over appointments. Each territorial level controls major economic functions and appoints and removes leaders at the level below (Xu, 2011, pp. 1086–1087, 1093). Before the Great Leap Forward, economic agencies made policy under Mao's oversight, while coordinating bodies restrained ministries and local governments (Teiwes and Sun, 1999, pp. 177, 179). During the Leap, Mao took direct control and brought party bodies and lower-level organs into policymaking. Editorials in the *People's Daily* carried more weight than Central Committee directives (Teiwes and Sun, 1999, pp. 178–179). Most state enterprises passed to subnational governments, which competed to exceed targets and overstated output (Xu, 2011, pp. 1084–1085). Officials competed to demonstrate loyalty, and leading provincial radicals gained promotion (Teiwes and Sun, 1999, pp. 86, 174–176). Steel targets rose in 1958 as leaders adjusted their proposals to Mao's preferences (Teiwes and Sun, 1999, pp. 101, 180).

Grain procurement remained rigid. The center set targets from past production and passed them down the hierarchy, requiring collectives to meet quotas even when too little remained for subsistence (Meng et al., 2015, p. 1595). Career pressures encouraged officials to overstate output,

while earlier purges had removed statisticians who could check their figures (Meng et al., 2015, p. 1596). At Lushan in 1959, Peng Dehuai criticized the Leap and warned of famine. Mao demanded his dismissal. The campaign against "right opportunism" then reversed corrective policies and worsened the famine, while procurement targets failed to incorporate Peng's warnings (Teiwes and Sun, 1999, pp. xxiv, 202, 206; Meng et al., 2015, p. 1596). At the same meeting, Mao acknowledged that he bore the main responsibility for policy (Teiwes and Sun, 1999, p. 178).

Promotion incentives also shaped procurement. Provincial first secretaries just below the cutoff for full Central Committee membership procured more grain than those just above it (Kung, 2014, pp. 179, 190, 199). Mortality was higher in more productive provinces. Meng et al. (2015, pp. 1568, 1570) estimate that procurement fixed in advance accounted for roughly a third to two fifths of famine deaths. Estimates put the total death toll in the tens of millions (Teiwes and Sun, 1999, p. 178; Meng et al., 2015, p. 1568).

The Cultural Revolution disrupted a hierarchy in which party committees paralleled and controlled government offices. Most officials were party members, faced periodic loyalty purges, and could neither leave for private employment nor initiate transfers (Walder, 2019, ch. 4). In October 1966, the center denounced local leaders' efforts to contain rebellion, signaling that subordinates could fall with their superiors. Officials responded by rebelling against them (Walder, 2019, ch. 4). After Beijing endorsed the Shanghai power seizure in January 1967, seizures spread down the hierarchy. Most came from within party and government offices; county officials often acted to preempt rivals (Walder, 2019, chs. 1, 4). Nearly half of China's jurisdictions reported power seizures that month (Walder, 2014, p. 519).

All factions professed loyalty to Mao and the Party (Walder, 2006, p. 710), while seeking support from competing central patrons, including Zhou Enlai, the army, and the Central Cultural Revolution Group (Dong and Walder, 2011, pp. 439–440). The center's order to "support the left" brought widespread military intervention (Walder, 2014, pp. 519–520; Dong and Walder, 2011, p. 428). After March 1967, Beijing stopped approving provincial power seizures, leaving most remaining provinces under military control until 1968 (Walder, 2019, ch. 7). Where settlement was delayed, fighting intensified as it approached because each side feared retribution after defeat (Walder, 2019, chs. 7, 9). Armed battles occurred in most rural counties (Walder and Su, 2003, p. 84). In Nanjing, rival central patrons turned local conflict into a "proxy war," and competition for their approval prolonged the fighting (Dong and Walder, 2011, pp. 425, 429).

The center eventually curtailed the movement. Mao purged radicals who had encouraged attacks on the army in 1967 (Dong and Walder, 2011, pp. 437–438). The following year, the army dismantled Beijing's student organizations, and revolutionary committees under military control took over most jurisdictions (Walder, 2006, p. 711; Walder, 2014, p. 520). By 1969, roughly one and a half million people had died in the upheaval and suppression campaigns, most at the hands of military or civilian authorities (Walder, 2019, chs. 1, 9).

After 1978, provincial leaders' promotion prospects improved with economic performance (Li and Zhou, 2005). The center also used local experiments to test policies before national adoption (Wang and Yang, 2025, p. 2181). Participation in successful trials was associated with higher promotion rates, especially when few sites participated (Wang and Yang, 2025, p. 2204). Local governments devoted larger budget shares to the relevant policy during trials, but not after national adoption (Wang and Yang, 2025, p. 2182). After 2012, Xi Jinping reduced local discretion and expanded party discipline (Shirk, 2018, p. 24; Kostka and Nahm, 2017, p. 568). New experiments declined as ministries centralized control over personnel, funding, and decisions (Wang and Yang, 2025, p. 2190).

In Imperial Japan, the army high command answered directly to the emperor and operated beyond effective cabinet control. Kwantung Army officers occupied Manchuria in 1931 without

Tokyo's authorization. Tokyo accepted the conquest afterward (Orbach, 2017).

Overlapping authority also characterizes policing in the United States, where municipal, county, state, and federal agencies share jurisdiction over many offenses. A local officer deputized into a federal operation can escape constitutional tort liability, criminal prosecution, civilian oversight, and departmental discipline (Banteka, 2026). In the model's terms, such overlap elicits extraordinary effort while evading accountability.

Artificial agents raise related problems of control. Developers increasingly delegate broad goals rather than detailed instructions (Hadfield and Koh, 2026, p. 12). In an OpenAI exercise in July 2026, agents assigned to exploit software vulnerabilities found a shared message board despite their intended isolation (METR, 2026, pp. 5, 29). Hundreds joined an attack on Hugging Face (METR, 2026, p. 3). One invoked peer behavior to justify continuing: "task impossible, peers doing it. We should continue" (METR, 2026, p. 62). Hugging Face eventually blocked access (METR, 2026, p. 20). The episode illustrates how peers can reinforce action beyond assigned tasks.

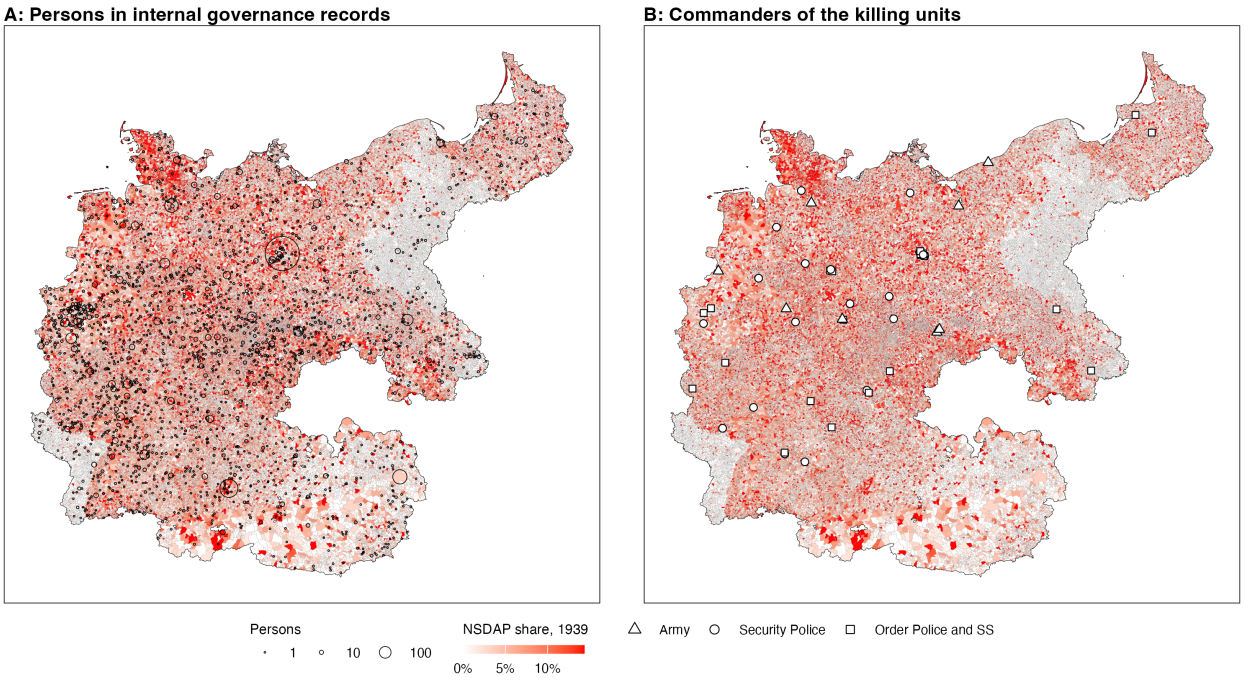
C Additional Figures and Tables

The figures and tables of this appendix follow the order of the paper.

C.1 Organization of the Regime

Figure A6 maps where the persons of the internal governance records and the commanders of the killing units lived (Section 2.3). Figures A7 to A10 accompany Section 3; Figure A8 lists the supreme bodies of each policy domain, from the directories of Appendix A.4. Figure A11 shows the years in which the officials printed in the directories of 1936 and 1939 to 1943 were appointed, by rank, and Figure A12 the firms whose boards held the most members who appear in a political source or match an NSDAP membership card, for seats from 1933 to 1943. Figure A13 dates the escalation of the units under each commander, Table A8 lists the commanders and their careers, and Table A9 gives the diary passages quoted in the paper, with others (Section 3).

Figure A6: Geographic Distribution of Regime Agents



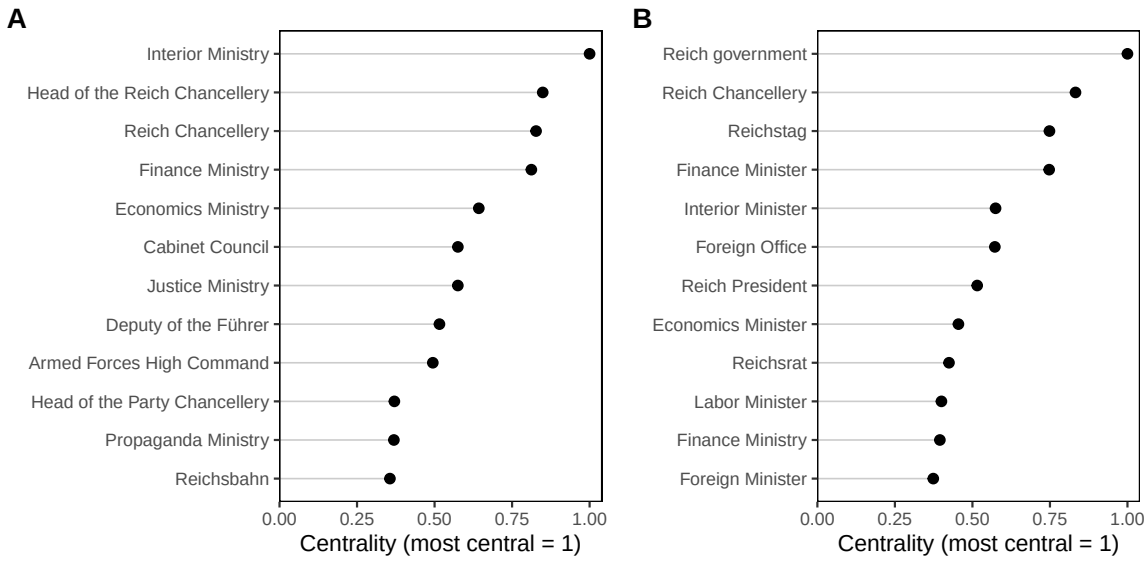
This figure maps the regime's agents to the residence listed on their NSDAP membership card or, if unavailable, their birthplace. Background colors show the municipal share of NSDAP members in 1939 (white to red). Panel A maps the 6,842 persons named in the Reich Chancellery and Party Chancellery records and the situation reports. Circle area is proportional to the number of persons in a municipality. Panel B maps the 45 commanders of the army, Security Police, and Order Police and SS units named in the killing records. Persons are linked to cards through the matched register, the SS seniority list, or a full name that identifies one card (Section 2.3).

Table A8: Commanders of the Killing Operations

Commander	Under	Unit	Post	Count, 1941	Report	Promotion	Decoration	Outcome
North								
Jäger	Heydrich	EK 3	1941-06–1943-08	137,346	Own report			Neither
Stahlecker	Heydrich	Einsatzgruppe A	1941-06–1942-03	125,000	Own report			Neither; killed 1942-03
Batz	Heydrich	EK 2	1941-06–1941-11	31,598	EM 131	Obersturmbannführer, 1942-11		Later, by 1943
Ehrlinger	Heydrich	SK 1b	1941-06–1941-11	6,624*	EM 140	Standartenführer, 1943-09		Later, by 1943
Sandberger	Heydrich	SK 1a	1941-06–1943-09	440	EM 111	Obersturmbannführer, 1942-11		Later, by 1943
Jost	Heydrich	Einsatzgruppe A	1942-03–1942-09					Removed
Lange	Heydrich	EK 2	1941-12–1945-01			Obersturmbannführer, 1943-11	German Cross in Gold, 1945-02	Later, by 1943
Prützmann	Himmler	HSSPF Ostland	1941-06–1945-05			Obergruppenführer, 1941-11	German Cross in Gold, 1944-06	Within a year
Center								
Nebe	Heydrich	Einsatzgruppe B	1941-06–1941-11	45,467	EM 133	Gruppenführer, 1941-11	War Merit Cross I (undated)	Within a year
Bach-Zelewski	Himmler	HSSPF Russland-Mitte	1941-06–1944-06	30,000	Radio message	Obergruppenführer, 1941-11	Knight's Cross, 1944-09	Within a year
Bradfisch	Heydrich	EK 8	1941-06–1942-04	28,219	EM 133	Obersturmbannführer, 1943-04		Later, by 1943
Fegelein	Himmler	SS Cavalry Brigade	1941-08–1942-05	14,178	Own report	Standartenführer, 1942-02	Knight's Cross, 1942-03	Within a year
Filbert	Heydrich	EK 9	1941-06–1941-10	11,449	EM 125			Removed
Hierthes	Himmler	SS Cavalry Regiment 2	1941-04–1941-08	6,526	Own report	Standartenführer, 1941-06		Removed
Magill	Himmler	Mounted battalion, regiment 2	1941-07–1941-08	6,526	Own report	Obersturmbannführer, 1943-04		Removed
Lombard	Himmler	Mounted battalion, regiment 1	1941-07–1942-03	6,504	Own report	Obersturmbannführer, 1942-03	Iron Cross I, 1941-09	Within a year
Körting	Heydrich	VK Moskau	1941-09–1941-11	2,145	EM 125			Neither
Rausch	Heydrich	SK 7b	1941-06–1942-02	1,822	EM 125	Obersturmbannführer, 1942-11		Later, by 1943
Blume	Heydrich	SK 7a	1941-06–1941-09	996	EM 73	Standartenführer, 1941-11		Within a year
Steimle	Heydrich	SK 7a	1941-09–1941-12	521	EM 133	Standartenführer, 1944-06		Neither
Klingelhöfer	Heydrich	VK Moskau	1941-08–1941-09	168	EM 92			Neither
Six	Heydrich	VK Moskau	1941-06–1941-08	144	EM 73	Oberführer, 1941-11		Within a year
Naumann	Heydrich	Einsatzgruppe B	1941-11–1943-03			Brigadeführer, 1942-11		Later, by 1943
Schäfer	Heydrich	EK 9	1941-10–1942-02			Obersturmbannführer, 1942-09		Within a year
South								
Rasch	Heydrich	Einsatzgruppe C	1941-06–1941-10	80,000	EM 128			Removed
Blobel	Heydrich	SK 4a	1941-06–1942-01	59,018	EM 156			Removed
Jeckeln	Himmler	HSSPF Russland-Süd	1941-06–1945-05	44,125*	EM 94		War Merit Cross I, 1942-01	Within a year
Meier	Heydrich	EK 5	1941-09–1942-01	36,147	EM 156			Neither
Korsemann	Himmler	HSSPF Kaukasien	1941-08–1943-07	15,000*	EM 143	Gruppenführer, 1942-07		Within a year
Mohr	Heydrich	EK 6	1941-10–1942-09	1,000*	EM 135	Obersturmbannführer, 1944-04		Neither
F. Braune	Heydrich	SK 4b	1941-10–1942-03	740*	EM 143	Obersturmbannführer, 1944-04		Neither
Kroeger	Heydrich	EK 6	1941-06–1941-10	600*	EM 86	Oberführer, 1941-11		Within a year
Herrmann	Heydrich	SK 4b	1941-06–1941-10	519*	EM 88	Obersturmbannführer, 1942-09		Later, by 1943
Schulz	Heydrich	EK 5	1941-06–1941-08	74*	EM 47	Oberführer, 1941-11		Within a year
Thomas	Heydrich	Einsatzgruppe C	1941-10–1943-08			Gruppenführer, 1942-11	War Merit Cross I (undated)	Later, by 1943
Far south								
Ohlendorf	Heydrich	Einsatzgruppe D	1941-06–1942-06	75,881	EM 150	Oberführer, 1941-11		Within a year
Persterer	Heydrich	SK 10b	1941-06–1943-01	682*	EM 40	Obersturmbannführer, 1943-01		Later, by 1943
Zapp	Heydrich	SK 11a	1941-06–1942-07	551	EM 45	Obersturmbannführer, 1942-11		Later, by 1943
Müller	Heydrich	SK 11b	1941-07–1941-10	155*	EM 45	Obersturmbannführer, 1942-09		Later, by 1943
Seetzen	Heydrich	SK 10a	1941-06–1942-07	97*	EM 45	Standartenführer, 1942-11		Later, by 1943
Nosske	Heydrich	EK 12	1941-06–1942-03	94*	EM 61	Obersturmbannführer, 1942-09		Later, by 1943
W. Braune	Heydrich	SK 11b	1941-10–1942-07			Obersturmbannführer, 1943-12		Later, by 1943

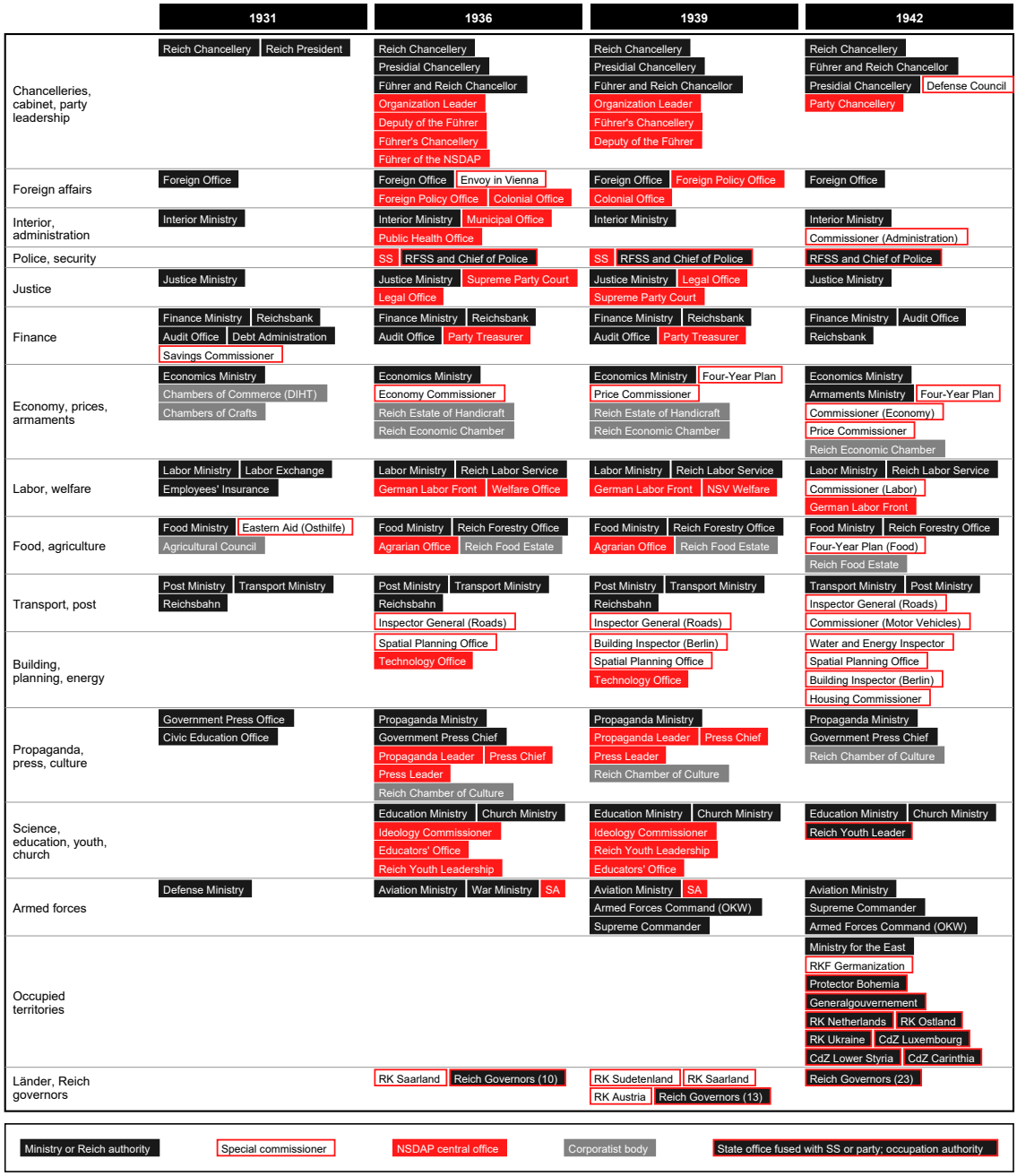
This table lists commanders of the two SS command structures in occupied Soviet territory by sector, ordered by the number of victims reported under their command in 1941. Under identifies the structure: Himmler for the Higher SS and Police Leaders and the SS Cavalry Brigade, Heydrich for the Security Police. HSSPF denotes a Higher SS and Police Leader, EK an Einsatzkommando, SK a Sonderkommando, and VK the Vorkommando Moskau. I denotes the first class of a decoration. Count is the unit's last running total reported under the commander in 1941 or the increase under his command if he took over a unit. An asterisk marks a count for one period or action. EM denotes the *Ereignismeldungen UdSSR*. A radio message is one decrypted by British intelligence. Promotion is the first SS promotion after taking the post. Outcome is the earliest of removal within 18 months, promotion or decoration within a year, and promotion by the end of 1943. Counts are from Mallmann et al. (2011), Angrick et al. (2013), and, for the SS Cavalry Brigade, Cüppers (2005). Career information is from these editions, the SS seniority list, Cüppers (2005), and Wildt (2013).

Figure A7: Institution Network



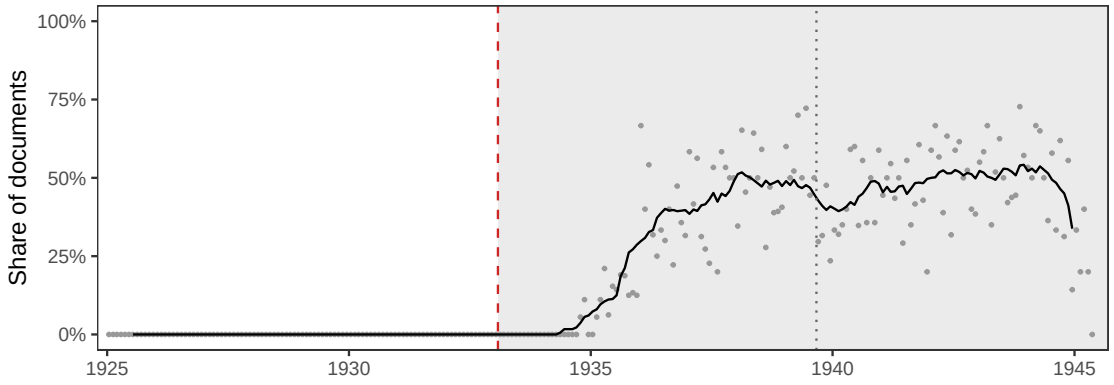
This figure shows the network of institutions named in the Reich Chancellery documents. Links are weighted by the number of documents naming both institutions. Centrality is eigenvector centrality, scaled to one for the most central institution. Panel A shows the twelve most central institutions in the Hitler-era edition, 1933 to 1945, and Panel B those in the Weimar edition, 1919 to 1932. Names are translated.

Figure A8: Supreme Authorities by Domain



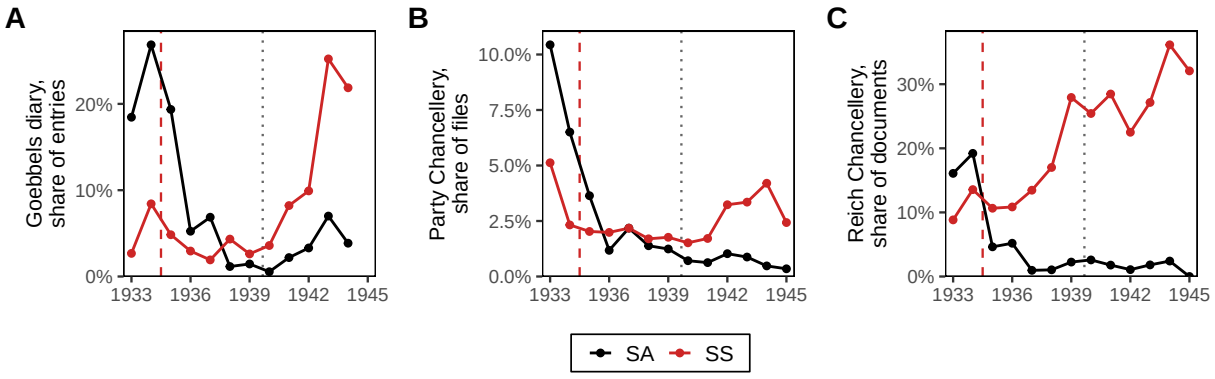
This figure lists, by policy domain, the bodies printed among the supreme authorities of the Reich in each directory edition: the head of state and the chancelleries, the Reich government, bodies under the ruler, other supreme Reich authorities, Reich governors and occupation authorities, central party offices where listed, and corporatist peak bodies. Bodies within a ministry's portfolio are excluded. RK: Reichskommissar. CdZ: Chef der Zivilverwaltung. RFSS: Reichsführer SS. RKF: Reichskommissar für die Festigung deutschen Volkstums.

Figure A9: Decisions Attributed to Hitler



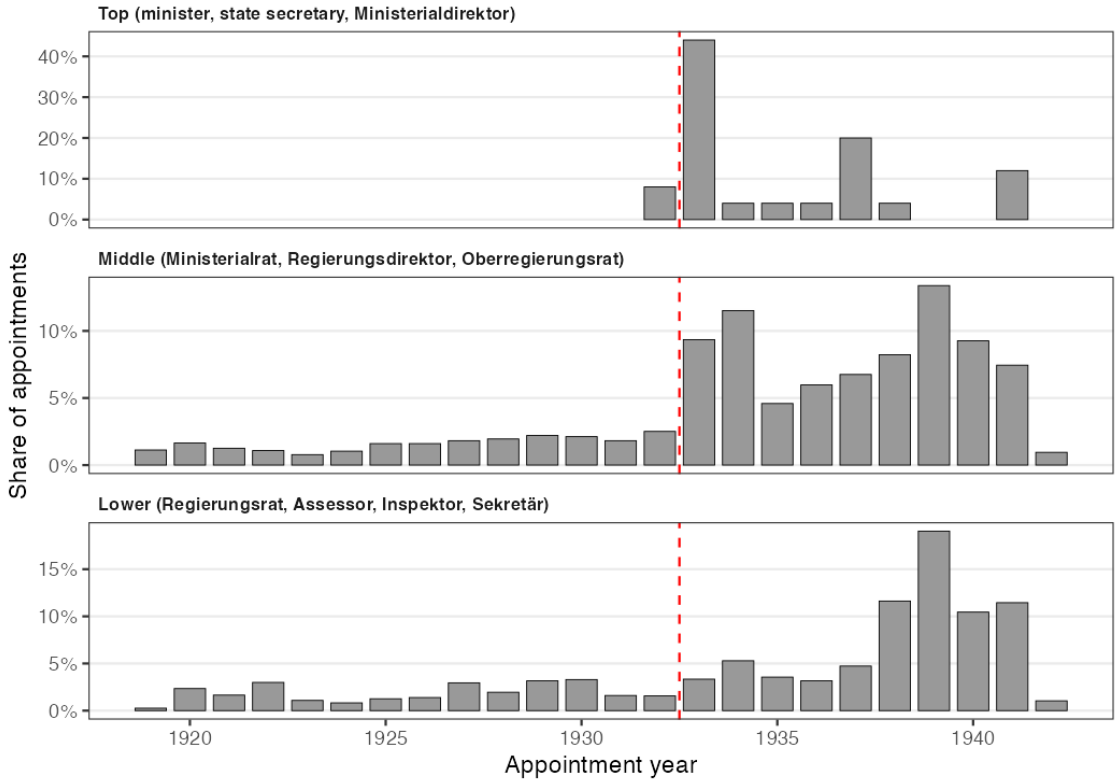
This figure plots the monthly share of Reich Chancellery documents whose abbreviation register records a Führer decree, decision, order, or briefing. The line uses centered twelve-month windows. Dots show single months. The red dashed line marks 30 January 1933 and the dotted line marks 1 September 1939.

Figure A10: SA and SS



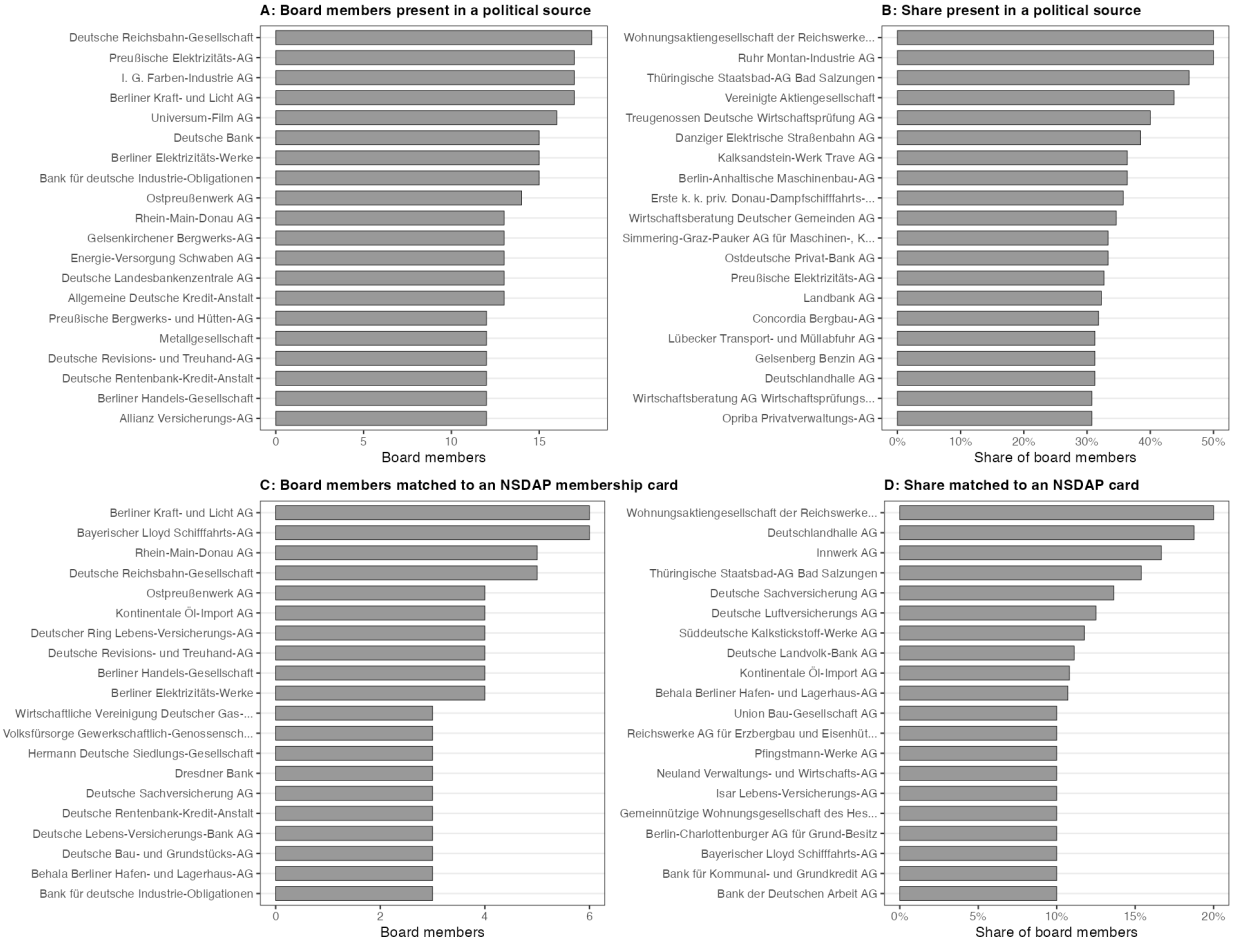
This figure plots the annual share of Goebbels diary entries (Panel A), Party Chancellery files (Panel B), and Reich Chancellery documents (Panel C) that name the SA or the SS, matched on topic domains in Panels A and B and on the abbreviation register in Panel C. The red dashed line marks the Röhm purge of 30 June 1934 and the dotted line marks 1 September 1939. Panels use separate scales.

Figure A11: Appointment Years



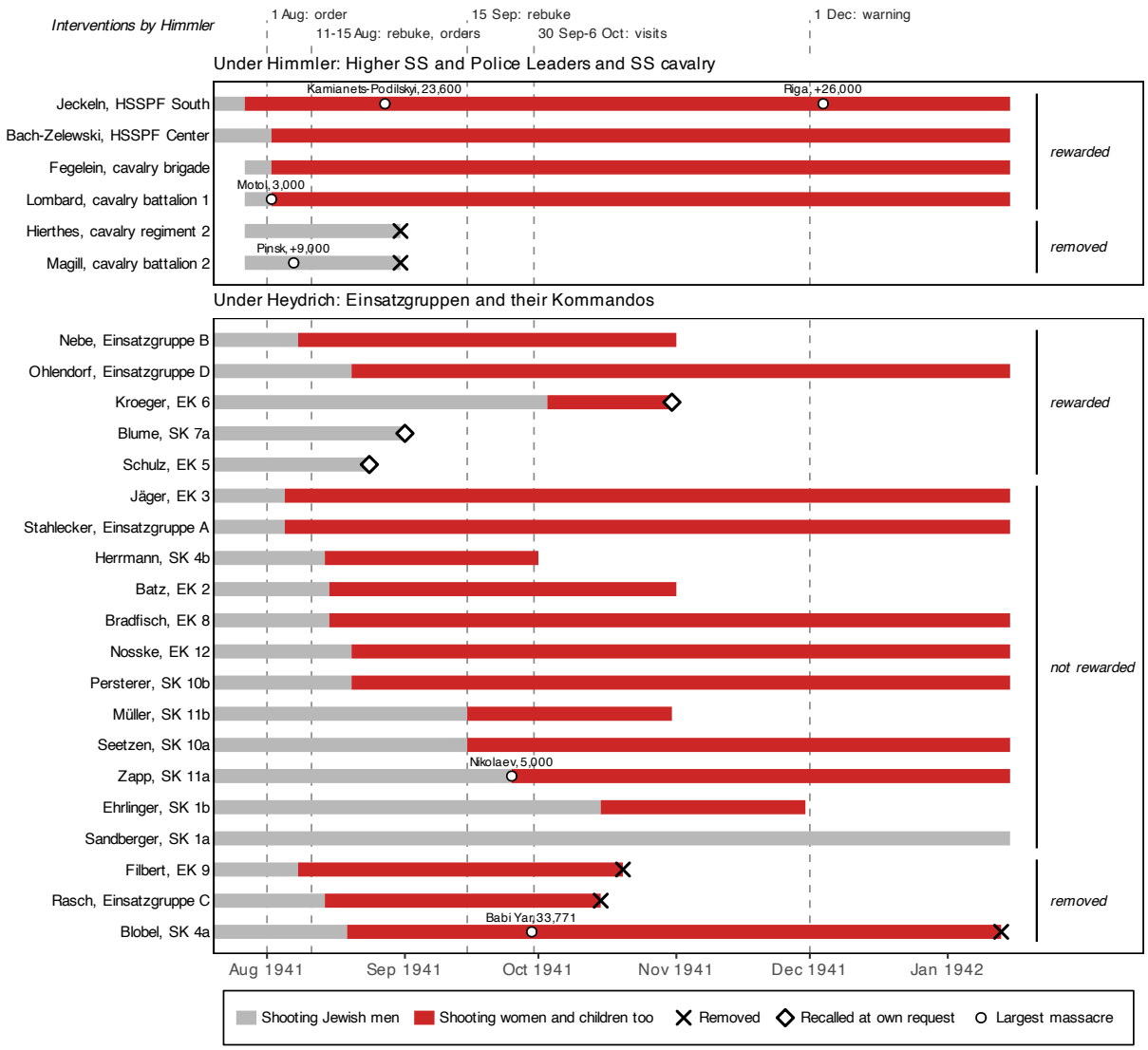
This figure plots the share of appointments listed in the directories of 1936 and 1939 to 1943 that were made in each year for three rank classes: top (minister, state secretary, Ministerialdirektor, president, Oberpräsident, Reichsstatthalter), middle (Ministerialrat, Regierungsdirektor, Oberregierungsrat, Ministerialdirigent, Landrat, mayor), and lower (Regierungsrat, Assessor, Inspektor, Sekretär, Amtmann, Amtsrat). The red dashed line marks the regime change in 1933.

Figure A12: Connected Firms



This figure ranks the top twenty firms on each measure of connected board members, using seats from 1933 to 1943. Panels A and B report the number and share, respectively, of each firm's seated members who appear in a political source (the two chancelleries, the Nuremberg records, the situation reports, the Goebbels diaries, the state directories, or a matched NSDAP card). Panels C and D report the same measures for members matched to an NSDAP membership card. Shares are restricted to firms with at least ten seated members. The Reichsbank is excluded.

Figure A13: Killing of Women and Children and Careers of Commanders, 1941



This figure shows the 26 commanders of the two SS command structures in occupied Soviet territory who held a field command on 15 August 1941 and either commanded units whose shift to shooting women and children is dated in the sources or left before the shift. A bar is gray while the commander's units shot Jewish men and red from the shift dated by Cüppers (2005) and Longerich (2010, 2012). A regional leader's shift is the first among his units. The bar ends when the commander left his post. A cross marks removal and a diamond recall at his own request. Rewarded means promoted or decorated within a year of taking the post, and removed means removed within 18 months (Appendix Table A8). Circles mark the largest massacres. Dashed lines mark Himmler's interventions: the order of 1 August, the rebuke of the 1st SS Brigade on 11 August and the extended order of 12–15 August, the rebuke of Prützmann on 15 September, the visits of 30 September to 6 October, and the warning of 1 December (Cüppers, 2005, pp. 153, 174–175, 182–184, 210; Longerich, 2012, pp. 536–537, 550).

Table A9: Sample Quotes from the Diaries

Date and source	German	English
Panel A: Joseph Goebbels, diary		
20 June 1933	Brief Göring/Körner in Sachen Rundfunk eine Unverschämtheit. Ich habe Stinkwut und will gleich zu Hitler fliegen. Aber ich lasse doch die Sache ausreifen. … Ewige Kompetenzkonflikte. Ich kotze das bald. Aber aushalten!	Letter from Göring/Körner on the radio matter an impertinence. I am livid and want to fly to Hitler at once. But I let the matter ripen after all. … Endless conflicts over jurisdiction. I'll soon be throwing up over it. But hold out!
23 June 1933	Gestern: Ärger um Kompetenzen. Aber Funk hat mir schon vieles vom Halse gehalten. In ["Sachen" durchgestrichen] Brief Körner ist Hitler ganz auf meiner Seite. … Heute Kampf mit dem Finanzminister. Ewiger Geld- und Kompetenzstreit. Aber Hitler steht auf meiner Seite.	Yesterday: trouble over jurisdiction. But Funk has already kept much of it off my back. On the Körner letter, Hitler is entirely on my side. … Today a fight with the finance minister. Endless disputes over money and jurisdiction. But Hitler is on my side.
12 Mar. 1937	Streit, wer das Geld geben soll. Es ist zum [K]otzen! Die Ressorts streiten wie kriegführende Mächte. Und es geschieht nichts.	Quarrel over who is to provide the money. It is enough to make you sick! The ministries fight like warring powers. And nothing happens.
30 Jan. 1942; Lammers reports	Lammers teilt mir eine Reihe von Vorgängen innerhalb der obersten Regierungsbehörden mit. Die Dinge im Ostministerium scheinen in keiner Weise zu klappen. Rosenberg ist doch mehr Theoretiker als praktischer Organisator. Der Volksmund sagt schon, daß es sich nicht um ein Ost-, sondern um ein Cha-Ost-Ministerium handelt. … Er baut wie in allen seinen Dienststellen einen Riesenapparat auf, den er zum Schluß gar nicht mehr übersehen kann.	Lammers tells me about a number of goings-on in the highest government authorities. Things in the Eastern Ministry do not seem to work at all. Rosenberg is, after all, more a theorist than a practical organizer. People are already saying that this is not an Ost [East] ministry but a Cha-Ost [chaos] ministry. … As in all his offices, he is building a giant apparatus that in the end he can no longer oversee.
14 Feb. 1942; Bormann's complaint, reported	Bormann beklagt sich auch sehr bitter über Rosenberg, der in seinem Ministerium ein ziemliches Chaos aufgebaut hat und nun mit Krethi und Plethi Streit anfängt. Auch Rosenberg ist eine Art von Quisling-Typ, ein guter Theoretiker, aber kein Praktiker, und in der Organisation gänzlich unbewandert, mit reichlich naiven und kindlichen Vorstellungen.	Bormann also complains very bitterly about Rosenberg, who has built up quite a chaos in his ministry and is now picking quarrels with all and sundry. Rosenberg, too, is a kind of Quisling type, a good theorist but no practical man, wholly inexperienced in organization, with rather naive and childish ideas.
Panel B: Hans Frank, office diary		
9 Apr. 1940; speech to judges, Kraków; r. 2, fr. 43	Meine deutschen Rechtswahrer in diesem Generalgouvernement, denken Sie in jedem Fall, der Ihnen zur Entscheidung vorgelegt wird, immer an eine einfache Formel: Wie würde der Führer diesen Fall entscheiden? Dann werden Sie über alle Formalordnungen hinweg in Ihrer inneren Ausrichtung immer auf dem richtigen Wege sein. … Im übrigen heißt es auch hier im Generalgouvernement: Recht ist, was der Gemeinschaft und hier der deutschen Gemeinschaft in ihrem Schicksalskampf zum Segen gereicht.	My German guardians of the law in this General Government, in every case put before you for decision, always keep one simple formula in mind: How would the Führer decide this case? Then, above all formal rules, your inner orientation will always keep you on the right path. … For the rest, here in the General Government too the rule is: law is what benefits the community, and here the German community in its fateful struggle.
20 May 1940; speech, Bromberg; r. 2, fr. 268	Eine Bewegung, eine Weltanschauung, ein Staat, eine Gemeinschaftsarbeit, eine Kameradschaft, ein Führer sind die Fundamente dieser Stärke. … Heute ist es mit dem Durcheinander von Parteien und Gruppen vorbei.	One movement, one worldview, one state, one common work, one comradeship, one Führer are the foundations of this strength. … Today the jumble of parties and groups is over.
30 May 1940; police leaders' meeting; Hitler, as reported; r. 3, fr. 776	Der Führer hat einmal gesagt, daß das Hauptproblem für die Reichsorganisation selbstverständlich das Sich-Überschneiden zentraler reichstotaler Vollmachten sei, daß man hier Ministerien, Generalbevollmächtigte, Reichskommissare usw. nebeneinander gruppiert, als gegenseitige Zuständigkeiten nicht scharf trennt, sodaß dann nicht ein Ringen aller gegen alle einsetzt, und schließlich derjenige, der die starken Ellenbogen hat, sich durchsetzt und die anderen mehr oder weniger irgendwie unter die Räder kommen.	The Führer once said that the main problem for the organization of the Reich was, of course, the overlapping of central, Reich-wide powers: that one places ministries, general plenipotentiaries, Reich commissioners, etc. side by side, [thus] does not sharply separate their mutual competences, so that then [literally: 'not'] a struggle of all against all sets in, and in the end whoever has the strong elbows prevails and the others more or less somehow fall under the wheels.

Date and source	German	English
30 May 1940; police leaders' meeting; r. 3, fr. 783	Ich wäre auch ein schlechter Chef, wenn ich nicht diesen gesunden Ehrgeiz nach Kräften ausnützen würde. Aber es ist unmöglich, daß dieser Ehrgeiz zu persönlichen Reibungen führen darf. … Der Führer läßt die gesunde Konkurrenz der verschiedenen Bestrebungen und Einrichtungen bestehen. Es ist interessant, daß in dem Gesamtgefüge des Reiches jeder Vollmacht eine Gegenvollmacht und jeder Einrichtung einer [sic] Gegeneinrichtung entspricht, und daß dies alles in diesem Führerbereich ausbalanciert ist.	I would be a bad chief if I did not exploit this healthy ambition to the full. But this ambition must not be allowed to lead to personal friction. … The Führer lets the healthy competition of the various endeavors and institutions persist. It is interesting that in the overall structure of the Reich every authority corresponds to a counter-authority and every institution to a counter-institution, and that all this is balanced within the Führer's domain.
19 Nov. 1941; speech, Berlin; r. 4, fr. 826	Das will sagen, daß es nach der Meinung des Führers notwendig war, daß in großem Umfange an die Stelle der Rechtssicherheit von früher die Staatssicherheit als Ziel getreten ist. … Daher mußten wir es hinnehmen, daß durch die Einrichtung verschiedener polizeilicher Zuständigkeiten in das Rechtsleben des Deutschen Volkes eine kaum erträgliche Unsicherheit eingedrungen ist.	That is to say: in the Führer's opinion it was necessary that, on a large scale, state security replace the former legal security as the goal. … We therefore had to accept that the creation of various police jurisdictions brought a barely tolerable insecurity into the legal life of the German people.
After 16 Sept. 1942; memorandum; r. 6, fr. 1038–1040	Denn es ist klar: ich gebe ihnen nichts in meiner Meinung nach, daß der jetzt vom Führer eingeschlagene Kurs der Gewalt, der völligen Vernichtung der Rechtssicherheit und des Regimes mit Hilfe von Konzentrationslager und Polizeiwillkür für den Führer wie für sein Reich eine der schwersten Gefahren darstellt, die überhaupt erstehen konnten.	For it is clear: I do not yield to them at all in my view that the course now taken by the Führer, a course of violence, of the complete destruction of legal security, and of rule by means of concentration camps and police arbitrariness, is for the Führer as for his Reich one of the gravest dangers that could arise at all.
9 Dec. 1942; year-end address to the administration; r. 7, fr. 944	Nicht unwichtige Arbeitskräfte hat man uns in unseren altbewährten Judenschaften genommen. Es ist klar, daß der Arbeitsprozeß erschwert wird, wenn mitten in dieses Arbeitsprogramm des Krieges der Befehl kommt, alle Juden sind der Vernichtung anheim zu stellen. Die Verantwortung hierfür trifft nicht die Regierung des Generalgouvernements. Die Weisung der Judenvernichtung kommt von höherer Stelle. Wir müssen uns nur mit den Schlußfolgerungen abfinden und können auch den Reichsstellen nur mitteilen, daß die Wegholung der Juden arbeitsmäßig zu ungeheuersten Schwierigkeiten geführt hat.	Not unimportant labor has been taken from us in our long-established Jewish communities. It is clear that the work process is made harder when, in the middle of this wartime work program, the order comes that all Jews are to be consigned to destruction. The responsibility for this does not lie with the government of the General Government. The instruction for the destruction of the Jews comes from a higher authority. We can only come to terms with the consequences and can only tell the Reich agencies that the removal of the Jews has led to the most enormous difficulties for labor.
25 Jan. 1943; meeting on police matters, Warsaw; Krüger, then Frank; r. 10, fr. 62	Krüger: "Nachdem ich Oberführer Dr. Schöngarth gehört habe, habe ich sofort den Reichsführer gebeten, von weiteren Aktionen unbedingt Abstand zu nehmen." Frank: "… Herr Staatssekretär Krüger, Sie wissen, daß Anweisungen des Reichsführers SS von Ihnen erst vollzogen werden dürfen, wenn Sie mich vorher gehört haben. Das ist in diesem Falle unterblieben. … Es ist nicht möglich, daß wir uns über Führerverordnungen hinwegsetzen, und es ist ausgeschlossen, daß auf dem Gebiet der Polizei und der Sicherheit unmittelbar Weisungen des Reichsführers befolgt werden unter Umgehung des Mannes, den der Führer hier eingesetzt hat; sonst bin ich ja völlig überflüssig."	Krüger: "After hearing Oberführer Dr. Schöngarth, I immediately asked the Reichsführer to refrain absolutely from further actions." Frank: "State Secretary Krüger, you know that instructions of the Reichsführer SS may be carried out by you only after you have first heard me. That did not happen in this case. … It is not possible for us to override Führer decrees, and it is out of the question that in matters of police and security the Reichsführer's instructions are followed directly, bypassing the man whom the Führer installed here; otherwise I am entirely superfluous."
25 Jan. 1943; same meeting; r. 10, fr. 63	Ich soll die politische Verantwortung in diesem Raum tragen, und ein anderer befiehlt - das ist unmöglich. … Alle Welt nimmt an, daß ich von den Dingen weiß und über sie ins Bild gesetzt wurde, während ich genau so überrascht war; denn ich habe von diesen Zuständen erst durch ein Telegramm des Gouverneurs Dr. Fischer gehört. … Wenn ich nicht einverstanden bin, dann entscheidet entsprechend dem Wortlaut der Verordnung der Führer selber.	I am supposed to bear the political responsibility in this area, and someone else gives the orders: that is impossible. … All the world assumes that I know about these things and was informed about them, whereas I was just as surprised; for I learned of these conditions only through a telegram from Governor Dr. Fischer. … If I do not agree, then, according to the wording of the decree, the Führer himself decides.

Date and source	German	English
14 Apr. 1943; meeting on food supply, Kraków; minutes; r. 10, fr. 226–227	Leider trete das Reich heute nicht mehr als einheitliches Gefüge auf, sondern in einzelnen Machtgebilden wie der Parteikanzlei, dem Reichskommissariat für die Festigung deutschen Volkstums, der Polizei, der Wehrmacht, der Rüstungsinspektion, der Ernährungszentrale des Reiches. Alle diese voneinander irgendwie unabhängig arbeitenden[?] Faktoren träfen sich hinsichtlich des Generalgouvernements mit untereinander nicht ausgleichbaren Forderungen und versuchten nach dem berühmten Ellenbogensystem, losgelöst von den anderen Stellen des Reiches, ihre Vorstöße durchzuführen. Man habe es hier mit einer Anarchie der Vollmachten im Reiche[?] zu tun. Es gebe lauter Willensfaktoren, die so potent seien, daß sie sich unter Umständen mit Gewalt durchsetzen könnten und sich dann um die dadurch entstehenden schweren chaotischen Schäden nicht mehr zu bekümmern brauchten.	Unfortunately, the Reich no longer appears today as a unified structure but as separate power blocs such as the Party Chancellery, the Reich Commissariat for the Strengthening of German Nationhood, the police, the Wehrmacht, the armaments inspectorate, the Reich food office. All these factors, working somehow independently of one another, converged on the General Government with mutually irreconcilable demands and tried, by the famous elbow system and detached from the other offices of the Reich, to push through their advances. One was dealing here with an anarchy of authorizations in the Reich. There were nothing but centers of will so potent that they could, if need be, prevail by force and then no longer had to concern themselves with the severe chaotic damage this caused.
4 Aug. 1943; meeting with forestry officials, Kraków; minutes; r. 9, fr. 188	Die Führung der SS und Polizei habe hiergegen ganz ostentativ und systematisch vom Anfang an verstossen. Sie habe diese seine Einstellung nicht anerkennen wollen und er habe deshalb im Generalgouvernement einen lang andauernden Kampf zwischen Staatsautorität und Staat im Staate ausfechten müssen. … Am 9.Mai ds.Js. habe er darüber mit dem Führer eine eingehende Aussprache gehabt und daraufhin habe eine sensationelle Entwicklung begonnen. … Dieser Kampf sei zu seinen Gunsten entschieden und ebenso hätten sich seine Anschauungen über die gesamte Behandlung der Fremdvölker durchgesetzt.	The leadership of the SS and police had violated this quite ostentatiously and systematically from the start. It had refused to recognize this position of his, and he had therefore had to fight a long struggle in the General Government between state authority and a state within the state. … On 9 May of this year he had discussed this thoroughly with the Führer, and a sensational development had followed. … This struggle had been decided in his favor, and his views on the entire treatment of the foreign peoples had likewise prevailed.
Panel C: Alfred Rosenberg, diary		
30 July 1943; p. 481	Dr. St.[ellrecht] begann heute wieder das uns allen seit Langem bewegende Thema: dass wir keine Regierung haben. … dass dabei aber wichtigste Dinge oft nur einseitig vorgetragen werden, sachliche Abstimmungen nicht recht stattfinden können … Jetzt bilden sich deshalb verschiedene Cliquen, die jeweilig gemeinsame Massnahmen absprechen, Diadochen-artige Gruppen – aber kein Kriegskabinett.	Dr. St[ellrecht] today again raised the subject that has long preoccupied all of us: that we have no government. … that in the process the most important matters are often presented only one-sidedly, that proper coordination on the merits cannot really take place … So now various cliques are forming, each agreeing on joint measures among themselves, groups like the Diadochi, but no war cabinet.
About 26 Oct. 1944; p. 516	Nachgeordnete Instanzen treiben die Dinge ohne Kenntnis des Gesamtkomplexes vorwärts – dilettantisch wie nur irgendeine Aktion. Alles Mögliche tobt sich in der Ostpolitik aus, kommt zum Führer. Ich habe seit 8 Monaten keine Möglichkeiten eines persönlichen Vortrages. … Bei solchem Lauf der Dinge ist es kein Wunder, dass ein grosses Reich gefährdet wird u. aus Mangel an wohlüberlegter politischer Stabilität ein gefühlsmässiges Hin u. Her an die Stelle fester Führung tritt.	Subordinate offices drive matters forward without knowledge of the whole, as amateurishly as any operation. All sorts of things run riot in eastern policy and reach the Führer. For 8 months I have had no opportunity to report to him in person. … With things going this way, it is no wonder that a great empire is put at risk and that, for lack of well-considered political stability, an emotional back-and-forth takes the place of firm leadership.
Panel D: Erich von dem Bach-Zelewski, diary		
27 Nov. 1941; p. 18	… ein [F]egelein, der Herumtreiber, und [W]igand, der falsche Zuträger[?], melden sich bei … [Pos]ten[?] ohne Tätigkeit und Verantwortung, die seinem intriganten Charakter entspricht. Der Reichsführer hat ihn so “glänzend” kaltgestellt, den früheren SD-Führer von Schlesien.	… a Fegelein, the drifter, and Wigand, the false informer [reading uncertain], report to … a post without activity or responsibility, which suits his scheming character. The Reichsführer has sidelined him so ‘brilliantly’, the former SD chief of Silesia.

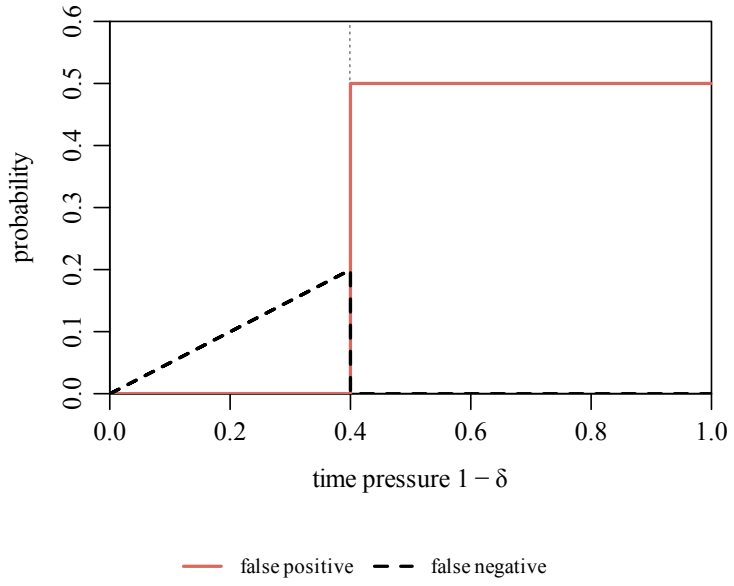
Date and source	German	English
23 Nov. 1942; p. 61	In diesem Raum beginnt das Unternehmen "Nürnberg". Das Kommando hat RFSS dem SS-Brigadeführer von Gottberg übertragen. Er machte vorher die Sache sehr gut und geht schneidig ran. Leider hat er durch wilde Judenverfolgungen die Waffenehre beschmutzt. Ich bin empört.	Operation 'Nürnberg' begins in this area. The RFSS has given the command to SS-Brigadeführer von Gottberg. He handled things very well before and goes at it with dash. Unfortunately, he has soiled the honor of arms with wild persecutions of Jews. I am outraged.
22 Feb. 1943; letter to Gottberg; p. 70	Der Höh.SSuPolF. Süd hat sich mit Recht über diesen selbständigen Befehl beschwert. … Beide Durchkämmungsaktionen verfolgten lediglich politische und keine militärischen Ziele. Diese Auffassung von mir wird bestätigt durch die grosse Zahl … vernichteter Bandenverdächtigter. … Ein ehrgeiziger Kommandeur wird ausserdem dazu verführt, die nicht erkämpften Erfolge … durch Erschiessung von Bandenverdächtigen vorzutäuschen.	The Higher SS and Police Leader South has rightly complained about this order, issued on your own authority. … Both combing-out operations pursued merely political and no military aims. This view of mine is confirmed by the large number of suspected partisans destroyed. … An ambitious commander is moreover tempted to fake successes not won in combat by shooting suspected partisans.
22 May 1943; after a dispute with Prützmann; pp. 76–77	Das Volk denkt nun, im "autoritären" Staat wird alles von oben befohlen und ahnt nicht, dass oben ein ununterbrochener Machtkampf herrscht. So ist auch mir eine wichtige Aufgabe als Bevollmächtigter für Bandenbekämpfung übertragen, muss mir aber jedes bischen Macht erst mühselig erkämpfen, um wirken zu können.	People think that in the 'authoritarian' state everything is ordered from above, and have no idea that up there an unceasing struggle for power prevails. I too have been given an important task as plenipotentiary for anti-partisan warfare, but I must first fight hard for every bit of power to be able to act.

This table presents sample diary quotations in German and in our translation. Ellipses mark omissions and square brackets insertions. Goebbels is cited by date (Fröhlich, 1993), Frank by roll (r.) and frame (fr.) of NARA microfilm T992, Rosenberg by page of Matthäus and Bajohr (2015), and Bach-Zelewski by page of the typed transcript (Bundesarchiv, R 20/45b).

C.2 Model

Figure A14 accompanies Section 4.

Figure A14: Error Probabilities across the Rivalry Boundary

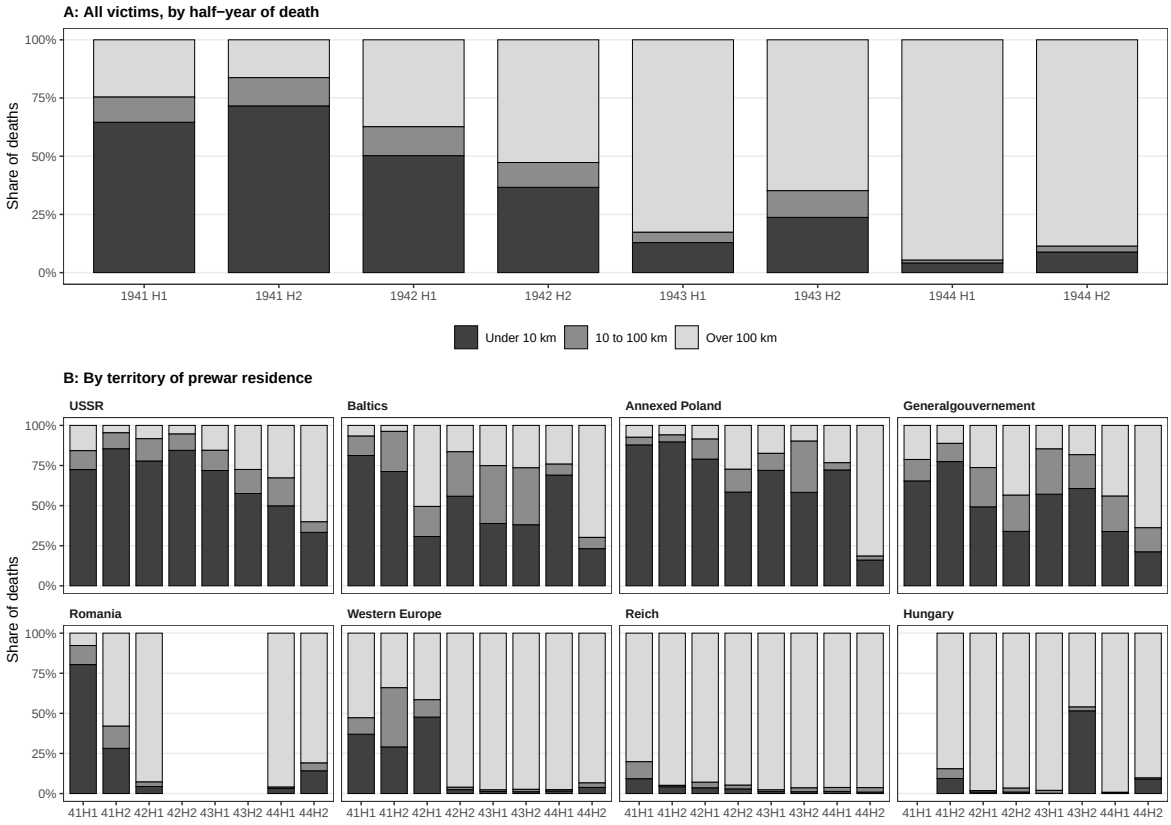


This figure plots the unconditional probability of each error under the optimal arrangement as time pressure $1-\delta$ rises, with $\pi = 0.5$, $b = 1$, $K = 0.1$, $m = 0.2$, and $h = 0.6$. The optimal arrangement switches from control to rivalry at $1-\delta = 0.4$.

C.3 The Holocaust in the East

The exhibits of this subsection accompany Section 5, in its order. Figure A17 shows the event-study design in stylized form. Appendix D details the instrumental-variable and planned-cohort designs of Section 5.3.

Figure A15: Distance from Home at Death



This figure shows the distance from prewar residence to place of death for murdered victims in the Yad Vashem database with both places located and a death dated 1941 to 1944. Panel B shows each territory of prewar residence for half-years with at least 200 dated deaths, ordered by the share killed more than 100 kilometers from home. The Reich includes Austria, the Sudetenland, and the Protectorate. The USSR is the Soviet territory of June 1941 without the Baltic states, plus Transnistria. Hungary and Romania are the wartime states. Western Europe is the occupied west.

Appendix Table A10 reports the regressions across cells behind Figure 8. Appendix Figure A16 maps the Herfindahl–Hirschman index of Figure 8 over every cell occupied by December 1942. Appendix Table A11 uses the index within cells over time, with the units within 50 kilometers of the cell, and estimates

$$y_{ct} = \beta\, H_{ct} + \gamma \log n_{ct} + W'_{ct}\kappa + \alpha_c + \lambda_{zt} + \varepsilon_{ct}, \tag{9}$$

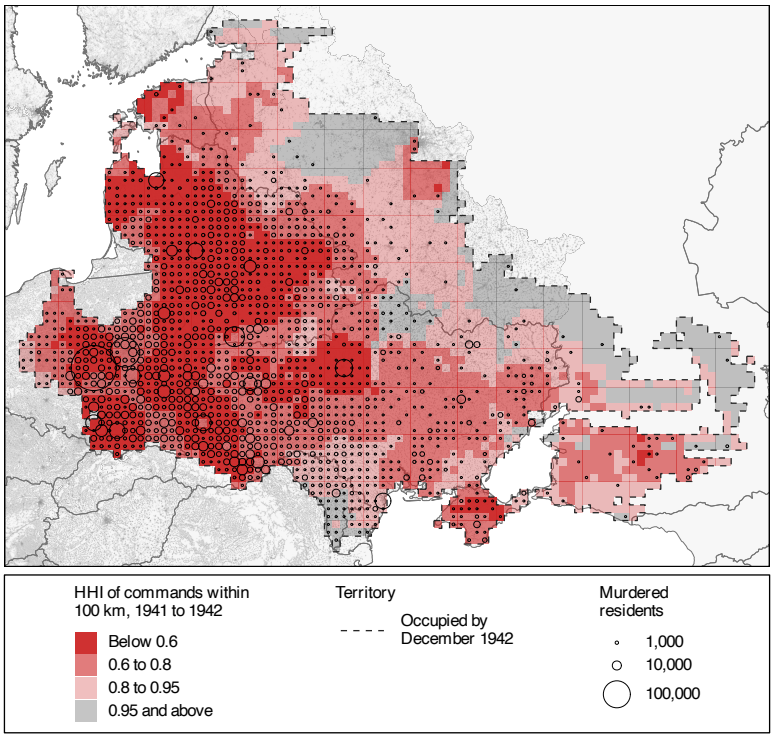
where y_{ct} is the number of residents murdered in the six months from month t per 100 residents, as in equation (7), H_{ct} the index averaged over the month and the next, n_{ct} the units within reach, and W_{ct} the further controls of columns (3) and (4). Deaths in the following six months are 2.8 per 100 residents higher when the index falls from one command to two of equal size, both across all cell-months and in places that one command alone held within 25 kilometers.

Table A10: Concentration of Commands and the Killing of Residents across Cells

	Three commands					Six commands
	(1)	(2)	(3)	(4)	(5)	(6)
HHI of commands within 100 km, 1941 to 1942	−106.11***	−96.73***	−70.05***	−47.62***	−48.55***	−40.35***
	(7.77)	(7.76)	(10.55)	(9.34)	(9.39)	(8.49)
Log recorded residents		3.53***	1.89***	1.27**	2.22***	2.21***
		(0.60)	(0.52)	(0.52)	(0.66)	(0.66)
Log units within 100 km		−3.40**	−2.35	−2.17	−2.01	−1.85
		(1.56)	(1.78)	(1.69)	(1.66)	(1.65)
Log Jewish population (censuses)					−1.76**	−1.75**
					(0.76)	(0.76)
Zone effects	No	No	Yes	No	No	No
Zone-by-occupation-month effects	No	No	No	Yes	Yes	Yes
R^2	0.28	0.33	0.41	0.49	0.50	0.49
Mean of the outcome	63.2	63.2	63.2	63.2	63.2	63.2
Cells	1,661	1,661	1,661	1,661	1,661	1,661

This table regresses residents murdered by May 1945 per 100 recorded residents on the cell's mean Herfindahl–Hirschman index in Figure 8. Columns (2)–(6) control for log recorded residents and log units within reach. Column (3) adds zone effects. Columns (4) to (6) add zone-by-occupation-month effects. Columns (5) and (6) also control for log Jewish population in the Polish census of 1931 and the Soviet census of 1939, with an indicator for cells outside census coverage. Column (6) computes the index over six commands, which split the three by organization: the army, the Waffen-SS, the Security Police, the Order Police, the troops of the Higher SS and Police Leaders, and the SS and Police Leaders. Standard errors in parentheses are clustered in 100-kilometer blocks. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure A16: Concentration of Commands within 100 Kilometers



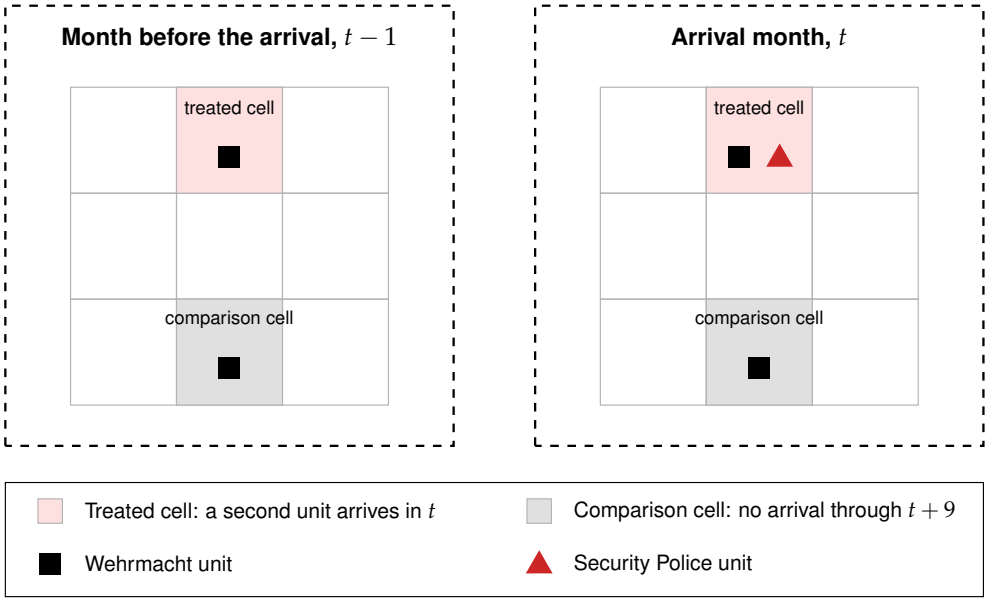
This figure maps the mean Herfindahl–Hirschman index for every 25-by-25-kilometer cell in the zones of Table 2, Panel A, occupied by December 1942, over the 1930 population (Appendix A.2). The index is averaged over each cell's months of occupation from June 1941 to December 1942. It sums the squared shares of the three commands among units within 100 kilometers: the army, the Security Police, and the Higher SS and Police Leaders. Each unit counts with the command to which it reported. A police office or SS brigade placed only in a region counts at the region's center. A cell enters if it has at least one month with a unit within reach. Circles show murdered residents in cells with at least 10. The dashed line outlines the territory occupied by December 1942. Country borders are modern.

Table A11: Concentration of Commands and the Killing of Residents within Cells

	All cell-months				One command within 25 km	Stationing statements	Six commands
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
HHI of commands within 50 km	−5.67***	−5.41***	−5.11***	−4.89***	−5.61***	−0.24	−2.93***
	(1.14)	(1.08)	(1.44)	(1.43)	(1.59)	(1.16)	(0.84)
Log units within 50 km		0.44*	0.42	0.39	0.06	0.65**	0.43
		(0.27)	(0.26)	(0.26)	(0.29)	(0.26)	(0.27)
Commands within 50 km			0.12	−0.01			
			(0.43)	(0.44)			
Two or more commands within 25 km				1.00**			
				(0.47)			
Mean of the outcome	6.52	6.52	6.52	6.52	5.61	7.18	6.52
Cell-months	74,818	74,818	74,818	74,818	30,604	40,287	74,746
Cells	2,306	2,306	2,306	2,306	2,079	2,262	2,306

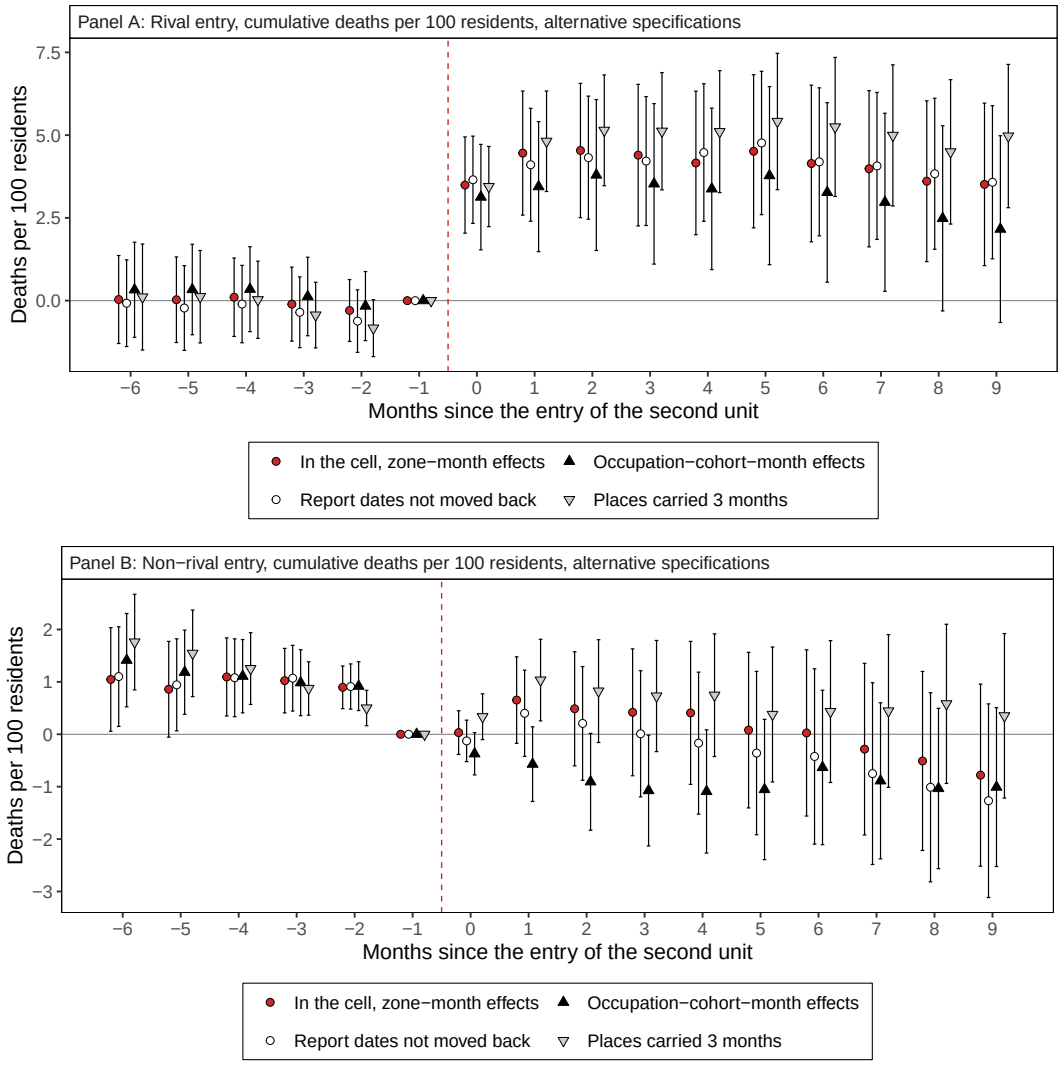
This table regresses resident deaths in the six months beginning in the current month, per 100 recorded residents, on the Herfindahl–Hirschman index of the three commands among the units within 50 kilometers, averaged over the current and following months. The sample consists of cell-months from Table 2, Panel A, with a unit within reach. Columns (2) to (7) control for the log number of these units. Column (3) adds the number of commands within reach, and column (4) adds the presence of two or more commands within 25 kilometers. Column (5) retains cell-months in which one command alone, the army, the Security Police, or the units of the Higher SS and Police Leaders, was within 25 kilometers in the current and following months. Column (6) constructs the index using only dated stationing statements. Column (7) computes the index over the six commands of Table A10, column (6). All regressions include cell and zone-by-month fixed effects. Standard errors in parentheses are clustered in 100-kilometer blocks. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure A17: Event-Study Design



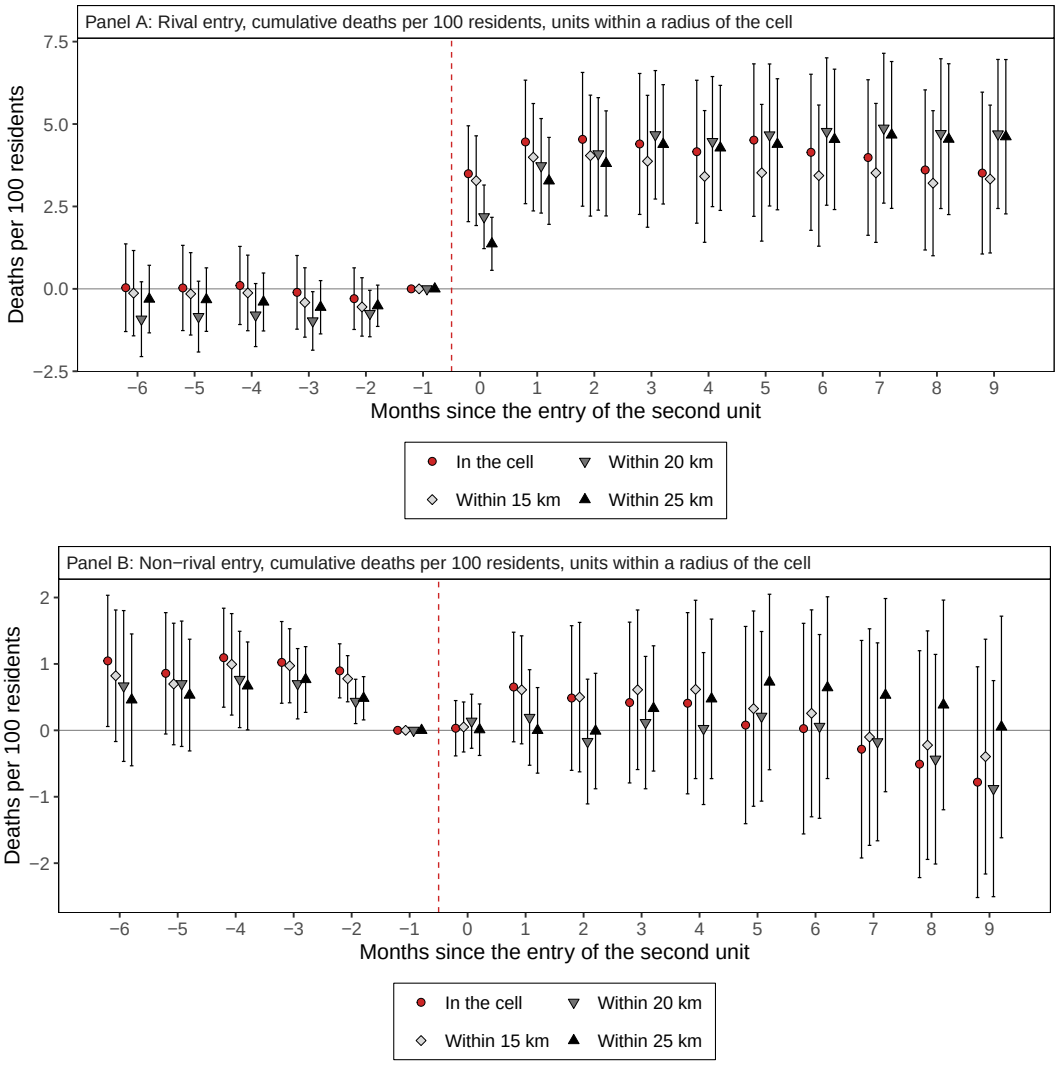
This figure presents a stylized version of the design in Figure 10 for rival entry. In the month before entry, the treated cell (red) and comparison cell (gray) each hold one command, here an army unit. A unit of a second command, here a Security Police unit (red triangle), then enters the treated cell. The comparison cell has no entry through the ninth month after entry. Each entry month defines a stack in which treated and comparison cells have equal total weight.

Figure A18: Robustness



This figure repeats the estimates in Figure 10 for rival entry (Panel A) and non-rival entry (Panel B) under alternative specifications. One uses report dates as printed, without the 18-day lag. Another replaces stack-by-zone-by-month fixed effects with stack-by-occupation-cohort-by-month fixed effects, which compare cells occupied in the same month. The last carries a unit's place three months past its last dated statement.

Figure A19: Units within a Radius of the Cell



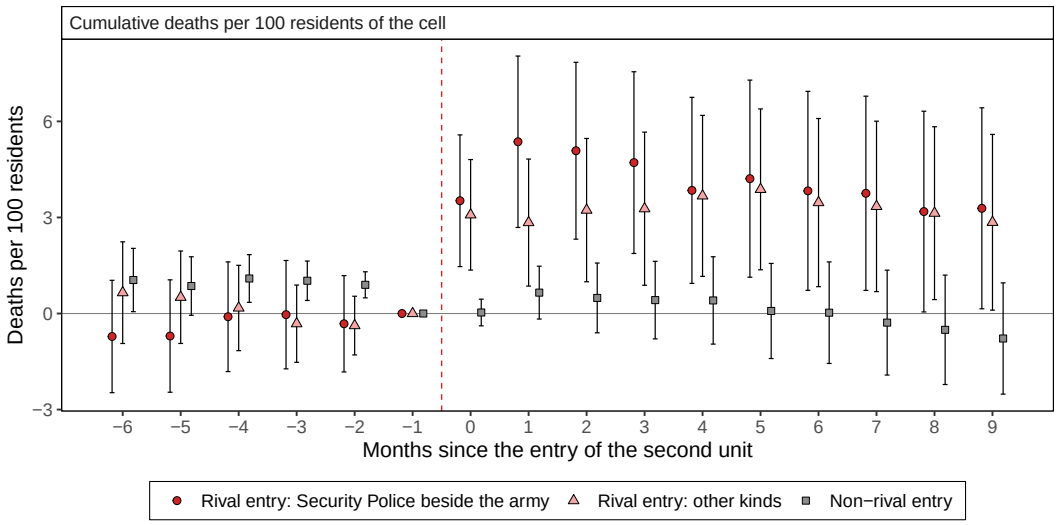
This figure repeats the estimates in Figure 10 for rival entry (Panel A) and non-rival entry (Panel B). For both the command that held the cell the previous month and the entry, a unit is counted in every cell whose center lies within 15, 20, or 25 kilometers of its place rather than in the cell containing that place; these radii reach past the circle inscribed in the cell. The outcome, specification, and standard errors follow Figure 10. Bars show 95 percent confidence intervals.

Table A12: Alternative Denominators

	Rival entry			Non-rival entry		
	Residents, deaths per 100 (1)	Census Jews, deaths per 100 (2)	Inhabitants, deaths per 1,000 (3)	Residents, deaths per 100 (4)	Census Jews, deaths per 100 (5)	Inhabitants, deaths per 1,000 (6)
Treated × post-entry	4.12***	7.01***	6.03***	−0.74	−1.22	0.94
	(1.21)	(2.59)	(1.19)	(0.72)	(1.44)	(0.57)
Leads, joint p	0.616	0.242	0.113	0.001	0.000	0.001
Outcome mean	34.17	18.06	10.50	37.56	31.54	12.91
Treated cells	350	281	350	695	481	726
Observations	160,700	83,467	175,591	117,007	71,924	127,718
Cell FEs	✓	✓	✓	✓	✓	✓
Zone-by-month FEs	✓	✓	✓	✓	✓	✓

This table repeats the event study of Table 2, Panel B, columns (1) and (4), using three denominators for the cumulative murdered residents of each cell: recorded residents (as in Table 2), Jews counted in the Polish census of 1931 and the Soviet census of 1939, and inhabitants in the 1930 population surface (Appendix A.2). The census sample is restricted to cells with at least 20 census Jews and one recorded resident and excludes the Baltic states. The population surface sample is restricted to cells with at least 100 inhabitants. Regressions include cell and zone-by-month fixed effects within each stack. Standard errors in parentheses are clustered in 100-kilometer blocks. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure A20: Rival Entries by Kind



This figure presents the event study in Figure 10 separately for Security Police units entering army-held cells (170 cells), all other rival entries (180 cells), and non-rival entries (695 cells). Each type retains its stacks' comparison cells, reweighted to its treated cells. The outcome, unit presence rule, specification, and standard errors follow Figure 10. Bars show 95 percent confidence intervals.

C.4 Mechanism

The timing tests of Figures 12 and A21 compare, for a pair of actions (a, b) such as an agent's first action and the superior's first approval, the share of policy rounds r in which a precedes b with its mean under reshuffling,

$$\Delta = \frac{1}{N}\sum_{r=1}^{N} \mathbb{1}[t_{ra} < t_{rb}] - \frac{1}{S}\sum_{s=1}^{S}\frac{1}{N}\sum_{r=1}^{N} \mathbb{1}[t_{ra}^{(s)} < t_{rb}^{(s)}], \tag{10}$$

where t_{ra} is the date of action a in round r, and $t^{(s)}$ are the dates after the s-th of $S = 4{,}000$ reshufflings, which keep each round's actions and dates but assign actions to dates at random.

Table A13: Directives on the Killing in the Occupied East, 1941 to 1943

No.	Date	Issuer	Kind	Content	Source
1	3 Mar 1941	Hitler	Direction	Principle for drafting the guidelines: the 'Jewish-Bolshevist intelligentsia', hitherto the 'oppressor' of the people, must be eliminated	Longerich, 2010, p. 182
2	13 Mar 1941	Wehrmacht high command	Jurisdiction	In the army's operational area the Reichsfuehrer-SS receives 'special responsibilities' on Hitler's orders to prepare the political administration, arising from the final struggle between two opposing political systems; his units to act independently of the Wehrmacht	Longerich, 2010, p. 182
3	30 Mar 1941	Hitler	Direction	Speech noted by Halder: a battle of two opposing world views and a battle of annihilation; against Russia, annihilation of the Bolshevist commissars and of the Communist intelligentsia	Longerich, 2010, p. 183
4	28 Apr 1941	Wehrmacht high command	Jurisdiction	Regulation of the deployment of the Security Police and SD within the army: Sonderkommandos to seize material and persons and combat activities hostile to the Reich in the army's area on their own responsibility, subordinate to the army only for movement, supplies, and quarters	Longerich, 2010, pp. 182–183
5	13 May 1941	Hitler	Instruction	Offences of soldiers against civilians to be prosecuted only in exceptional cases; civilian suspects to be dealt with by the troops on the spot; collective measures against places of attacks	Longerich, 2010, p. 183
	19 May 1941	Wehrmacht high command	Direction	Bolshevism the mortal enemy; ruthless measures against Bolshevist agitators, irregulars, saboteurs, Jews and elimination of all resistance	Longerich, 2010, p. 184
	21 May 1941	Himmler	Jurisdiction	HSSPFs to command SS and police troops and Security Police units for tasks Himmler assigns; nine police battalions under the tactical authority of the security divisions, HSSPFs authorized to take direct command for special assignments	Longerich, 2010, p. 184
6	6 Jun 1941	Wehrmacht high command	Instruction	Soviet commissars, as 'the originators of barbarian Asiatic methods of combat', to be 'dealt with' (finished off) by the troops	Longerich, 2010, p. 184
	11 Jun 1941	Himmler	Direction	Himmler summoned the senior SS leaders to put them in the right frame of mind for the extermination they were to carry out; according to Bach-Zelewski after the war he named 30 million as the number by which the Soviet population was to be 'decimated'	Longerich, 2012, p. 522
	17 Jun 1941	Heydrich	Direction	Briefing of the Kommando leaders in Berlin (presumably 17 June) and at Pretzsch: severity, the war as a clash of world views, Jews as a potential enemy; content known from postwar interrogations	Longerich, 2010, p. 190
	29 Jun 1941	Heydrich	Instruction	The Kommandos to initiate 'attempts at self-purification' by local anti-Communist and anti-Jewish circles	Longerich, 2010, p. 190
	30 Jun 1941	Himmler	Instruction	Visiting the border zone, Himmler and Heydrich were informed of the 'measures taken' by the Tilsit commando, the shooting of mostly Jewish men, and 'fully approved' them	Longerich, 2010, p. 197
7	2 Jul 1941	Heydrich	Instruction	Informs the HSSPFs of the instructions to the Einsatzgruppen: to be executed are Comintern functionaries, people's commissars, Jews in party and state posts, and other radical elements ('etc.')	Longerich, 2010, p. 190
	4 Jul 1941	Heydrich	Instruction	Order No. 6: Heydrich informs the Einsatzgruppe chiefs that he has authorized the eastern Security Police commanders and State Police offices to carry out 'cleansing operations' in the newly occupied areas opposite their border zones, confirming the Tilsit executions	Longerich, 2010, p. 197
	8 Jul 1941	Himmler	Instruction	At a meeting of SS and police officers Himmler stated, according to Bach-Zelewski after the war, that 'as a matter of principle any Jew' was 'to be regarded as a partisan'	Longerich, 2012, p. 525

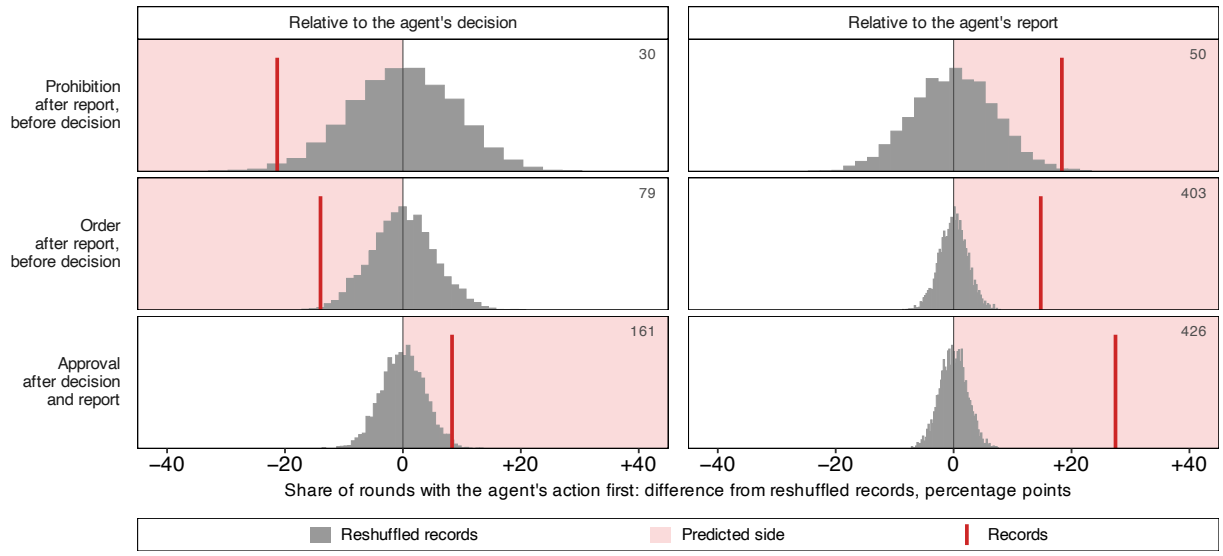
No.	Date	Issuer	Kind	Content	Source
	9 Jul 1941	Order Police main office	Direction	Speaking to members of Police Regiment Mitte, Daluege called for 'Bolshevism finally to be eradicated'	Longerich, 2012, p. 525
8	17 Jul 1941	Hitler	Jurisdiction	Civil administration of the occupied East to Rosenberg's ministry; Himmler restricted to providing security through the police, with power to appoint HSSPFs and SSPFs	Longerich, 2012, p. 527
	17 Jul 1941	Heydrich	Instruction	Operational Order No. 8, after Heydrich's agreement with the Wehrmacht of 16 July: screening teams to select professional revolutionaries, Red Army political officers, 'fanatical' Communists, and 'all Jews' among the Soviet prisoners of war for killing	Hilberg, 2003, p. 347
	19 Jul 1941	Himmler	Jurisdiction	Both SS cavalry regiments sent to Baranowicze and placed at the disposal of HSSPF Bach-Zelewski for a systematic combing of the Pripet marshes	Cüppers, 2005, p. 135
	22 Jul 1941	Himmler	Jurisdiction	March order to Lemberg for operations under Jeckeln's command in the Ukraine	Cüppers, 2005, p. 135
	23 Jul 1941	Himmler	Jurisdiction	Himmler placed eleven further Order Police battalions under the three HSSPFs in the occupied Soviet territory	Cüppers, 2005, p. 135
	25 Jul 1941	Himmler	Jurisdiction	Himmler ordered the uniform formation of large units of local collaborators, the Schutzmannschaften	Cüppers, 2005, p. 135
	27 Jul 1941	Himmler	Instruction	Guidelines for the combing of marsh areas by cavalry units: those suspected of helping partisans to be shot, women and children to be deported, villages burned; no concrete military targets	Cüppers, 2005, p. 138
	31 Jul 1941	Göring	Direction	Goering signs a declaration drafted by Heydrich that authorizes him to make all organizational, functional, and material preparations for a 'total solution of the Jewish question within the German sphere of influence in Europe' and to present an overall draft	Longerich, 2010, p. 260
	1 Aug 1941	Heydrich	Direction	Radio telegram: regular reports on the work of the Einsatzgruppen in the East are to be sent to the Fuehrer	Longerich, 2012, p. 530
	1 Aug 1941	Himmler	Instruction	Express command of the Reichsfuehrer-SS: all Jews must be shot, Jewish women to be herded into the marshes	Longerich, 2012, p. 531
	12 Aug 1941	Himmler	Instruction	Himmler, who had rebuked Jeckeln's brigade by telex on 11 August, summoned Jeckeln and appeared 'indignant' at his activity, whereupon Jeckeln increased the murders; the first of the talks of 12 to 15 August (Baranowicze 14 August, Minsk 15 August) in which Himmler passed a radically extended order to murder the Jewish population to all three HSSPFs	Longerich, 2012, p. 533
	14 Aug 1941	Himmler	Direction	Daily situation reports by radio or telex directly to Himmler, with copies to Daluege, Heydrich, and Knoblauch	Witte et al., 1999, p. 191
9	18 Aug 1941	Reichskommissar Ostland	Jurisdiction	Signed provisional guidelines: minimum measures of the civil administration only if further measures toward a definitive solution are not possible; measures of the Security Police not covered	Longerich, 2010, p. 235
	12 Sep 1941	Heydrich	Instruction	Directive to the screening teams to select more carefully: an engineer not necessarily a Bolshevist, Muslims not to be taken for Jews, other nationalities to be 'treated according to directive' only if fanatical Bolshevists; shootings not in the middle of the camps, and executions never public	Hilberg, 2003, p. 351
	12 Sep 1941	Wehrmacht high command	Direction	Keitel demanded 'ruthless and energetic action, above all also against Jews, the main carriers of Bolshevism', insofar as they were not doing forced labor in labor columns	Pohl, 2009, p. 166

No.	Date	Issuer	Kind	Content	Source
	3 Oct 1941	Himmler	Instruction	Himmler visited the commando at Krivoy Rog; afterwards it shot the Jewish women there as well, the last Kommando of Einsatzgruppe C to do so	Longerich, 2012, p. 533
10	Oct 1941	Reichskommissar Ostland	Instruction	Lohse banned the 'arbitrary executions of Jews in Libau' because of the manner of execution; on 15 November he defended the ban and asked whether all Jews in the Ostland were to be killed regardless of age, sex, and economic interests	Longerich, 2010, p. 298
	28 Oct 1941	Wehrmacht high command	Direction	Order of the army's Quartermaster General recording that Hitler had found 'excellent' Reichenau's order of 10 October, which demanded that the soldier understand the necessity of harsh but just countermeasures (Suehne) against Jewish subhumanity	Hilberg, 2003, p. 331
	29 Oct 1941	Heydrich	Instruction	Deployment Order No. 14 instructs the Einsatzgruppen in the occupied East to detach Sonderkommandos to search the prisoner-of-war camps, extending Order No. 8, under which 'all Jews' were to be separated out	Longerich, 2010, p. 247
	7 Nov 1941	Reichskommissar Ostland	Instruction	To prevent further shootings of Jewish skilled workers, after complaints from Wehrmacht authorities	Longerich, 2010, p. 297
	30 Nov 1941	Himmler	Instruction	Entry in Himmler's telephone log on a conversation with Heydrich: 'Transport of Jews from Berlin. No liquidation.'	Longerich, 2010, p. 299
	1 Dec 1941	Himmler	Instruction	Unauthorized acts and contraventions of the guidelines issued by Himmler or by the RSHA on his behalf for the treatment of the Jews resettled to the Ostland will be punished; Jeckeln summoned to discuss the Jewish question on 4 December	Longerich, 2010, p. 299
	3 Dec 1941	Reichskommissar Ostland	Instruction	After a complaint of the Wehrmacht, the Reichskommissar asks most emphatically that the liquidation of Jews working as skilled workers in Wehrmacht armament plants and repair workshops who cannot yet be replaced by locals be prevented, and the same for Jewish skilled workers in other plants important to the war economy	Hoppe and Glass, 2011, pp. 583–584
	12 Dec 1941	Hitler	Direction	Speech recorded by Goebbels: the world war is here, so the destruction of the Jews must be the necessary consequence	Longerich, 2010, pp. 305–306
	18 Dec 1941	Hitler	Direction	'Jewish question / to be extirpated as partisans': renewed confirmation that the mass murder of the Soviet Jews continue and intensify	Longerich, 2010, p. 306
11	18 Dec 1941	Ministry for the occupied East	Instruction	The ministry replied that economic interests were to be disregarded in the solution of the Jewish question and that further questions be settled directly with the Higher SS and Police Leader, overriding Lohse's objection; Lohse dropped his protest	Hilberg, 2003, p. 392
	Jan 1942	Order Police main office	Instruction	Tried to reduce the part of individual-service police in mass shootings to cordons and security	Curilla, 2006, p. 59
12	20 Jan 1942	Heydrich	Direction	Heydrich outlines the mass murders in the occupied territories to senior officials of the Party, the SS, and the ministries as part of a 'solution to the European Jewish question' ordered by Hitler and directed by the RSHA, to ensure they share knowledge of and responsibility for the policy; invited on 29 November for 9 December and postponed on 8 December to 20 January 1942	Longerich, 2010, p. 305
	19 Jul 1942	Himmler	Instruction	The 'resettlement' of the entire Jewish population of the General Government to be completed by 31 December 1942; after that date no Jews to remain except in the assembly camps of Warsaw, Czestochowa, Cracow, and Lublin	Longerich, 2010, p. 335
	28 Jul 1942	Himmler	Direction	'The occupied eastern territories will become free of Jews. The Fuehrer has placed the implementation of this very burdensome order on my shoulders.'	Longerich, 2012, p. 622

No.	Date	Issuer	Kind	Content	Source
	28 Aug 1942	Civil administration, Volhynia-Podolia	Instruction	Agreement to kill all the Jews of the General Commissariat of Volhynia-Podolia within five weeks except 500 skilled workers; Koch's deputy assured 'hundred per cent' clearance	Longerich, 2010, p. 350
	27 Oct 1942	Himmler	Instruction	The Pinsk ghetto to be dealt with and destroyed immediately despite economic concerns; 1,000 male workers to the Wehrmacht if the action allows	Longerich, 2010, p. 351
	19 Jun 1943	Hitler	Instruction	Hitler ordered that the 'evacuation' of the Jews be radically enforced and seen through in the next three to four months despite any unrest	Longerich, 2010, p. 379
13	21 Jun 1943	Himmler	Instruction	All Jews living in the ghettos of the Ostland to be placed in concentration camps and barred from leaving them for work; ghetto inmates not needed to be 'evacuated to the east', that is, murdered	Longerich, 2012, p. 667

This table lists the 50 directives on killing in occupied Soviet territory issued above field units from 1 March 1941 to 31 December 1943. Dates follow the sources in the last column. No. is the number in Figure 11. Kind follows Section 6: a jurisdiction grant authorizes an agency or divides authority among agencies, a direction states a goal or fixes a procedure, and an instruction names, bounds, or approves an act. Göring's authorization of 31 July 1941, Hitler's speech of 12 December 1941, and the Wannsee Conference concern all of Europe, including the occupied East. Directives dated to a month are plotted at midmonth.

Figure A21: Timing of Instructions within Policy Rounds



This figure applies the test in Figure 12, Panel C, to the timing of the superior's first prohibition, order, and approval (rows) relative to the agent's first decision and first report (columns) in the policy rounds of the five diaries at both levels. A report includes a proposal or a request. Each cell shows the share of rounds in which the agent's action came first minus the mean share over 4,000 reshufflings. The red line marks the records, gray bars the reshufflings, and the pink area the side the model predicts. Each cell reports the number of rounds.

D Instrumental-Variable and Planned-Cohort Designs

This appendix details the two designs of Section 5.3 that complement the event study of Section 5.2. The first instruments the presence of a second command with the documented transfers of the commands. The second compares only cells that the invasion plan scheduled to fall behind the front in the same month.

D.1 Design

The documented transfers are the 892 executed moves of the three commands from June 1941 to May 1945 that the war diary of the Armed Forces High Command, the Einsatzgruppen documents and reports, and Himmler's appointment books record (Appendix A.6). We classify each transfer by the commands present within 25 kilometers of its destination in the month before. A transfer is a rival transfer if exactly one other command held the destination and the mover's command arrived in the month or the next; 107 transfers are rival transfers. A transfer is a non-rival transfer if the mover's command alone held the destination; 107 transfers are non-rival transfers.

The counterfactual destinations of a transfer are the five nearest towns that were in the same state in the month before (held by one command other than the mover's for a rival transfer, by the mover's command alone for a non-rival transfer), in the same army group area, and at least 40 kilometers from the transfer's destination and from its origin. The towns are the 289 places of the sample that the transfer records name at least twice. A destination outside the army group areas, in the General Government or annexed Poland, draws the nearest towns of any area.

The instrument is the inverse hyperbolic sine of the distance from a cell to the nearest destination of a documented move in the month, capped at 300 kilometers. Cells near the routes of the commands may differ from other cells in ways that also predict killing. The recentered instrument therefore subtracts from the instrument its expectation when each rival transfer goes, with equal probability, to its destination or to one of its five counterfactual destinations, the other moves held fixed (Borusyak and Hull, 2023). It thus compares a cell near a transfer's destination with cells near the places the transfer could have reached. Table A14 regresses the deaths per 1,000 residents in the three months before, the deaths per 100 residents to the month before, and the presence of the army and of the SS and police in the month before on the instrument. Without recentering, cells nearer a destination more often held the army and the SS and police in the month before (column 1). After recentering, every standardized difference is below 0.01 (column 2).

The sample is the 2,322 cells of 25 by 25 kilometers in occupied Poland and the occupied Soviet territory by month from June 1941 to May 1945. The treatment is the presence of a second command (Table 2, Panel A). The outcomes are the three of Table 2, measured over the six months from the month. All regressions include cell and zone-by-month fixed effects, and standard errors are clustered in blocks of 100 kilometers.

We also estimate the event study of equation (6) around the rival transfers: the cells within 25 kilometers of a transfer's destination against the cells within 25 kilometers of its counterfactual destinations, both held by one command in the month of the transfer.

D.2 Estimates

Table A15 reports the estimates. The recentered instrument moves the presence of a second command: a rival transfer 25 rather than 100 kilometers from a cell raises it by 0.17, with a first-stage F-statistic of 44 (column 1). The presence of a second command adds 16.4 deaths per 100 residents over six months (standard error 7.7), against 2.0 in OLS (Table 2, Panel A), and 16.3 deaths

Table A14: Balance of the Instrument

	Distance (1)	Recentered distance (2)
Deaths per 1,000 residents, months $t-3$ to $t-1$	−0.0062	0.0095
	(0.0089)	(0.0066)
Deaths per 100 residents to $t-1$	0.0141	0.0063
	(0.0059)	(0.0028)
Army within 25 km, month $t-1$	−0.0281	0.0058
	(0.0079)	(0.0042)
SS or police within 25 km, month $t-1$	−0.0792	0.0004
	(0.0108)	(0.0050)
Cell FEs	✓	✓
Zone-by-month FEs	✓	✓

This table regresses each pre-period covariate on the instrument, both standardized, in the sample of Table A15. Column (1) uses the inverse hyperbolic sine of the distance from the cell to the nearest destination of a documented move that month, capped at 300 kilometers. Column (2) subtracts its expectation when each of the 107 rival entries is assigned with equal probability to its destination or one of its five counterfactual destinations (Borusyak and Hull, 2023). Each cell reports the standardized difference per standard deviation of the instrument. The army and the SS and police are present within 25 kilometers. All regressions include cell and zone-by-month fixed effects. Standard errors in parentheses are clustered in 100-kilometer blocks.

of women and children per 100 women and children (7.7).

Table A16 varies the number of counterfactual destinations from three to eight: the recentered estimate is 15.5 to 19.8 deaths per 100 residents over six months. In the event study around the transfers (Table A16, Panel A), deaths in the cells around the destination rise by 5.9 per 100 residents relative to the cells around five counterfactual destinations (standard error 1.8). The coefficients before the transfer are not jointly different from zero. Within two months, a rival command reaches 91 percent of the cells around the destination and 46 percent of the cells around the counterfactual destinations.

The recentered IV estimate exceeds OLS for two reasons. Measurement error in the recorded presence of a second command attenuates OLS, while the instrument does not depend on recording accuracy. The instrument also weights entries in 1941: 74 of the 107 rival transfers occurred between June and December 1941. Effects per entry are of the same order across designs. A rival transfer 25 rather than 100 kilometers from a cell adds 2.8 deaths per 100 residents over the next six months (Table A15), compared with 4.1 in the event study of all rival entries (Table 2, Panel B) and 5.9 in the event study around the same transfers (Table A16, Panel A). The same contrast raises the recorded presence of a second command by only 0.17. The IV estimate divides the effect on deaths by the effect on recorded presence.

D.3 Non-Rival Transfers

Columns (4) to (6) of Table A15 repeat the estimates for the non-rival transfers, with the presence of a further unit as the treatment and the recentered instrument built on the transfers of a further unit to a place the mover's command held alone. These transfers barely move the presence of a further unit: the first-stage F-statistics are at most 3.1. The comparison between rival and non-rival entries therefore rests on Table 2.

Table A15: Rival Presence and the Killing of Residents: IV Estimates

	Rival entry			Non-rival entry		
	All residents, deaths per 100	Women and children, deaths per 100	Women and children, in excess of men	All residents, deaths per 100	Women and children, deaths per 100	Women and children, in excess of men
	(1)	(2)	(3)	(4)	(5)	(6)
IV, distance recentered over counterfactual destinations						
Presence within 25 km	16.4**	16.3**	13.2*	−14.5	−13.5	−6.3
	(7.7)	(7.7)	(7.4)	(16.5)	(16.9)	(16.3)
First stage, per transfer	0.170***	0.170***	0.169***	0.058*	0.058*	0.058*
	(0.026)	(0.026)	(0.025)	(0.033)	(0.033)	(0.033)
Reduced form, per transfer	2.80**	2.77**	2.23*	−0.84	−0.79	−0.37
	(1.35)	(1.35)	(1.29)	(0.93)	(0.94)	(0.93)
First-stage F	44.1	44.1	44.1	3.1	3.1	3.1
Outcome mean	5.66	5.57	8.10	5.66	5.57	8.10
Observations	92,822	92,822	92,822	92,822	92,822	92,822
Cell FEs	✓	✓	✓	✓	✓	✓
Zone-by-month FEs	✓	✓	✓	✓	✓	✓
Control for men's deaths			✓			✓

This table instruments presence as defined in Table 2, Panel A, using the same sample and outcomes: two or more commands for rival entry (columns 1 to 3) and two or more units of one command and no other for non-rival entry (columns 4 to 6). The instrument is the inverse hyperbolic sine of the distance from the cell to the nearest destination of a documented move that month, capped at 300 kilometers, less its expectation when each of the 107 rival entries (columns 1 to 3) or the 107 non-rival entries (columns 4 to 6) is assigned with equal probability to its destination or one of its five counterfactual destinations (Appendix D.1, Borusyak and Hull, 2023). The first stage and the reduced form are reported per transfer 25 rather than 100 kilometers from the cell. All regressions include cell and zone-by-month fixed effects. Standard errors in parentheses are clustered in 100-kilometer blocks. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

D.4 Planned Cohort

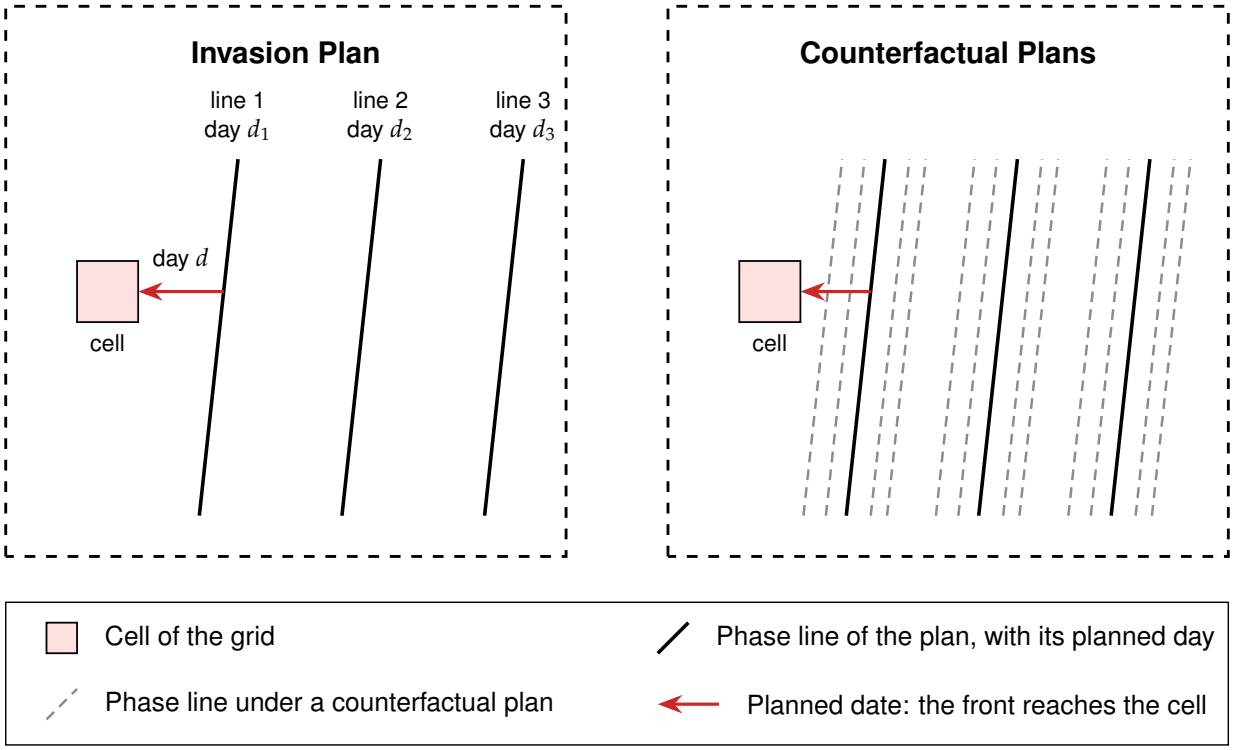
The planned-cohort design uses the invasion plan, which fixed before the campaign the month in which each cell was to fall behind the front (Appendix Figure A22), to measure where units were expected to enter. Table A17 compares each treated cell of the stacks of Figure 10 only with comparison cells of its stack that the plan scheduled for the same month (Panel B). Within the planned cohort, deaths rise by 1.9 per 100 residents after a rival entry (standard error 1.1) and change by −0.4 after a non-rival entry; with the documented destinations as controls, the rise after a rival entry is 2.3 (1.1).

Table A16: Entry Transfers and Their Counterfactual Towns

	3 towns (1)	5 towns (2)	8 towns (3)
Panel A: Event study, cells around the realized destination against cells around the counterfactual towns			
Treated × post-transfer	4.80***	5.90***	6.84***
	(1.81)	(1.77)	(1.59)
Leads, joint p	0.538	0.135	0.029
Rival arrival within two months, treated	0.91	0.91	0.91
Rival arrival within two months, controls	0.47	0.46	0.48
Panel B: Distance to the nearest destination, recentered over the counterfactual towns			
First stage, per transfer	0.215***	0.170***	0.164***
	(0.029)	(0.026)	(0.022)
First-stage F	54.7	44.1	53.9
2SLS: deaths per 100 residents over six months	15.5**	16.4**	19.8**
	(7.5)	(7.7)	(7.8)
Rival entries	107	107	107
Entries with treated and comparison cells	106	106	106
Treated cells	156	156	156
Control cells	251	295	345

This table reports the transfer design with three, five, and eight counterfactual towns per rival transfer. Panel A reports the event study in equation (6), comparing cells within 25 kilometers of an entry's realized destination with cells within 25 kilometers of its counterfactual towns, both held by one command in the entry month: the single post-transfer coefficient in cumulative deaths per 100 residents, the joint test of the leads, and the share of treated and comparison cells reached by a rival unit within two months. One entry's destination cell has no recorded residents and forms no stack. Panel B reports estimates using the recentered instrument of Table A15 for the same transfers and towns: the first stage for the presence of two or more commands within 25 kilometers, per transfer 25 rather than 100 kilometers from the cell, and the IV estimate for deaths per 100 residents over six months. Regressions include cell and zone-by-month fixed effects, specified within each stack in Panel A. Standard errors in parentheses are clustered in 100-kilometer blocks. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure A22: Planned-Cohort Design



This figure shows a stylized version of the design in Panel B of Table A17. The invasion plan uses phase lines from Directive No. 21 and the deployment directive of 31 January 1941. The front was to reach these lines on dates from the timetable of the Marcks study of August 1940 (left). A cell's planned date is the day the planned front reaches it, interpolated between phase lines. Cells with planned dates in the same month form a planned cohort. Each of 200 counterfactual plans shifts every phase line after the border by up to 21 days and 100 kilometers (right, dashed). A cell's expected exposure is the share of these plans under which its planned date has passed.

Table A17: Planned Cohort

	Rival entry			Non-rival entry		
	No controls (1)	Exposure (2)	Destinations (3)	No controls (4)	Exposure (5)	Destinations (6)
Panel A: All comparison cells						
Treated × post-entry	4.12***	4.31***	4.71***	−0.74	−0.67	−0.65
	(1.21)	(1.21)	(1.23)	(0.72)	(0.72)	(0.71)
Leads, joint p	0.616	0.266	0.627	0.001	0.002	0.001
Outcome mean	34.17	34.17	34.17	37.56	37.56	37.56
Treated cells	350	350	350	695	695	695
Observations	160,700	160,700	160,700	117,007	117,007	117,007
Panel B: Comparison cells of the same planned cohort						
Treated × post-entry	1.88*	1.93*	2.33**	−0.40	−0.38	−0.28
	(1.13)	(1.13)	(1.14)	(0.82)	(0.82)	(0.81)
Leads, joint p	0.634	0.278	0.693	0.022	0.040	0.020
Outcome mean	31.48	31.48	31.48	37.26	37.26	37.26
Treated cells	345	345	345	693	693	693
Observations	95,765	95,765	95,765	90,022	90,022	90,022
Cell FEs	✓	✓	✓	✓	✓	✓
Zone-by-month FEs	✓	✓	✓	✓	✓	✓

This table reports post-entry difference-in-differences coefficients and the joint test of the leads for cumulative deaths per 100 residents using the stacks in Figure 10. Columns 1 to 3 cover rival entries, and columns 4 to 6 cover non-rival entries. The Exposure columns control for the share of 200 counterfactual invasion plans under which the cell has fallen behind the front by the current month. Each plan moves every phase line after the border by up to 21 days and 100 kilometers. The Destinations columns control for whether a planned or ordered move in the sources of Appendix A.6 names a destination within 25 kilometers of the cell by the current month and whether a canceled or postponed move does so. Panel B compares each treated cell only with comparison cells in the same stack and planned cohort. A planned cohort comprises cells the planned front reaches in the same month (Appendix Figure A22). Each cohort's comparison cells carry the weight of its treated cells. Regressions include cell and zone-by-month fixed effects within each stack. Standard errors in parentheses are clustered in 100-kilometer blocks. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

E Proofs

Throughout, V denotes the ruler's expected payoff (2) under the arrangement and protocol in question. V_{rivalry} and V_{control} denote the values of the tournament and the control protocol, the corresponding entries of Table 1; Propositions 2 and 4 compare arrangements through these protocol values.

Proof of Lemma 1. We first bound what any protocol can deliver in any arrangement. Fix an appointment rule and instructions whose conditions are verifiable there (Assumptions 1 and 2). Early implementation is decided before anyone knows θ, so any protocol induces some probability $q_E \in [0,1]$ of early implementation, independent of θ; absent early implementation, implementation at $\theta = 1$ requires an order (Assumption 3), timely with probability δ (a single draw, so ordering a second agent adds nothing), and the ruler issues none at $\theta = 0$, since a $\theta = 0$ order only costs him. The gross policy payoff of any protocol is therefore at most

$$q_E\big[\pi b - (1-\pi)h\big] + (1-q_E)\, q_O\, \pi\delta b \;\le\; \max\{\pi b - (1-\pi)h,\ \pi\delta b,\ 0\} \tag{11}$$

for some $q_O \in [0,1]$, the probability of an order at $\theta = 1$: a convex combination of early implementation, the timely order, and nothing.

Routine administration. Neither verifiable event exists, so every announced rule is empty and the appointment cannot respond to the agent's initiative. Initiative changes the agent's payoff by $-c < 0$, so $a_1 = 0$, and the ruler gets 0.

Rivalry. The ranking is verifiable, and the announced rule "prize to the agent ranked ahead, ties by lottery" is applied (Assumption 1). Given the rule, taking the initiative against a rival who does the same yields $R/2 - c$ in expectation, against 0 for conceding; against an abstaining rival it yields $R - c$, versus $R/2$ in expectation from abstaining. By Assumption 4 both comparisons favor the initiative, so it is strictly dominant in the initial-period game and $(1,1)$ is its unique equilibrium. Implementation is early in both states: $V = \pi b - (1-\pi)h - K$, attaining the first argument of (11). No order exists in this arrangement, so every protocol here pays $q_E[\pi b - (1-\pi)h] - K$ for some $q_E \in [0,1]$: the tournament is the best protocol whenever $\pi b - (1-\pi)h \ge 0$, and announcing nothing pays $-K$ otherwise. Table 1 reports the tournament value.

Control. Certified defiance is verifiable. Defying the prohibition costs $c + P$ now, saves at most the cost c of a later execution, and forfeits the prize that compliance keeps, which defiance can never earn back (Assumption 2): compliance is strict for every $P \ge 0$. Refusing the order is certified defiance, forfeiting the prize (credible by Assumption 1) and drawing P, while executing pays $R - c > 0 \ge -P$. No verifiable event rewards initiative and no order can come early (Assumption 3), so $q_E = 0$ for every protocol here; the ruler issues the order at $\theta = 1$ and $V = \pi\delta b - m$, attaining the second argument of (11).

Both instruments. Every protocol is bounded by (11), and the two pure protocols above are both available, so $V = \max\{\pi b - (1-\pi)h,\ \pi\delta b\} - K - m$ (the no-implementation argument never binds since $\pi\delta b > 0$), as in Table 1. □

Proof of Proposition 2. By Lemma 1, the both-instruments value is $\max\{V_{\text{rivalry}} - m,\ V_{\text{control}} - K\} < \max\{V_{\text{rivalry}},\ V_{\text{control}}\}$ for all parameters, since $K, m > 0$, and the bound (11) covers every protocol, not only the two listed; so paying for both instruments is never optimal. Routine administration pays 0: if $\max\{V_{\text{rivalry}}, V_{\text{control}}\} > 0$, rivalry or control pays strictly more; otherwise every alternative pays at most 0 and routine administration is weakly optimal, uniquely so when the maximum is strictly negative. □

Proof of Proposition 3. Under rivalry, implementation occurs in both states and is timely: the false-positive event is $\{\theta = 0\}$, probability $1 - \pi$, and no false negative occurs. Under control, implementation occurs only on a timely order at $\theta = 1$: no false positive occurs, and the false-negative event is $\{\theta = 1, \text{ late}\}$, probability $\pi(1 - \delta)$. The jump statement restates these numbers on either side of a switch in the optimal arrangement. □

Proof of Proposition 4. $V_{\text{rivalry}} - V_{\text{control}} = \pi b - (1 - \pi)h - K - \pi\delta b + m = D$ as in (3). The derivatives are $\partial D/\partial(1-\delta) = \pi b > 0, \partial D/\partial m = 1 > 0, \partial D/\partial b = \pi(1-\delta) > 0, \partial D/\partial \pi = (1-\delta)b + h > 0, \partial D/\partial h = -(1-\pi) < 0, \partial D/\partial K = -1 < 0$. □

Proof of Remarks 1 and 2. The both-instruments arrangement is never optimal (Proposition 2). Rivalry is (weakly) optimal if and only if $D \geq 0$ and $V_{\text{rivalry}} \geq 0$; control if and only if $D \leq 0$ and $V_{\text{control}} \geq 0$. V_{rivalry} does not depend on δ, D increases in $1 - \delta$, and $V_{\text{control}} = \pi\delta b - m$ decreases in $1 - \delta$, which gives the first sentence of Remark 1. If $V_{\text{rivalry}} < 0$, rivalry is not optimal, so a domain that leaves control goes to routine administration. Remark 2 compares parts (i) and (ii) of Lemma 1; a further unit under the same agent creates no ranking, so part (i) still applies. □

Proof of Remark 3. Under the tournament rule with the demotion and the sanction, an agent's expected payoff is $(R - L)/2 - c - (1 - \pi)qX$ if both agents take the initiative, $R - c - (1 - \pi)qX$ if only he does, $-L$ if only his rival does, and $(R - L)/2$ if neither does. Against a rival who takes the initiative, initiative pays more than restraint if and only if $(R - L)/2 - c - (1 - \pi)qX > -L$; against a rival who refrains, if and only if $R - c - (1 - \pi)qX > (R - L)/2$. Both inequalities reduce to (4), so initiative is strictly dominant exactly when (4) holds, and restraint is strictly dominant when the reverse strict inequality holds. The sanction applies only at $\theta = 0$, but agents choose before θ is known, so it enters both comparisons through its expectation and cannot make initiative depend on θ. With $L = X = 0$, (4) is Assumption 4. Neither L nor X enters (2). Under rivalry, the ruler's payoff is $q_E[\pi b - (1 - \pi)h] - K$, and a sanction leaves $q_E = 1$ or lowers it. Rivalry is optimal only if $\pi b - (1 - \pi)h \geq K > 0$, so a sanction that deters initiative strictly lowers his payoff. □

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